Intro to Geometry

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In this pdf, we'll just introduce language of homological algebra, not going to the applications e.g. geometry. In category theory, an abelian category

generates a category of chain complexes, that contains homological and homotopical information, whose calculation is done by transformating category to homotopic/derived/triangulated/dg/∞-category, and all of them is distinct and very interesting, and the key idea of homological algebra is that homology is a linearization of homotopy, so generally speaking homology is easier to calculate, while homotopy is not, but we are missing many data in the process of linearization. On the other hand, ∞-algebra is another linearization of homotopy without losing data, but this is rather complicated. I'll also introduce several concrete homology including $A/L/E-\infty/BV$-algebra and coalgebra.

Contents

The major purpose of this pdf is to introduce homological technique to study geometry both algebraically and differentially. What is homological algebra? It depends, but homological algebra is a linearlization of geometry or its abelianization, which can measure some geometric invariants.

The alternating sum of dimension of homology classes of each degree is Euler characteristic. This is a topological invariant, but in sheaf cohomology, it helps us define Hilbert polynomial, the flatness condition of the algebraic variety.

If the given geometry is complex geometry, homological algebra has extra structures e.g. Hodge structures.

vector bundles. First, the question is how vector/fiber bundle of a geometry arises homological structures. I will introduce what is vector bundle and fibre bundle of a smooth real manifold. These defines global/local sections, where we can find de Rham cohomology. If it further looks carefully, the algebraic structure (complex or almost complex) generates extra data on de Rham cohomology group called Hodge decomposition. Other than vector bundle, Sheaf is a nice generalization of vector bundles, for at least, it has a functorial property, and it can generalize notion of algebraic variety to any moduli space. I'll introduce stack(orbifold) and groupoid theory.

What kind of structures does homological algebra has? As shown in de Rham cohomology, homological algebra is a linearlization of geometry. Or from a categorical viewpoint, it is derived category, having triangulated structures.

1 Preliminary

1.1 Linear Algebra

The very question in linear algebra is what is the specific applications in algebra and geometry. First of all, how trace is used in Lagrangian, or trace incharacter theory. Exterior algebra for volume form and Hodge star operator.

Definition 1.1.1 *(Inner Product)*
Inner product $< \cdot, \cdot >$ *is a biliner form* $V \times V \to \mathbb{R}$ *for some vector space* V.

Definition 1.1.2 *(Hermitian Inner Product)*
Hermitian inner product is

- *bilinear,*
- $< \alpha u, v >= \alpha < u, v >$
- $< u, \alpha v >= \overline{\alpha} < u, v >$
- $\overline{< u, v >} =< v, u >$
- $< u, u > \geqq 0$ *where* $< u, u >= 0$ *iff* $u = 0$.

A basic example if $h(z, w) = \Sigma z_i \overline{w_i}$

Definition 1.1.3 $< A^* u, v >=< u, Av >$

Definition 1.1.4 *(Trace)*

Example 1.2 *(Trace)*

- *(Character)*
  A character of ρ *is* $\chi(g) = Tr(\rho(g))$

- *(Hermitian metric)*
  $< A, B >= Tr(A^\dagger B)$

- *(Idempotent Matrix)*
  If $A = A^2$, $Tr(A) = dim(A)$ *because eigenvalue of* A *is either* 1 *or* 0. *So if* $A = Q^{-1} \Lambda Q$ *where* Λ *is a diagonal matrix whose value is either* 1 *or* 0.

- *(Exponential)*
  $det(exp(tA)) = 1 + t(Tr(A)) + O(t^2)$

1.3 Homology

Definition 1.3.1 *(Monomorphism and ker/coker)*
We generalize homological algebra to category theory from abstract algebra, so we can extend this notion to sheaf theory, while this generalization requires us reinterpretation of basic operations as subobject/quotient and kernel/image etc.

In a category $\mathscr{C}$, *suppose* $f \circ g_1 = f \circ g_2$ *means* $g_1 = g_2$ *for all* $g_1, g_2 \in Mor(\mathscr{C})$, *then the morphism* f *is monomorphism. By using this monomorphism, we can generalize notion of kernel into abelian category. In the language*

of category, kernel is defined $ker(f) = (K, i)$ as a pair where i is monomorphism $i : K \to X$ such that $f \circ i = 0$ (this is equivalent expression with $f \circ i = g \circ i$ in abelian category), and this $ker(f)$ is a subobject $ker(f) \subset X$, where subobject is defined by the pair (A, i) where $i : A \to X$ is a monomorphism. (just notice that on some cases monomorphism has the equal meaning with injectivity. For example, in category of Sets, monomorphism is equivalent to injectivity, thus $ker(f)$ is just equivalent to ordinary kernel in set theory). Similarly, for $A \subset X$, the quotient is defined by cokernel $X \xrightarrow{p} X/A = coker(i)$ so that $p \circ i = 0$, and in particular, $im(d_{n+1}) \subset ker(d_n) \subset X$ is a sequence of subobjects, thus $ker(d_n)/im(d_{n+1})$ is defined.

Example 1.4 *(Abelian Category)*
$\mathcal{A}\lfloor$, $Coh(X)$ where X is noetherian, and $QCoh(X)$ where X is quasi-separated, not necessarily notherian.

Definition 1.4.1 *(Homology)*
In theory of homology, the chain complex is a sequence $\cdots C_{n+1} \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} C_{n-1} \xrightarrow{d_{n-1}} \cdots$ defined over objects C_n of the abelian category. However, the object C_n doesn't need to be an abelian group e.g. coherent sheaf.

Definition 1.4.2 *()*

1.5 Algebraic Curve

Complex Analysis:

We'll consider algebraic interpretation of complex analysis, where the holomorphic functions are calculated by their singularities, and the integral pertaining this has monodromy, or it has a Laurant expansion to calculate residue theorem, or in diffyq, singularities is more than power series solution, and we'll study algebraic geometry and representation theory for this.

The first question is the inverse problem: given the complex domain(plane), with some singularities, can we determine all the possible function whose domain is it? In 1-dimensional complex analysis, if the domain of the funcion $f(z)$ is $\mathbb{C}\backslash\{0\}$, then the function $f(z)$ could be $f(z) = \frac{1}{z}$ but also $f(z) = log(z)$, This is called monodromy, and we use monodromy representation to determine all the possibilities. This problem is reduced to the combinational problem of hypergeometric functions or KZ equation in general.

Moduli Problem:

Answering for what is moduli space is a very ambiguous problem, but we should approach from the concrete examples, that can be generated by GIT quotient of some geometries: for example, it might be M_g moduli space i.e. a

set of stable algebraic curves of genus g, or in particular, $M_{g,n}$ a moduli space of genus g of n-marked points as M_g consists of their disjoint union $M_g = \coprod M_{g,n}$, and this idea is closely related to complex analysis, and our first goal is to describe the moduli spaces explicitly.

Another argument that sounds is why we should use the technique of algebraic geometry for complex analysis, and I believe it's because algebraic variety could be considered to be "hard" geometry, compared with smooth manifold in general, lacking diversity of their shapes, and especially the automorphism of the geometry could be a finite group, also called stability condition. For example, $\mathbb{CP}^1$ is the only algebraic curve, which is 1-dimensional compact projective geometries of genus 0, and similarly an elliptic curve is the only genus 1 compact algebraic curve.

The moduli problem can be further generalized by the language of stacks. Or perhaps derived algebraic geometry and so on.....

1.6 GIT Quotient

Definition 1.6.1 *(GIT Quotient)*

By context of moduli problem, GIT quotient is modularization. For an algebraic variety X, if a reductive group G acts on X algebraically, it will define quotient $X//G$ of good properties. For example,

$$M_{0,n} = Conf_n(C)/SL2(C)$$

Why G needs to be reductive group because if a polynomial ring A is fixed by G, A^G is finitely generated (or in other words, if G is not reductive, GIT is undefiend, or it doesn't have a good property).

Proposition 1.7 *(Keel-Mori Theorem)*

For any schemes, GIT quotient exist as Deligne-Mumford stack.

Remark 1.8 *(Compactness)*
Compactness is also a very important notion in the context of algebraic geometry, since compactness guarantees the bounray problem in geometry, and it the geometry satisfies compactness, it'll enables us to discuss intersection theory, including Poincare duality. If the geometry if compact, then the automorphism is beautiful,

$$Aut(\mathbb{P}^1) = PGL_2(\mathbb{C})$$

while for non-compact space, the automorphism

$Aut(\mathbb{C}) = \mathbb{C} \rtimes \mathbb{C}$

is ∞-dimensional, and it's very dirty.

1.9 Module Theory

Definition 1.9.1 *(Tensor product as module)*

Let M and N be R-module. The tensor product as R-module is tensor product as a vector space with quotient as follows:

$M \otimes_R N = M \otimes N/(m, r \cdot n) \sim (m \cdot r, n)$

ex:

$\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} = \mathbb{Z}/gcd(m,n)\mathbb{Z}$

Definition 1.9.2 *(Flat module)*

R-module M is flat if any injection $K \hookrightarrow L$ maps to $K \otimes_R M \to L \otimes_R M$ is injective.

Especially if $- \otimes_R M$ is faithfully flat if and only if short exact sequence maps to short exact sequence.

- *(non-ex I)*
  $\mathbb{Z}/2\mathbb{Z}$ is not flat. For example an injection $\phi : \mathbb{Z} \to 2\mathbb{Z}$ doesn't make $\phi \otimes \mathbb{Z}/2\mathbb{Z}$ injective.
- *(non-ex II)*
  R/I cannot be flat, since it's torsion, but I could be flat.
- *(ex I)*
  If M and N are flat over R, then $M \otimes N$ is flat.
- *(ex II)*
  If $S \to S'$ is flat and M is flat, then $M \otimes_S S'$ is flat.
- *(ex III)*
  If M is R-module, its localization $M_{\mathfrak{p}}$ is flat $R_{\mathfrak{p}}$ module.

Another Characterization :

For all linear combinations $\Sigma_{i=1}^{m} r_i x_i = 0$ where $r_i \in R$ $x_i \in M$, we have change of basis

there exists an element $y_j \in M$ and $a_{i,j} \in R$ such that

$\Sigma_{i=1}^{m} r_i a_{i,j} = 0$ *for* $0 \leqq j \leqq n$

$x_i = \Sigma_{j=1}^{n} a_{i,j} y_j = 0$ *for* $0 \leqq i \leqq m$

Why is this definition equivalent to $- \otimes_R M$ is an exact functor? Apparantly, $- \otimes_R M$ is right exact, so if the second morphism of $0 \to A \otimes_R M \to B \otimes_R M \to C \otimes_R M \to 0$ is injective, it makes short exact sequence.

Example 1.10 *(Flat which is not projective)*

If $R = C^{\infty}(\mathbb{R})$*, and* $M = R_{\mathfrak{m}} = R/I$ *where* $\mathfrak{m} = \{f \in R | f(0) = 0\}$ *and* $I = \{f \in R | \exists \epsilon > 0 : f(x) = 0, \forall x, |x| < \epsilon\}$.

Note: R is not integral domain.

Definition 1.10.1 *(Projective Module)*
A module P is projective iff every short exact sequence $0 \to A \to B \to P \to 0$ *that splits.*

ex:

R is a Dedekind domain $R = \mathbb{Q}[\sqrt{-6}]$ *and* $P = (2, \sqrt{-6})$*, then if* $A = (3, \sqrt{-6})$*, then* $B \cong A \oplus P$.

Projecitve module is flat.

Definition 1.10.2 *(Flat Ring Extension)*

A ring homomorphism is flat extension if it's a flat morphism.

ex:

$R \to R[x]$ *is flat.*

2 Concrete Homology

2.1 Simplicial Homology

Definition 2.1.1 *(Simplicial Homology)*
obvious.

Definition 2.1.2 *(Relative Homology)*
To step forad to Borel-Moore, we introduce relative homology. For $A \subset X$*, the relative homology* $H_i(X, A)$ *is i-th homology of X ignoring A part, which is simply* $C_i(X, A) := C_i(X)/C_i(A)$*. This results the long exact sequence* $\cdots H_{i+1}(X, A) \to$

$H_i(A) \to H_i(X) \to H_i(X, A) \to \cdots$. In other words, using $A = X \backslash K$, the relative homology $H_i(X, X\backslash K)$ is called K-localized homology.

Definition 2.1.3 *(Borel-Moore Homology)*
Borel-Moore homology is basically the same as simplicial but generalization to noncompact case. It's defined by $H_i^{BM}(X) = \varinjlim_{K \subset X} H_i(X, X\backslash K)$ for K compact subset of X.

Proposition 2.2 *(Intuition of Homotopy)*
For a topology map $f : X \to Y$, its chain map $C.f : C.(X) \to C.(Y)$ has mapping $(\sigma : \Delta^n \to X) \to (f \circ \sigma : \Delta^n \to Y)$ for each n, whose homology is $H_n f : H_n(X) \to H_n(Y)$. Consider the homotopy is a map $H : X \times [0, 1] \to Y$, that can define homotopy relation $f \sim g$ where $f(x) = H(x, 0)$ and $g(x) = H(x, 1)$, then it naturally adds 1-dimension to the geometry, thus the homotopy H can be divided into $H_n : X^n \times [0, 1] \to Y^{n+1}$, so $H = \Sigma_n H_n$. This is not surprising argument, as we can see in homotopy lifting, and now $\partial \circ H + H \circ \partial : X^n \to Y^n$ is a morphism between the same degrees. Here consider $H : X \times [0, 1] \to Y$ in two dimensional, if $a \subset X$ is a 1-chain, thus $f(a)$ and $g(a)$ are also 1-chain, that can continuously changed in $[0, 1]$, and $\partial(H(a)) = g(a) - f(a) - H(\partial(a))$, thus removing a, $f - g = \partial \circ H + H \circ \partial$.

2.3 de Rham Cohomology

Definition 2.3.1 *(Vector Field)*
A vector field X is a linear map $X : C^\infty(M) \to C^\infty(M)$ that satisfes product rule.

$$X(fg)p = fX(g) + X(f)g \text{ for all } f, g \in C^\infty(M)$$

Vector bundle is an abstraction of differential forms on a smooth manifold M. In particular, vector bundle is tangent bundle TM, whose tensor or direct sum $\bigwedge^* TM$ and $TM \oplus TM$ and dual TM^*, and trivially line bundle $M \to \mathbb{R}$. In differential form, differential of 1-form becomes 2-forms, and from 0-form to 1-form, so in analogy, we could define a morphism from the global section of a vector bundle to another global section of a vector bundle, which is called connection $\nabla : \Gamma(E \otimes \bigwedge^* TM) \to \Gamma(E \otimes \bigwedge^* TM)$ (what is differential? It is any morphism what satisfies product rule).

Definition 2.3.2 *(Vector Bundle)*
A vector bundle is a morphism $\pi : E \to B$ such that the inverse $\pi^{-1}(x) = E_x$ is a vector space. A section is an inverse map $\sigma : B \to E$ such that comosition $\pi \circ \sigma = id$ is identity. We will denote $\Gamma(E, U)$ for $U \subset B$ as section.

Remark 2.4 *(Zero Section)*
Zero section $\sigma : B \to E$ is local/global section of vector bundle, and the morphism maps to zero as $\sigma(x) = 0 \in E_x$, for $x \in B$.

Remark 2.5 *(Existence of Global Section)*
The global section of vector bundle always exists. At least zero section is a global section, and every vector bundle has it. On the other hand, global section of principal bundle exists iff it's trivial bundle. Of course, we are interested in any non-zero sections. There exists nowhere vanishing global section in a vector bundle if its Euler class is zero.

Definition 2.5.1 *(Connection Form)*
$\nabla : \Gamma(E) \to \Gamma(E \otimes T^*M) = \Gamma(E) \otimes \Gamma(T^*M)$

$\nabla(fv) = \nabla(v) \otimes df + f\nabla(v)$

where ∇ *is the exterior derivative of* f.

In general, the connection form is generalized to n*-th power product*

$\nabla : \Gamma(E) \to \Gamma(E \otimes \bigwedge^n T^*M) = \Gamma(E) \otimes \Gamma(\bigwedge^n T^*M)$

$\nabla(v \wedge \alpha) = \nabla(v) \wedge d\alpha + (-1)^{deg(v)} v \wedge d\alpha.$

Definition 2.5.2 *(Flat Connection)*
If the curvature derived from the connection vanishes, the connection ∇ *is flat, and* $\nabla^2 = 0$*, so it can form the chain complex.*

Definition 2.5.3 *(Curvature Form)*
The problem is curvature from derived from the connection ∇ *has a two different way of representations.*

First, the curvature form of connection is $F_\nabla = dA + A \wedge A$ *or* $A \wedge A = \frac{1}{2}[A, A]$*. Or alternatively,* $\Omega = d\omega + \omega \wedge \omega$*. These two forms have actually the same Riemannian curvature tensor.*

Explicitly, $F_\nabla(s) = \Sigma_{i,j=1}^{k} \Sigma_{p,q=1}^{n} R_{pqi}^{j} s^i dx^p \wedge dx^q \otimes e_j$

where R_{pqi}^{j} *is a Riemann curvature tensor of* F_∇.

Definition 2.5.4 *(Elliptic Operator)*
Elliptic operator is a generalization of Laplacian. Let L *be a linear differential operator of order* m *in the domain* $\Omega \subset \mathbb{R}^n$ *given by*

$Lu = \Sigma_{|\alpha| \leqq m} a_\alpha(x) \partial^\alpha u$

L *is called Elliptic if for all* $x \in \Omega$ *and every non-zero* $\xi \in \mathbb{R}^n$,
$\Sigma_{|\alpha| \leqq m} a_\alpha(x) \xi^\alpha \neq 0$

Definition 2.5.5 *(Elliptic Complex)*
Let $E_0, E_1, \cdots, E_k$ be vector bundles on a smooth manifold M.

For the sequence of vector bundles $0 \to E_0 \to E_1 \to \cdots \to E_k \to 0$.

For the sequence of vector bundles

$$0 \to \Gamma(E_0) \to^{P_1} \Gamma(E_1) \to^{P_2} \cdots \to^{P_k} \Gamma(E_k) \to 0$$

such that $P_{i+1} \circ P_i = 0$.

$$0 \to \pi^* E_0 \to^{\sigma(P_1)} \pi^* E_1 \to^{\sigma(P_2)} \cdots \to^{\sigma(P_k)} \pi^* E_k \to 0$$

is exact outside of zero section. Here π is the projection of cotangent bundle $T^*M \to M$ and π^* is the pullback of a vector bundle.

Proposition 2.6 *(Property of Elliptic Complex)*
In Elliptic complex, the kernel is of finite dimension, and analytically nice wrt smoothness and Fredholm. Therefore, the cohomology is of finite dimensional, and it can be idenfied with Laplacian.

2.7 Hodge Theory

Definition 2.7.1 *(Hodge Star Operator)*
A Hodge star is a linear operation in the exterior algebras $\star : \bigwedge^k(V) \to \bigwedge^{n-k}(V)$, and Hodge star maps an object $w \in V$ to its dual $\star w \in V$, and twice the operations is plus-minus identity as $\star \circ \star = (-1)^{k(n-k)} Id$. Hodge star operator is multi-linear map.

Let's see an explicit illustration. The domain $\bigwedge^k(V)$ consists of k-dim bivector space. If $k = 1$, an element is a vector, and vector is an arrow, vector is geometrically interpreted as 1-dim bounded line. For higher dimension, we let bivector given by wedge product $v_1 \wedge v_2 \wedge \cdots \wedge v_k$ as a k-dimensional cube as a subset of V.

For example, if $n = 3$, the hodge dual of a vector $e_1 \in \mathbb{R} = V$ is $e_2 \wedge e^3 \in \mathbb{R} \wedge \mathbb{R}$, so the Hodge dual corresponds to the complement space.

Definition 2.7.2 *(Hermitian Metric)*
$h_p(\eta, \bar{\zeta}) = \overline{h_p(\zeta, \overline{\eta})}$ for all $\zeta, \eta \in E_p$ and

$h_p(\zeta, \zeta) > 0$ for all $\zeta \neq 0 \in E_p$

Hermitian metric is a complex valued function, but we can also define Riemannian metric if it we add conjugate, or we subtract to make (1,1)-form on M.

$g = \frac{1}{2}(h + \overline{h})$
$\omega = \frac{i}{2}(h - \overline{h})$

Definition 2.7.3 *(Hermitian Connection (or Dolbeault Operator))*
A Hermitian connection ∇ is a connection on a Hermitian vector bundle E over a smooth manifold M compatible with a Hermitian metric $< \cdot, \cdot >$ on E, meaning that

$$v < s, t > = < \nabla_v s, t > + < s, \nabla_v t >$$

for all smooth vector field $v \in C^\infty(M, E)$ and all smooth sections s, t of E. Then there is a unique Hermitian connection whose $(0,1)$ part is Dolbeault operator $\overline{\partial}_E$ on E associated to the holomorphic structure. Formally, Dolbeault operator is a projection $\overline{\partial}_E = \pi^{(0,1)}\nabla$. This ∇ is called Chern connection on E.

More explicitly, any section $s \in \Gamma(E)$ is written as $s = \Sigma s^i e_i$ for some function $s^i \in C^\infty(U_\alpha, \mathbb{C})$. Define an operator locally by

$\overline{\partial}_E(s) := \Sigma \overline{\partial}(s^i) \otimes e_i$ where $\overline{\partial}_E$ is a Cauchy-Riemann operator of the base manifold. That is,

$\overline{\partial}(\alpha) = \Sigma \frac{\partial f_{IJ}}{\partial \overline{z}^l} d\overline{z}^l \wedge d\overline{z}^I \wedge d\overline{z}^J$
(Curvature)
The curvature $\Omega = d\omega + \omega \wedge \omega$ be the curvature form of ∇. Since $\pi^{0,1}\nabla = \overline{\partial}_E$ is vanishing if squared, so $\Omega^{2,0}$ is trivial, so ∇ is skew-symmetric $\Omega^{0,2}$. Thus Ω is a $(1,1)$-form given by

$$\Omega = \overline{\partial}_E \omega$$

Definition 2.7.4 *(Holomorphic Vector Bundle)*
Over a complex manifold X, the projection map $E \to X$ is holomorphic, which means the local trivialization is a biholomorphic map $\phi_U : \pi^{-1}(U) \to U \times \mathbb{C}^k$.

Example 2.8 *(Holomorphic Tangent Bundle)*
For a complex manifold M, the real $2n$-dimensional real vector bundle TM on M, and endomorphism $J : TM \to TM$ s.t. $J^2 = -Id$.

If complexified, $J : TM \otimes \mathbb{C} \to TM \otimes \mathbb{C}$ where $J(X + iY) = J(X) + iJ(Y)$ where $X, Y \in TM$

Since $J^2 = -Id$, eigenvalues are i and $-i$, and it splits

$$TM \otimes \mathbb{C} = T^{1,0}M \oplus T^{0,1}M$$

The holomorphic tangent bundle of M is vector bundle $T^{1,0}M$, and anti-holomorphic tangent bundle is $T^{0,1}M$.

Definition 2.8.1 *(Hermtian Yang-Mills Connection (or Instanton))*
For a principal bundle $P \to E$, a connection $A_\alpha \in \Omega^1(U_\alpha, ad(P))$. If the connection A satisfies $d_A \star F_A = 0$, then A is Yang-Mills connection (also called instanton). This is Hermitian YM connection, if in addition, the connection A is connection on a Hermitian vector bundle E over a Kähler manifold X of dimension n.

Then the Hermitian YM equaitons are:

$F_A^{0,2} = 0$
$F_A \cdot \omega = \lambda(E) Id$ *for some* $\lambda(E) \in \mathbb{C}$

Recall the curvature F_A is skew-Hermitian, so $F_A^{0,2} = 0$ implies $F_A^{2,0} = 0$. If X is compact, then $\lambda(E)$ can be computed using Chern-Weil theory. Namely

$$\begin{aligned} deg(E) &= \int_X c_1(E) \wedge \omega^{n-1} \\ &= \frac{i}{2\pi} \int_X Tr(F_A) \wedge \omega^{n-1} \\ &= \frac{i}{2\pi} \int_X Tr(F_A \wedge \omega) \omega^n \end{aligned} \tag{1}$$

Hence,

$\lambda(E) = -\frac{2\pi i}{n! Vol(X)} \mu(E)$

$\mu(E) = \frac{deg(E)}{rank(E)}$

Definition 2.8.2 *(Adjoint Operator)*
(Adjoint Operator)
Given an operator is a $T = \Sigma a_k(x) D^k$ where D is differential, adjoint operator T^* is an operator that satisfies $< Tu, v >=< u, T^* v >$ where $< \cdot, \cdot >$ is an inner product.

(Laplacian)
Given a de Rham cochain complex $(\Omega^* * (X), d)$, let adjoint operator be $\delta : \Omega^{k+1}(X) \to \Omega^k(X)$, and the Laplacian $\Delta = d\delta + \delta d : \Omega^k(X) \to \Omega^k(X)$

(Harmonic)
A form on X is harmonic if $H^k_\Delta(M) = \{\alpha \in \Omega^k(X) | \Delta\alpha = 0\}$

Proposition 2.9 *(Hodge Decomposition)*
There exists a unique decomposition

$$\omega = d\alpha + \delta\beta + \gamma$$

in which γ is harmonic $\Delta\gamma = 0$ In terms of L^2 metric on differential forms, this gives an orthogonal direct sum decomposition

$$\Omega^k(M) \cong im(d_{k-1}) \oplus ker(d_{k+1}) \oplus H^k_\Delta(M)$$

Hodge theorem states that $H^k_\Delta(M) \cong H^k(M, \mathbb{R})$

Proposition 2.10 *(Hodge Conjecture)*
Let X be a non-singular complex projective variety, and let Hodge classes be

$$Hdg^k(X) = H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X)$$

We claim every Hodge class $Hdg^k(X)$ on X is a linear combination with rational coefficients of the cohomology clases of complex subvarieties of X.

Definition 2.10.1 *(Pullback Bundle)*
Let $\pi : E \to B$ be a fiber bundle and $B' \to B$ be a continuous map. Define a pullback bundle is

$$f^*E = \{(b', e) \in B' \times E | f(b') = \pi(e)\} \subset B' \times E$$

and the projection map $\pi' : f^*E \to B'$ given by the projection onto the first factor, i.e. $\pi'(b', e) = b'$ and another projection $h : f^*E \to E$. These maps creates a commutative diagram

$$\begin{tikzcd} f^*E \arrow[r, "h"] \arrow[d, "\pi^*"] & E \arrow[d, "\pi"] \\ B' \arrow[r, "f"] & B \end{tikzcd}$$

Definition 2.10.2 *(Hodge Theory on Elliptic Complex)*
Let elliptic complex be $L_i : \Gamma(E_i) \to \Gamma(E_{i+1})$ be linear differential operators, and denote it by

$$E^{\cdot} = \oplus\Gamma(E_i)$$

$$L = \oplus L_i : E^{\cdot} \to E^{\cdot}$$

and let $L^{\cdot}$ is the adjoint of L, and $\Delta = LL^{\cdot} + L^{\cdot}L$. As usual, this yields the harmonic sections

$$\mathfrak{H} = \{e \in E^{\cdot} | \Delta e = 0\}$$

Let $H : E^{\cdot} \to \mathfrak{H}$ be the orthogonal projection, and let G be the Green's operator for Δ. Then, Hodge theorem states

- *H and G are well-defined*
- *$Id = H + \Delta G = H + G\Delta$*
- *$LG = GL$ and $L^{\cdot}G = GL^{\cdot}$*
- *$H(E_j) \cong \mathfrak{H}(E_j)$ (cohomology correspondence)*

Definition 2.10.3 *(Dolbeault Complex)*
From the above definition of $\overline{\partial}_E$, Dolbeault complex is a complex if $\overline{\partial}_E^2 = 0$. Note in a complex manifold, $\overline{\partial}_E^2 = 0$ always exists, but in general (e.g. symplectic manifold), it doesn't always work.

First a warning that de Rham cohomology might be merely a real analytic notion, so in complex geometry, complexity has an extra symmetry in the homological algebra, and the adjoint operator of the differential operator defines Laplacian, thus Hodge decomposition. A generalization of de Rham cocomplex is elliptic complex, and "elliptic" sounds real analytic (indeed in complex geometry, we cannot distinguish elliptic and hyperbolic, or if we mention elliptic, it might be real analytic). In elliptic comlex, the differential operator is elliptic, meaning that the differential is non-vanishing (what do I mean by that).

Definition 2.10.4 *(Mixed Hodge Theory)*
Mixed Hodge theory generalizes pure Hodge theory to geometry of singularity or non-compact case. In mixed Hodge theory, the Hodge decomposition is not directly calculated, but there exists weight filtration and Hodge filtaration, and we can recover the Hodge decomposition by combining them. $0 = W_{-1} \subset W_0 \subset W_1 \subset \cdots W_n = H$ where W_k is a weight space that contains all weight k component or less than k, which simply means $W_k = \bigoplus_{p \leqq k} H^p$.

Definition 2.10.5 *(Variation of Hodge Theory)*
For a fibration $\pi : X \to S$ we define cohomology $H^k(X_s)$ for each fiber $X_s = \pi^{-1}(s)$ on $s \in S$, and Gauss-Manin connection is defined by $\nabla : H^k_{dR}(X_s) \to H^k_{dR}(X_s) \otimes \Omega^1_S$, and as usual it make a Hodge decomposition for each fiber $H^k_{dR}(X_s) = \bigoplus_{p+q=k} H^{p,q}(X_s)$ (assuming the fiber is always complex).

Hodge filtration is defined by $F^p H^k_{dR}(X) := \bigoplus_{r \geqq p} H^{r,k-r}(X)$, and indeed $H^k = F^0 \supset F^1 \supset F^2 \cdots F^k \supset F^{k+1} = 0$ is inclusion relation, and Gauss-Manin connection ∇ shifts Hodge filtration $\nabla F^p \subset F^{p-1} \otimes \Omega^1_S$ by no more than 1 by Griffith transversality.

Definition 2.10.6 *(The Case of Log Scheme)*
For non-compact scheme X, the divisor taing from compactification $D = \overline{X} \backslash X$ defines logarithmic complex $\Omega^{\cdot}_{\overline{X}}(log(D))$, and for example if $D = D_1 \cup D_2$, D_1 D_2 are weight 1, $D_1 \cap D_2$ are weight 2, more precisely W_1 consists of log differential form of either D_1 or D_2.

Definition 2.10.7 *(Perverse Sheaf)*
For an open embedding $j : U \hookrightarrow X$, there is an intermediaate extension $j_{*!}\mathbb{Q}_U[n]$ where $n = dim(X)$. $D = X \backslash U$. Log de Rham cohomology is the same as sheaf cohomology of perverse sheaf, and $H^i_{log}(X) \cong H^i(X, j_{!*}\mathbb{Q}_U[n])$. However, log scheme is defined by the pair (U, D), but it depends on the compactification.

2.11 Lie Algebra Cohomology

I'd suppose that the purpose of Lie algebra cohomology is to compare cohomology of a geometry M and its quotient $M//G$ where G is a Lie group.

Definition 2.11.1 *(Lie Algebra Cohomology)*

Let $\mathfrak{g}$ be a Lie algebra over R, and M be a representation of $\mathfrak{g}$. If M is trivial representation,

$H^n(\mathfrak{g}; M) = Ext^n_{U_{\mathfrak{g}}}(R, M)$

Definition 2.11.2 *(Chevalley-Eilenberg Complex)*

Chevalley-Eilenberg complex is $Hom_k(\bigwedge^\bullet \mathfrak{g}, M) \cong M \otimes \bigwedge^\bullet \mathfrak{g}^*$

Definition 2.11.3 *(Hamiltonian Vector Field)*

Moment map is similar to Hamiltonian vector field.

Definition 2.11.4 *(Symplectic Reduction)*

Let (M, ω) be a symplectic manifold where a Lie group G acts on it, and $\Phi : M \to \mathfrak{g}^*$ is a momentum map. Let $M_0 = \Phi^{-1}(0)$ and $0 \in \mathfrak{g}^*$ is a regular value. Then, there exists a symplectic quotient $\tilde{M} = M_0/G$ (often denoted as $M//G$), and this $\tilde{M}$ is also a symplectic manifold whose differential form is induced by the pullback as $\pi^*\tilde{\omega} = \iota^*\omega$ where $\pi : M_0 \to \tilde{M}$ and $\iota : M_0 \to M$.

Then the functional space is Lie algebra cohomology.
$C^\infty(\tilde{M}) = C^\infty(M_0)^{\mathfrak{g}} = H^0(\mathfrak{g}, C^\infty(M_0))$

where $C^\infty(M_0) = C^\infty(M)/$
since every smooth function on M_0 can be extended to M. coincides with the ideal generated by momentum map $I[\Phi]$.

Definition 2.11.5 *(Koszul Resolution)*
From symplectic reduction $\tilde{M}$ of a geometry M, we'll construct a Koszul resolution

$$K^\bullet = \bigwedge^\bullet C^\infty(M) \to C^\infty(M_0).$$

For a derivation $\delta : K^q \to K^{q-1}$,

$\delta f = 0$ and $\delta X = \phi_X$. Let X_i be a basis of $\mathfrak{g}$.

Similarly, the higher degree $\delta : \mathfrak{g} \otimes C^\infty(M) \to C^\infty(M)$ is

$$\delta(\Sigma_i X_i \otimes f_i) = \Sigma_i f_i \phi_i$$

For $\bigwedge^2 \mathfrak{g} \otimes C^\infty(M) \to \mathfrak{g} \otimes C^\infty(M) \to C^\infty(M) \to 0$

$$\delta(X \wedge Y \otimes f) = Y \otimes \phi_X f - X \otimes \phi_Y f$$

where δ is extended to an odd derivation. Notice that $\delta^2 = 0$.

Definition 2.11.6 *(BRST Cohomology)*
For a functional operator $Q : \mathfrak{H} \to \mathfrak{H}$ is an endomorphism, and particularly if $Q^2 = 0$, it'll be a chain complex, called BRST cochain complex. C^i consists of i-th ghost, 0-th ghost is a wave function of the actual physics state.

Definition 2.11.7 *(BRST Double Complex)*
Furthermore BRST has both gauge symmetry and anti-gauge symmetry structure, so we can have two distinct differential operators, and accordingly the wave function can be classified by the p-ghost and q-anti ghost, and we let $C^{p,q}$ consists of it. To sum it up, $(C^{\cdot,\cdot}, Q_{BRST}, d_{other})$ is a double complex. Or by using the newly defined differential operator $D = Q_{BRST} + d_{other}$, BRST double complex becomes just a complex $(C^{\bullet,\bullet}, D)$

Definition 2.11.8 *(BRST Cohomology With Lie Algebra)*
The BRST cochain complex has an explicit construction using Chevalley-Eilenberg cohomology with the gauge group $\mathfrak{g}$,

$$C^{p,q} = C^p(\mathfrak{g}; K^q) = \bigwedge^p \mathfrak{g}^* \otimes \bigwedge^p \mathfrak{g} \otimes C^\infty(M)$$

where $D = \{Q, -\}$ is the differential, and $Q = \alpha^i \phi_i - \frac{1}{2} f^i_{jk} \alpha^j \wedge \alpha^k \wedge X_i$, where X_i is a basis of $\mathfrak{g}$ and $[X_i, X_j] = f^i_{jk} X_k$ and α^j is dual basis of $\mathfrak{g}^*$.

2.12 Floer Homology

Definition 2.12.1 *(Hessian and Morse Index) For a geometry M, if $f : M \to \mathbb{R}$ is a smooth function, let $p \in M$ be a critical point $f'(p) = 0$, then Hessian $Hf_p(u, v)$ is*

$$Hf_p(u, v) = \Sigma_{i,j=1}^{n} \frac{\partial^2 f}{\partial x_i \partial x_j}(p) a_i b_j$$

where $u = \Sigma_{i=1}^{n} a_i \frac{\partial}{\partial x_i}_p$ and $v = \Sigma_{j=1}^{n} b_j \frac{\partial}{\partial x_j}_p$. If f is a Morse function (that is definable from any smooth functions), the Hessian Hf_p is nondegenerate.

Finally, we define Morse index $\mu(p)$ as the number of negative eigenvalues of the Hessian $(\frac{\partial^2 f}{\partial x_i \partial x_j}(p))$.

Definition 2.12.2 *(Morse Homology)*
Let's define Morse chain complex. For each k, the C_k is generated by critial points of degree k as $C_k := \oplus_{\mu(p)=k} \mathbb{Z}p$.

Let's define Morse differential.

For two critical points $p, q \in M$, the set

$M(p, q) = \{x : \mathbb{R} \to M | \dot{x} = -grad(f), \lim_{\tau \to -\infty} x(\tau) = p, \lim_{\tau \to \infty} x(\tau) = q, \}$

descibes path which follows the gradient. Further we take quotient as $\hat{M}(p, q) = M(p, q)/\mathbb{R}$ by translation action.

The derivative at critical point $p \in M$ is

$\partial : C_k \to C_{k-1}$
$\partial p = \Sigma_{\mu(q)=k-1} \# \hat{M}(p, q) q$

Indeed $\partial^2 = 0$, so the Morse chain complex is a chain complex.

In fact, Morse homology is isomorphic to singular homology.

Definition 2.12.3 *(Singular Homology)*

Remark 2.13 *(Moment Map and Morse Theory)*

Definition 2.13.1 *(Lagrangian Intersection)*
If M is a symplectic manifold, a smooth function $H_t : M \times [0, 1] \to \mathbb{R}$ has a vector field X_t such that $\omega(\cdot, X_t) = dH_t$ called Hamiltonian vector field. Further, we define a Hamiltonian isotopy ϕ_t such that

- $d\phi_t = X_t \circ \phi_t$
- $\phi_0 = id$

$\phi_t : M \to M$ is a symplectomorphism, and if $L \subset M$ is a Lagrangian submanifold, $\phi_t(L)$ is also a Lagrangian submanifold. For example, 0_X is a zero section of the cotangent bundle T^*X and its Lagrangian submanifold.

For a function $H = f \circ \pi : T^*M \to \mathbb{R}$ where $\pi : T^*M \to M$ is a projection, Its Hamiltonian isotopy ϕ_t is, in particular, $\phi_1(0_X)$ is the graph of df. If f is a Morse function, 0_X and $\phi_1(0_X)$ intersects transversally.

Floer homology is an infinite dimensional analogue of finite dimensional Morse theory.

Definition 2.13.2 *(Floer Homology)*
For ϕ_t, we define

$$\Omega = \{l : [0,1] \to M | l(0) \in L, l(1) \in \phi_1(L), l \text{ is homotopic to } \phi_t(x_0)\}$$

where $x_0 \in L$ is a fixed point. The universal covering space $\tilde{\Omega}$ of Ω is

$$\tilde{\Omega} = \{u : [0,1]\times[0,1] \to M | u(\tau, 0) \in L, u(\tau, 1) \in \phi_1(L), u(0,t) = \phi_t(x_0)\}/\text{homotopy}$$

We introduce a function $F : \tilde{\Omega} \to \mathbb{R}$

$$F = \int_0^1 d\tau \int_0^1 dt \omega(\tfrac{\partial u}{\partial t}, \tfrac{\partial u}{\partial \tau})$$

Now taking two intersection points $p, q \in L \cap \phi_1(L)$, the set

$$M(p,q) = \{u : \mathbb{R} \to \Omega | \tfrac{\partial u}{\partial \tau} = -grad(F), \lim_{\tau\to-\infty} u(\tau, [0,1]) = p, \lim_{\tau\to\infty} u(\tau, [0,1]) = q, \}$$

Now Ω is infinite dimensional smooth manifold with Riemannian metric, we are similarly allowed to define Morse theory.

2.14 Hochschild Homology

Definition 2.14.1 *(Hochschild Complex)*
$C^*(A, A) = \bigoplus_{n\geq 0} Hom(A^{\otimes n}, A)$ with differential d and cup product. Its differential d is $d\phi(a_1, \cdots, a_{n+1}) = a_1 + \phi(a_2, \cdots, a_{n+1}) + 2\sum_{n=1}^{n}(-1)^i\phi(a_1, \cdots, a_i a_{i+1}, \cdots, a_{n+1}) + (-1)^{n+1}\phi(a_1, \cdots, a_n)a_{n+1}$, and cup product is $f \cup g(a_1, \cdots, a_{m+n}) = f(a_1, \cdots, a_m)g(a_{m+1}, \cdots, a_{n+m})$, and $d(f \cup g) = (df) \cup g + (-1)^m f \cup (dg) +$ higher homotopy terms. $HH^1(A)$ shows derivation, $HH^2(A)$ shows deformation, and $HH^3(A)$ shows obstruction.

Definition 2.14.2 *(Operad)*
An operad is a collection $\{O(n)\}_{n\in\mathbb{N}}$ with multi-array operation $O(k)\times O(l_1)\times O(l_2)\times\cdots\times O(l_k)\to O(l_1+l_2+\cdots+l_k)$, where $O(n)$ could be any objects, and if each $O(k)$ is a topological space, it's topological operad. An operad can describe algebra by acting on Hochschild cochain complex $C^*(A,A)$, and O-algebra is generated by an operad O together with how that acts on it by the action maps $\gamma_n : O(n)\otimes V^{\otimes n}\to V$. However, depending on the action γ_n, the O-algebra should be constructed differently, and if $O = Ass$, O-algebra can be strictly associative, DGA, or even A_∞-algebra.

One of the applications of operad is Deligne conjecture, which claims Hochschild cochain complex is Gerstenhaber algebra, and it's also E_2-algebra, or Koszul duality claims the duality of A_∞-L_∞ conversion. A_∞ can be understood without using operad, so as L_∞-algebra, since associative operad can generate A_∞ algebra as trivial case, and it doesn't need to be A_∞ operad.

For transformation of different operads, operad morphism $\phi : O_1\to O_2$ where $O_i\to O_i-algebra$ maps functorially $\phi^* : O_1-algebra\ toO_2-algebra$.

Example 2.15 *(Operad Algebra)*
List of algebra operad acting on Hochschild cochain complex.

- *(A_∞)*
  A_∞-operad is a topological operad, since $A_\infty(n) = K_n$ for each $n\in\mathbb{N}$ is assigned to Stasheff polytope K_n, working as an associahedron. A_∞-algebra is generated by A_∞ operad acting on Hochschlild chain complex $A_\infty(n)\times C^*(A,A)^{\otimes n}\to C^*(A,A)$. If $n = 2$, homotopy means the existence of continuous transformation from $(fg)h$ to $f(gh)$. In particular, if $m_k = 0$ where $k > 2$, it's dg algebra.

- *(L_∞)*
  L_∞-operad is an algebraic operad, also called dg-operad. l_1 is differential, l_2 is Lie bracket, and l_n for $n\geqq 3$ is higher Jacobi identity. L_∞ algebra if L_∞ operad acts on Gerstenhaber algebra. In particular, if $l_k = 0$ where $k > 2$, it's dg Lie algebra.

- *(E_2)*
  E_2-operad (also called little disks operad) is a topological operad $\{E_2(n)\}$ where $E_2(n) = \{(D_1,\cdots,D_n)\subset D^2|D_i\cap D_j = \emptyset\}$, that can be interpreted as a moduli space of circles inside of a unit circle D, which is in other words, a configuration space $Conf_n(D^2) = \{(x_1,x_2)|x_1\neq x_2\}$. E_2-algebra is Gerstenhaber algebra. The Fulton-Mcpherson operad $FM_2(n) = Conf_n(\mathbb{R}^2)/\mathbb{R}\rtimes\mathbb{R}_{\leqq 0}$ is a compactification of E_2-operad, but it generates the same algebra.

- *(E_n, $n\geqq 3$)*
  E_n-algebra could be $Poisson_n$ algebra.

- *(Poisson Operad)*
  Poisson operad is just one colored operad, since there are two operations but it's defined on the same space.

- *(dg-Operad)*
  An operad $O(n)$ is a complex.

- *(PaB-Operad)*
  PaB consists of braid group B_n with projection $B_n \to S_n$, and $PaB(n) \cong E_2(n)$ is a weak equivalence by means of homotopy, and E_2 is a topological operad, and PaB operad is a group operad not topological operad, but its classifying BB_n is a topological space, that's given by $\pi_1(BB_n) = B_n$, and $BB_n \cong Conf_n(\mathbb{R}^2)$.

- *(Swiss Cheese-operad)*

Definition 2.15.1 *(Algebras)*

- *(DGA)*
  For de Rham cochain $(\Omega^*(M), d)$, wedge product $\wedge : \Omega^p \times \Omega^q \to \Omega^{p+q}$ where $\alpha \wedge \beta = (-1)^{pq}\beta \wedge \alpha$, being associative with unit $1 \in \Omega^0$. Its differential is $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^p \alpha \wedge d\beta$ where $\alpha \in \Omega^p$. Other DGA could be Chevalley-Eilenberg and BRST.

- *(DGLA)*
  For de Rham cochain $(\Omega^*(M), d)$, DGLA is $\Omega^*(M, \mathfrak{g}) = \Omega^*(M) \otimes \mathfrak{g}$, whose bracket is $[\alpha \otimes X, \beta \otimes Y] = (\alpha \wedge \beta) \otimes [X, Y]_{\mathfrak{g}}$, and its differential is $d(\alpha \otimes X) = d\alpha \otimes X$, denoted by $(\Omega^*(M), d, [\cdot, \cdot])$.

- *(BV)*
  BV algebra A consists of operator $\Delta : A \to A$ s.t. $\Delta^2 = 0$ with BV bracket $\{a, b\} = (-1)^{|a|}(\Delta(ab) - (\Delta a)b - (-1)^{|a|}a(\Delta b))$. This BV algebra has a chain structure since $\Delta^2 = 0$, but it doesn't need to be dga.

- *(A_∞-algebra)*
  Loop space $\Omega X = \{\gamma : [0, 1] \to X | \gamma(0) = \gamma(1) = x_0\}$, whose coposition $\gamma_1 \star \gamma_2(t) = \gamma_1(2t)$ if $(0 \leqq t \leqq 1/2)$ $\gamma_1 \star \gamma_2(t) = \gamma_2(2t - 1)$ if $(1/2 \leqq t \leqq 1$. $(\gamma_1 \star \gamma_2) \star \gamma_3 \cong \gamma_1 \star (\gamma_2 \star \gamma_3)$ isn't the same, but equivalent up to homotopy.

- *()*

Definition 2.15.2 *(Colored Operad)*
An colored operad is a collection $\{O(c_1, \cdots, c_k; c)\}$ where c_i, c describe types of an operad, not number or variables with morphisms $f \circ_i g \in O(c_1, \cdots, c_i, d_1, \cdots, d_m, c_{i+1}, \cdots, c_m; d)$ for $f \in O(c_1, \cdots, c_n; c)$ and $g \in O(d_1, \cdots, d_m; d)$ An easy example is a colored operad with just one color, since the Jacobian bracket is 3-ary operation

$O(2) \times O(2) \to O(3)$, whose colored operad version is simply replacing to dots $O(*,*;*) \times O(*,*;*) \to O(*,*,*;*)$.

Definition 2.15.3 *(PROP)*
PROP is a category with objects named $0, 1, 2, \cdots$ of countable cardinality, whose morphism $P(n,m)$ is an operad if $m = 1$ as $P(n,1) = O(n)$, or more roughly PROP describes n inputs and m outputs, where an operad is just n inputs and one output, so PROP is simply a generalization of Operad. PROP is symmetric monoidal category. Furthermore, dg-PROP is a category, whose morphism $O(m,n) : \cdots \to O(m,n)^i \xrightarrow{d} O(m,n)^{i+1} \to \cdots$ itself is a chain complex. PROP can be used to define bialgebra and HOPF algebra, which aren't necessarily one-output.

Definition 2.15.4 *()*

Definition 2.15.5 *()*

Definition 2.15.6 *()*

Definition 2.15.7 *()*

Definition 2.15.8 *()*

Definition 2.15.9 *()*

2.16 Other Homologies

Definition 2.16.1 *(Quantum Cohomology)*
Skipped as already mentioned in Marc Gross.

Definition 2.16.2 *(Group Cohomology)*

Definition 2.16.3 *(L^2 Cohomology)*

Definition 2.16.4 *(Étale Homology)*

Definition 2.16.5 *(Cyclic Homology)*

Definition 2.16.6 *(Crystal Homology)*

Definition 2.16.7 *(Kolmogorov Homology)*

Definition 2.16.8 *(Borel Cohomology)*

3 Abstract Homology

We disucss the property of homology for itself and how homology connects to ∞-category problem. Or trig category, dg-category to name a few.

3.1 Simplicial Method

In this section, we define cotangent complex, which is a derived language of cotangent bundle T^*M, that can be defined by an extension of Kähler differential form $\Omega_{X/S}$ for morphism of scheme $X \to S$. Also, simplicial resolution.

Definition 3.1.1 *(Simplex Category)*
A simplicial category Δ is a category whose objects are $[n] = \{0, 1, 2, \cdots, n\} \in \Delta$, and its morphisms are non-strictly increasing map of finite order, which might be generated by coface and codegeneracy maps.

- *(coface)*
  $\delta^{n,i} : [n-1] \to [n]$

  is a morphism that misses i namely, $\delta^{4,2} : [3] \to [4]$ maps to $(0, 1, 2, 3) \mapsto (0, 1, 3, 4)$.

- *(codegeneracy)*
  $\sigma^{n,i} : [n+1] \to [n]$

  is a morphism that counts i twice: namely, $\sigma^{4,2} : [5] \to [4]$ maps to $(0, 1, 2, 3, 4, 5) \mapsto (0, 1, 2, 2, 3, 4)$.

Definition 3.1.2 *(Simplicial Object)*
A simplicial object U is a contravariant functor

$$U : \Delta \to \mathscr{C}$$

esp if $\mathscr{C}$ is a Set, then U is simplicial set. Or if $\mathscr{C}$ is a category of commutative ring, then U is simplicial commutative ring.

Now, denote $U_n = U([n])$, $d_j^n = U(\delta_j^n) : U_n \to U_{n-1}$ and $s_j^n = U(\sigma_j^n) : U_n \to U_{n+1}$

Definition 3.1.3 *(Skeleton)*
$sk_n : Simp(\mathscr{C}) \to Simp_n(\mathscr{C})$ $cosk_n : Simp_n(\mathscr{C}) \to Simp(\mathscr{C})$

Definition 3.1.4 *(Simplicial object and fiber product)*
Let V and W be simplicial objects of a category $\mathscr{C}$, and morphism $a : V \to U$ and $b : W \to U$. Assume product $V_n \times_{U_n} W_n$ exists in $\mathscr{C}$, then

- $(V \times_U W)_n = V_n \times_{U_n} W_n$
- $d_i^n = (d_i^n, d_i^n)$
- $s_i^n = (s_i^n, s_i^n)$

In other words, $U \times V$ is a product of presheaves on U and V on Δ.

In fact, for any ring map $A \to B$, the simplicial resolution is of finite type, and in fact a trivial Kan fibration.

Example 3.2 *(Linear algebra on Fiber product)*
Consider the tensor product of modules

$$k[x, y] = k[x] \otimes_k k[x]$$

$$k[x, y, z] = k[x, y] \otimes_{k[x]} k[x, y]$$

and the dual of tensor product is fiber product. Consider $Spec(k) = pt$, $Spec(k[x]) = \mathbb{A}^1$, $Spec(k[x, y]) = \mathbb{A}^2$, and $Spec(k[x, y, z]) = \mathbb{A}^3$, and the tensor product of modules above will be

$$\mathbb{A}^2 = \mathbb{A}^1 \otimes_{pt} \mathbb{A}^1$$

$$\mathbb{A}^3 = \mathbb{A}^2 \otimes_{pt} \mathbb{A}^2$$

Now in general, Let $X_i = k[x_0, x_1, \cdots, x_i]$, then $X_{i+1} = X_i \otimes X_{i-1} X_i$.

Definition 3.2.1 *(Simplicial Resolution)*

Let A be a ring, and B be A-algebra. We'll construct a simplicial resolution $\epsilon : P^{\cdot} \to B$.

where $P_0 = B$, $P_1 = A[B]$, $P_2 = A[A[B]]$, and $P_n = A[A[A[\cdots A[B] \cdots]$ and so on.

This can be alternatively $P_{\cdot} = X_{\cdot}(B)$.

Here, $X_{\cdot}$ is defined as follows.

Consider functors $\mathscr{C} \leftrightarrows F, U\mathscr{D}$, where $\mathscr{D} = Set$, $\mathscr{C} = k - Mod$ and forgetful functor U and some functor F which maps a set to generating object, that is left adjoint to for. We let composition $X_0 = F \circ U$, and $X_1 = U \circ F \circ U \circ F$, $X_n = U \circ F \circ \cdots \circ F$ of n times of iterations. Notice X_n is an endfunctor $X_n \in Fun(\mathscr{C}, \mathscr{C})$.

Write $L = FU$ and $\delta = F\eta U$, so that we define counit and comultiplication

- $\epsilon : L \to I$
- $\delta : L \to L^2$

The pair (L, ϵ, δ) is called comonad.

- $d_i = L^{n-i}\epsilon L^i : L^{n+1} \to L^n$
- $s_i = L^{n-1-i}\delta L^i : L^n \to L^{n+1}$

for $i = 0, 1, 2, \cdots, n-1$.

Definition 3.2.2 *(Kähler differential)*
Kähler differential form $\Omega_{S/R}$ on a commutative ring homomorphism $\phi : R \to S$ can be defined as $\Omega_{S/R} = I/I^2$ where $I = ker(S \otimes_R S \to S)$ generated by $ds = s \otimes_R 1 - 1 \otimes_R s$. Otherwise, $\Omega_{S/R}$ is defined that for $s, t \in S$ and $r \in R$

- $dr = 0$
- $d(s + t) = ds + dt$
- $d(st) = sdt + tds$ *(product rule)*

Remark 3.3 *()*
If R and S are local ring, this ideal I is the maximal ideal, and $\Omega_{S/R} = I/I^2$ is also called a cotangent space. Also, let's consider cotangent space is dual of tangent space, since cotangent space is generated by differential of geometry, while tangent space is differential of functions over the geometry, exactly dual.

Definition 3.3.1 *(Cotangent Complex)*
Given a ring morphism $B \to A$, we have resolution $P. \to A$, where $P_{n+2} = P_{n+1} \otimes_{P_n} P_{n+1}$, and the cotangent complex is $L^n_{B/A} = \Omega_{P_n/A} \otimes P_n/\epsilon B$ for $n \geqq 0$, so that its homology is $H^0(L_{B/A}) = \Omega_{B/A}$. This resolution $P.toA$ is not unique, but one of the easy interpretations is the standard resolution $\cdots P_3 = B \otimes_A B \otimes_A B \to P_2 = B \otimes_A B \to P1 = B \to A \to 0$.

Example 3.4 *(The Case $k[x]$)*
The cotanget complex has alredy a geometric property. If in particular the resolution $P. \to A$ has $A = k$ and $B = k[x]$ for some field k, each P_n corresponds to just an affine plane, since $P_2 = k[x, y] = k[x] \otimes_k k[x]$, thus inductively $P_n = k[x_1, \cdots, x_n] = k[x_1, \cdots, k_{n-1}] \otimes_{k[x_1, \cdots, k_{n-2}]} k[x_1, \cdots, k_{n-1}]$, whose geometry corresponds to $Spec(P_n) = \mathbb{A}^n$. This is just a single simplice, and Dold-Kan correspondence $[n] \mapsto P_n^{\otimes n}$ is applicable.

Definition 3.4.1 *(Koszul Complex)*
For a commutative ring A, and $s : A^r \to A$ is a linear map where $f_1, \cdots, f_n \in A$ are functions, so that the standard basis $e_i \in A^r$ becomes $s(e_i) = f_i$, Koszul complex is a sequence $\Lambda^i A^r \to \Lambda^{i-1} A^r \to \cdots \to \Lambda^1 A^r \to \Lambda^0 A^r \cong A$, or equivalently given by the graded structure $\Lambda^{\cdot} = \oplus_i \Lambda^i A^r$ whose endomorphism is defined by $\alpha^1 \wedge \cdots \wedge \alpha_k \mapsto \Sigma_{i=1}^k (-1)^{i+1} s(\alpha^i) \alpha^1 \wedge \cdots \hat{\alpha^i} \cdots \wedge \alpha_k$ for each $0 \geqq k \geqq r$.

Definition 3.4.2 *()*

Definition 3.4.3 *(Chaotic/Indiscrete Topology)*
A category $\mathscr{C}$ has a canonically a site if $\{f : V \to U | f$ is an isomorphism$\}$ is a covering of U.

This corresponding topology is called chaotic or indiscrete topology.

Definition 3.4.4 *(Derived Lower Schriek)*
For category $\mathscr{C}$ is Derived category, ex:

$\mathscr{C}^{op}_{B/A}$ is a category with object $\alpha : P \to B$ or $\alpha : P' \to B$ and morphism is $s : \alpha \to \alpha'$.

Let $A \to B$ be a ring map.

$L_{B/A} = L_{\pi!}(Li^* \Omega_{\mathscr{O}/A}) = L_{\pi!}(i^* \Omega_{\mathscr{O}/A}) = L_{\pi!}(\Omega_{\mathscr{O}/A} \otimes_{\mathscr{O}} \underline{B})$

in $D(B)$ where $Li^* : D(\mathscr{O}) \to D(\underline{B})$ and $L_{\pi!} : D(\underline{B}) \to D(B)$.

Definition 3.4.5 *(Polynomial Differential Form)*
$\Omega^{\cdot}_{poly}([n]) = \mathbb{Q}[t_1, \cdots, t_n, dt_1, \cdots, dt_n]/(\Sigma t_i - 1, \Sigma dt_i)$

and morphism $[n] \to [m]$ induces functorially $\Omega([m]) \to \Omega([n])$, and $t_i \mapsto \Sigma_{u(j)=i} t_j$.

3.5 Categorification of Chain Complexes

Definition 3.5.1 *(Category of Chain Complexes)*
For defining the category of chain complexes $Ch(\mathscr{A})$, its objects $C. \in Ob(Ch(\mathscr{A}))$

can be a chain complex $\cdots C_{n+1} \overset{d_{n+1}}{\to} \cdots C_n \overset{d_n}{\to} \cdots C_{n-1} \cdots$, *where* $d_n \in Mor(Ch(\mathscr{A}))$ *is a differential operator such that* $d_n \circ d_{n+1} = 0$ *for all* n, *and dually, the category of cochain complex* $Ch(\mathscr{A})^{op}$. *The abelian category* $\mathscr{A}$ *could be* Ab, $Coh(X)$, *or* $FinVect_{\Bbbk}$.

I'll introduce many distinct definitions derived from category of chain complexes e.g. homotopy category, derived category, triangulated category, dg category, ∞-category etc, and they are all seems to be used for different purposes, while equally important question is how these distinct definitions are related. The diagram below shows the life span of homotopy category ends by $infty$-theory, and it roughly shows all kinds of categories can be related.

$$(HomotopyCategory) \overset{localization}{\Rightarrow} (DerivedCategory) \overset{=}{\Rightarrow} (TriangulatedCategory) \overset{dg-enhancement}{\Leftarrow} (DGCategory) \overset{dg-nerve}{\Rightarrow} (\text{infty}-Category) \Rightarrow (HomotopyCategory)$$

Definition 3.5.2 *(Variety of Consequences of Category of Chain Complex)*

- *(Homotopy Category)*
  The homotopy category induced by the category of chain complexes is $K(\mathscr{A}) = Ch(\mathscr{A})/\sim$ *where* $f \sim g$ *if* $f - g = hd + dh$.
- *(Derived Category)*
  Derived category is genenrated from either homotopy category or category of chain complexes, hence $D(\mathscr{A}) = K(\mathscr{A})[qis^{-1}]$ *where making all the quasi-isomorphism inversible, while similarly* $D(\mathscr{A}) = Ch(\mathscr{A})[qis^{-1}]$, *which makes the same result.*
- *(Triangulated Categeory)*
  Triangulated category consists of 4 axioms. First, (TR1) says that the existence of distinguished triangle $A \overset{u}{\to} B \overset{v}{\to} C \overset{w}{\to} A[1]$ *for any* $f : A \to B$ *where* $C = Cone(f)$ *is a mapping cone, where* $[1]$ *is the shifting or translation functor just adding suspension denoted* $[1]X = \Sigma X$. *In (TR2) if* $A \overset{u}{\to} B \overset{v}{\to} C \overset{w}{\to} A[1]$ *is distinguished triangle, then* $B \overset{v}{\to} C \overset{w}{\to} A[1] \overset{u[1]}{\to} B[1]$ *is also distinguished triangle. In (TR3), For a morphism of two triangles* $X \to Y \to Z$ *and* $X' \to Y' \to Z'$, *if existence of two morphism* $f : X \to X'$ *and* $g : Y \to Y'$ *makes the complement* $h : Z \to Z'$, *makeing the whole diagram commutes. (TR4) as octahedral axiom says the* 3 *distinguished triangles generate another distinguished triangle.*

  Its obvious consequence is that short exact sequence can be defined in the category of chain complexes, since it's an abelian category, while in category of chain complexes, we can also define distinguished triangle, namely derived version of short exact sequence, and triangulated category never becomes an abelian category, since it's already taken its quotient by the homotopy equivalence, and the kernel/cokernel is undefined.

- *(Model Category)*
  Model categories are, for example, category of simplicial sets $sSet$ and category of topological spaces Top, and it consists of fibration, cofibration, and weak equivalences, which can be constructible anywhere where homotopy theory is definable. In $sSet$, fibration is Kan fibration, cofibration is injection, weak equivalence if its geometric realization $|X| \to |Y|$ is weak equivalence in topological space, which means its homotopy $\pi_n(f) : \pi_n(|X|) \to \pi_n(|Y|)$ is isomorphism for all $n \in \mathbb{Z}_{\geqq 0}$.

- *(DG Category)*
  In short, DG category is a category whose $Hom(X, Y)$ itself is a chain complex. DG category $\mathscr{C}$ consists of data $Ob(\mathscr{C})$ with morphisms $Hom_{\mathscr{C}}(X, Y) = \oplus_{i \in \mathbb{Z}} Hom^i_{\mathscr{C}}(X, Y)$ as $\mathbb{Z}$-graded k-vector space, and the differential is $d : Hom^i_{\mathscr{C}}(X, Y) \to Hom^{i+1}_{\mathscr{C}}(X, Y)$ such that $d \circ d = 0$. The composition $\circ : Hom_{\mathscr{C}}(Y, Z) \otimes Hom_{\mathscr{C}}(X, Y) \to Hom_{\mathscr{C}}(X, Z)$ needs to be the chain map, which means $d(f \circ g) = (df) \circ g + (-1)^{|f|} f \circ (dg)$. Also, id_X $inHom^0_{\mathscr{C}}(X, X)$. Listing few examples of DG category as a category of chain complexes $Ch(k)$ and category of elliptic complexes, and category of de Rham complexes etc.

- *(∞-Category)*
  ∞-category is a quasi-category consists of objects, 1-morphisms, 2-morphisms, and n-morphisms where $n \in \mathbb{Z}$ in general. Strictly this is not a category, since associativity failed but equivalent up to homotopy $(f \circ g) \circ h \cong f \circ (g \circ h)$.

Definition 3.5.3 *(Why Derived Categry is Triangulated category)*
What do you think?

Definition 3.5.4 *(Model Category the Same as Triangulated Category?)*
No, but it's somewhat common.

Definition 3.5.5 *(DG-Enhancement)*
Intuitively, if localization makes a category of chain complexes to a derived category(triangulated category), DG enhancement is the opposite operation. DG-enhancement is a process of finding a DG-category $\mathcal{A}$ equivalent to the triangulated category $\mathcal{T}$, so that $H^0((A)) \cong \mathcal{T}$.

Definition 3.5.6 *((∞,1)-Category)*
(∞,1)-Category is ∞-Category but all n-morphism of higher degree $n \geqq 2$ can be invertible.

Definition 3.5.7 *(dg-Nerve)*
dg-Nerve is a tool to generate (∞,1)-Category from DG category ((∞,1)-Category is already ∞-Category), and its idea is to take the positive part $Hom_{\mathcal{C}}(X, Y)^{\geqq 0}$, that can be transformed to simplicial complex. By Lurie's definition, for dg-category $\mathcal{C}$, the nerve $N_{dg}(\mathcal{C})$ is a simplicial set.

For $N_{dg}(\mathcal{C})$, the object is $Ob(\mathcal{C})$, 1-morphism is $Hom^0(X,Y)$ in some dg-category, 2-morphism is H such that $d(H) = g \circ f - h$, and $g \circ f \sim h$ needs to be homotopy, and $f, g, h \in Hom^0$ and $H \in Hom^1$, and n-morphism is just repeating the homotopy.

Definition 3.5.8 *(Pretriangulated Category)*
$\mathcal{C}$ is pretriangulated category is if shift $[1]$ is defined, and it's closed under this shift operation, and the cone $Cone(f)$ exists for any morphisms $f : X \to Y$, and any twisted complex is closed under $\mathcal{C}$.

Twisted complex C is $C = \bigoplus_i X_i[k_i]$ whose differential is $d_C = \begin{bmatrix} d_{X_1} & q_{12} & \cdots \\ 0 & d_{X_2} & \cdots \\ \cdots & & \cdots \end{bmatrix}$,
so that $d_C^2 = 0$.

Definition 3.5.9 *(Pretriangulated Hull)*
DG category $\mathcal{C}$ is not pretriangulated but pretriangularizable. DG category already defines the shifting operator and cones, and they are closed under operations, and the only remaining problem is the existence of the twisting complexes. If we accept twisting complex generated from the category, so that the newly generated category pretriangulated category, then it's called pretriangulization.

Definition 3.5.10 *(Triangulization)*
The triangulization of pretriangulated category $\mathbb{C}$ is possible, and it's simply taken to the categorical homotopy $H^0(\mathcal{C})$. If the morphism of $\mathcal{C}$ $Hom_{\mathcal{C}}(X,Y)$ is a chain complex, then its homology $H^0(Hom_{\mathcal{C}}(X,Y))$ is the triangulted category.

Definition 3.5.11 *(DG-Enhancement)*
dg-enhancement is the opposite operation from the triangulization, means to find a category $\mathcal{D}$ such that $H^0(\mathcal{D}) \cong \mathcal{T}$. In particular, if the triangulated category $\mathcal{T} = D(\mathcal{A})$ is derived category, then its dg-enhancement is $C(\mathcal{A})$. Or if $\mathcal{T}$ is small category, we can use dg-Yoneda embedding to make dg-enhancement $\mathcal{T} \hookrightarrow Fun(\mathcal{T}^{op}, Ch(\Bbbk))$.

3.6 Triangulated Category

Definition 3.6.1 *(Exact Couple)*
Exact couple is a tool to calculate spectral sequence. In triangulated category $\mathcal{T}$, an exact couple is a pair (D, E, i, j, k) for $D, E \in \mathcal{T}$, defining the triangle $D \xrightarrow{i} D \xrightarrow{j} E \xrightarrow{k} D[1]$ where i is filtration inclusion, k is connecting morphism, j is quotient.

Definition 3.6.2 *(Postnikov Tower)*
Consider Postnikov tower for reasoning why long sequence in triangulated category is needed. Postnikov tower says the existence of tower $\cdots P_n(X) to P_{n-1}(X) \to \cdots \to P_1(X) \to P_0(X)$ for $\pi_i(P_i(X)) \cong \pi_i(X)$ for $i \leqq n$ and $\pi_i(P_i(X)) = 0$ for $i > n$, so that $X \cong \varprojlim_n P_n(X)$.

Definition 3.6.3 *(Tensor Triangulated Catgory)*
Tensor triangulated category $\mathcal{T}$ is both a triangulated category and a monoidal category, satisfying preservation of distinguished triangle $A \to B \to C \to A[1] \to$ by $X \otimes -$ for all objects $X \in \mathcal{T}$. This tensor product $- \otimes -$ is an operation in monoidal category. For example, derived category $(D(R), \otimes_R^L, R)$, perfect complex $Perf(R)$, stable homotopy category $(SH, \wedge, \mathcal{S})$.

In tensor triangulated category, Balmer spectrum $Spc(\mathcal{T})$ can discuss the reconstruction of $Spec(R)$ from $Perf(R)$. Balmer spectrum $Spc(\mathcal{T})$ is a collection of prime thick $\otimes$-ideals, that is, $I \subset \mathcal{T}$ is a $\otimes$-ideal, if for all $X \in \mathcal{T}$ and $Y \in I$, $X \otimes Y \in I$, and it's thick $\otimes$-ideal, if it's closed under triangle and direct summand, and it's prime if $X \otimes Y \in P$ if $X \in P$ or $Y \in P$, and we define $Spc(\mathcal{T}) = \{P \subset \mathcal{T} |\ P$ are prime thick $\otimes$-ideals$\}$. As a result, the $Spec(R)$ can be reconstructed from the Balmer sequence as $Spc(Perf(R)) \cong Spec(R)$.

Definition 3.6.4 *(Triangulated Category)*

- *(Compact Object)*
  Let $\mathcal{T}$ be a triangulated category which contains direct sums. $C \in \mathcal{T}$ is compact if for any family $\{X_i\}_{i \in I}$, the natural morphism $\bigoplus_{i \in I} Hom_{\mathcal{T}}(C, X_i) \to Hom_{\mathcal{T}}(C, \bigoplus_{i \in I} X_i)$ is isomorphism.
- *(Center)*
  The center $Z(\mathcal{T})$ of the triangulated category $\mathcal{T}$ is $Z(\mathcal{T}) := End(Id_{\mathcal{T}})$.
- *(Lefschetz decomposition)*
  If the triangulated category $\mathcal{T}$ has a self-isomorphism, the orthogonal decomposition $< A_0, A_1(1), A_2(2), \cdots, A_{m-1}(m-1) >$ is called Lefschetz decomposition, where $A_k \subset \mathcal{T}$ is a triangulated subcategory, and $A_k(k) := F^k(A_k)$ where $F : \mathcal{T} \to \mathcal{T}$ is an autoisomorphism.

3.7 ∞-World

Definition 3.7.1 *(A-∞ Algebra)*

Definition 3.7.2 *(A-∞ Category)*

Definition 3.7.3 *(Fukaya Category)*

Definition 3.7.4 *(Moduli of A_∞-Category)*
Consider A_∞ category has a fixed operation m_i for all i, so moduli space of A_∞ means the different varieties of m_i.

Definition 3.7.5 *(Higher Stack)*
Stack is a category whose objects are base manifold, and its morphisms are the fibers. Higher stack is a 2-category whose objects are base manifold, and its morphisms are the stacks.

3.8 Equivalence of Categories

Two categories $\mathcal{C}$ and $\mathcal{D}$ are equivalence of categories if the functor $F : \mathcal{C} \to \mathcal{D}$ has an inverse $G : \mathcal{D} \to \mathcal{C}$ such that the composisions $F \circ G \cong Id_{\mathcal{D}}$ and $G \circ F \cong Id_{\mathcal{C}}$ are naturally isomorphic to the identities. At one sight, equivalence of categories looks the same as isomorphism of categories, but they are actually many variety of difference, as we can see in Fourier-Mukai and Morita equivalence, so it's very rich in applications.

Example 3.9 *(Derived Categories)*
Equivalence of categories can be used for derived category by means of homological algebra, and $D(\mathcal{A}) \cong D(\mathcal{B})$ where $\mathcal{A}$ and $\mathcal{B}$ are abelian categories, which in particular, $D^b(A - mod) \cong D^b(B - mod)$ where $A = k[x]/(x^2)$ and $B = k \oplus k$ for both satisfy SES $0 \to S \to A \to S \to 0$ where $S = k$.

Definition 3.9.1 *(Fourier-Mukai Transform)*
Fourier-Mukai is a paradigm to interpret algebraic variety by language of derived categories.

$$\Phi^{\mathcal{P}} : D^b(Coh(X)) \to D^b(Coh(Y))$$

Definition 3.9.2 *(Morita Equivalence)*
Morita equivalence is an equivalence of categories $A - Mod \cong B - Mod$ of two rings A and B. For example,

- *(Product)*
  $A - Mod \cong A \times A - Mod$
- *(Matrix)*
  $A - Mod \cong M_n(A) - Mod$
- *(Scheme)*
  $QCoh(X) \cong A{-}Mod$ *for an affine scheme* $X = Spec(A)$, *thus* $QCoh(X) \cong QCoh(Y)$.
- *(Quantum Group)*
  $A{-}Mod \cong U_q(\mathfrak{g}){-}Mod$. *In particular,* $k[x]/(x^2){-}Mod \cong U_q(\mathfrak{sl}_2){-}Mod$
- *(Progenerator Module)*
  $RHom_A(T, -) : D^b(A - mod) \to D^b(End_A(T) - mod)$ *T is a tilting module. Tilting module if finite projective dimension and* $Ext^{>0}(T, T) = 0$.

- *(Quasi-Frobenius Algebra)*
  For $A - Mod \cong B - Mod$, if A is QF, then B is also QF.

Definition 3.9.3 *(Topological Space)*
If $X \sim Y$ is a homotopical equivalence of topological spaces X and Y, then there is a categorical equivalence $Top/X \cong Top/Y$.

3.10 Resolution Problem

Definition 3.10.1 *(Perfect Complex)*
$E^{\cdot} \in D(QCoh(X))$ is perfect iff $\forall x \in X$, $\exists U \ni x$, $E^{\cdot}|_U \cong$ free complex of finite length where $E^{\cdot}$ is generated by some finite amount of free $\mathscr{O}_U$-module. $E^{\cdot}$ is perfect iff $E^{\cdot} \in D(QCoh(X))$ is compact object. The subcategory $Perf(X) \subset D(QCoh(X))$ is a category of perfect complexes.

Example 3.11 *(Perfect Complex)*
Cotangent complex of moduli stack is perfect.

Definition 3.11.1 *(Model Category)*
Model category consists of fibration, cofibration, and weak-equivalences. An example of naturally created model category is $Ch(R)$, Set, and Top, or for a Grothendieck site $\mathcal{C}$, let $sPsh(\mathcal{C})$ be a category of simplicial presheaves, and we can add model structure on it, where morphism is weak equivalence for all $U \in \mathcal{C}$, and fibration is Kan fibration, cofibration is just inclusion morphism.

Definition 3.11.2 *(Cofibrant)*
The notion of cofibrant can be defined only $Ch(R)$. For a model category, an object X is cofibrant if $0 \to X$ is cofibration, where 0 is an initial object of the model category, and X is a chain. It's cofibrant only if degreewise projective. Cofibration is a map, that can lift trivial fibration. Now furthermore, let's consider L_∞ is a chain complex as an object of $Ch(R)$.

Definition 3.11.3 *(Quasi-Isomorphism)*
Let's visualize quasi-isomorphism, injective/projective object, which are all ignored in derived category by localization. In model category, weak-equivalence is quasi-isomorphism. Acyclic if .

Definition 3.11.4 *(Interpretation of Injective Resolution)*
Injective module lifts a morphism with universal property. It's similar to triangle in additive category, where one part is quasi-isomorphism, and if quasi-isomorphism vanishes by localization, injective resolution becomes trivial.

More geometrically, sheaf of module $\tilde{M}$ is vector bundle over the strcture sheaf, so injective resolution of $\tilde{M}$ is connection.

3.12 Algebraization

Definition 3.12.1 *(Ext and Tor)*

Definition 3.12.2 *(Yoneda Algebra)*

Definition 3.12.3 *()*

Definition 3.12.4 *()*

Definition 3.12.5 *()*

Definition 3.12.6 *()*

3.13 Sheaf

The idea of sheaf is to systematically classify the functions defined on the domain X, and for the structure, sheaf is defined to be a presheaf with extra structures on it. Presheaf is merely a contravariant functor, but it's already possible to construct simplicial category, while sheaf is a necessary step to define sheaf cohomology. But sheaf cohomology could be different from de Rham cohomology(or whatever non-sheaf theoretic), since sheaf can grasp the geometry from global perspective, while de Rham is just cohomology of differential forms, requiring smoothness, but most of algebraic geometry including moduli space contain singularity.

Example 3.14 *(Presheaf)*
For example, consider a functor $F : Top \to Ab$ where $Top = \mathscr{O}(X)$ is a category of topological space over X whose objects are open subsets $U \subset X$ by some topology, and their morphisms are inclusion morphisms, and Ab is a category of abelian groups, which in other words is to correspond each open neighborhood $U \subset X$ by set of functions $Hom(U, \mathbb{C})$ (of course it could be anything like $Hom(U, \mathbb{C}^n)$)

Why do we need to do this? Because not all the functions share the same domains. For example, if $X = \mathbb{C}$ and $f(z) = \frac{1}{z}$, then the function f has a singular point at $\{0\}$, which means the domain is $U = \mathbb{C} - \{0\}$, and $f \in Hom(U, \mathbb{C})$ but $f \notin Hom(X, \mathbb{C})$. This set of functions defines a vector bundle. Just for the sake of terminology, in algebraic geometry, the subscheme of codimension 1 is often called divisor, and in this example, singular points are divisors. Divisors in algebraic geometry implicitly means vector bundles. Alright, we have already talked many in presheaf, but we haven't reached sheaf yet.

My interpretation is that sheaf is a classification tool of function of some topological space, not algebraic geometry itself. In deed, sheaf is a broader subject, and for example, not all the functions needs to be analytic/holomorphic, it could be smooth functions or else.

Definition 3.14.1 *(Sheaf)*
obvious.

Definition 3.14.2 *(Cosheaf)*
Cosheaf is a copresheaf $G : Open(X) \to \mathscr{C}$ wtih a gluing condition with a pushout. The gluing condition can be coequalizer given by

$$G(U_1) \cap G(U_2) \rightrightarrows G(U_1) \oplus G(U_2) \to G(U) \text{ for all } U = U_1 \cup U_2$$

If sheaf makes sheaf cohomology, cosheaf makes sheaf homology

Definition 3.14.3 *(A-∞ Category)*
A-∞ category $\mathscr{A}$ consists of objects $Ob(\mathscr{A})$ such that the composition maps m_n satisfies the A-∞ conditions $\sum_{r+s+t=n} (-1)^{r+st} m_{r+1+t}(1^{\otimes r} \otimes m_s \otimes 1^{\otimes s}) = 0$.

Here is the list of A-∞ properties. A-∞ functor $F^n : A^{\otimes n} \to B$ whose composition satisfies the higher homotopy. It's A-∞ equivalent, if F_1 is quasi-isomorphism, in other words, $Hom_A(X, Y) \cong Hom_B(F_1X, F_1Y)$, and A-$\infty$ enhancement of T is an existence of A-∞ category s.t $H^0(A) \cong T$.

Example 3.15 *(Fukaya Category)*
Fukaya category $Fuk(Y)$ is an A-∞ category with symplectic structure with Floer homology. $Fuk(Y)$ only applies if Y is compact symplectic, since if it's non-compact, the Floer homology could be infinite-dimensional, while wrapped Fukaya category $W(Y)$ is a generalization of $Fuk(Y)$ to the non-compact case. The given symplectic geometry Y can be decomposed to each local data, and if its Fukaya category of each of the local data can be pushed out and glued, it will be the whole wrapped Fukaya category. Considered $Y \mapsto W(Y)$ is a morphism mapping geometry to its wrapped Fukaya category $Symp \to Cat_\infty$, which is atually a cosheaf.

Definition 3.15.1 *(Homological Mirror Symmetry)*
HMS Conjecture claims A-∞ equivalence of $D^b(Coh(X)) \cong Fuk(Y)$, as convention $D^b(Coh(X))$ is interpreted as A-∞ enhancement of itself. In addition, Verdier duality on A-model and Grothendieck duality on B-model correspond each other. Grothendieck duality on B-model side $RHom(-, \omega_X[dim(X)])$ corresponds to Verdier duality on A-model sides, which is a mirror of Fukaya categories, and Verdier duality is just inverse of Lagrangian as $SS(D\mathscr{F}) = -SS(\mathscr{F})$.

Definition 3.15.2 *(Derived Stack and MC Equation)*
For the moduli space M, L_∞ structure consists of $L_x = RHom(F,F)[1]$, $\frac{L_\infty}{gauge} \cong$ formal nbhd of $x \in M = \hat{M}_x$ from perspective of derived stack.

Example 3.16 *(Structure sheaf)*
Structure sheaf in algebraic geometry is a contravariant functor from Top to Ab as $F : Top \to Ab$. In particular, Top is a category whose objects are all open subset $U \subset X$ with Zariski topology, if scheme theory is considered (and it could be alternately étale or fppf or fpqc etc).

It's clear the definition is, but for pedagogistic purpose, a sheaf doesn't define a scheme, but the sheaf is a structure defined over the scheme.

Example 3.17 *(Constant Sheaf)*
Another famous example of sheaf is constant sheaf, which means that the stalk is always having the same object, which means for all the point $p \in X$, the stalk $F_p = L$ constant.

The stalked value doesn't seem to be interesting, but in general for the open sets, a constant sheaf is a sheaf that maps $U \subset X$ to a set of functions, which each function is constant on U.

There are some variations of constant sheaf. For example, if X is connected and the codomain is category of abelian groups, then the locally constant sheaf is constant sheaf. This is called local system. Also, perverse sheaf is created by simply shifting a constant sheaf.

Definition 3.17.1 *(Local System)*
Now here is the tip. Constant in constant sheaf is that it's geometrically trivial but the topology might not be trivial, and local system as a constant sheaf bijectively corresponds to monodromy representations (which is group homomorphisms) $\rho : \pi_1(X,x) \to Aut(L)$.

Example 3.18 *(Local System)*
From horizontal sections of vector bundle with a flat connection ∇, we can define local system.

$E_U^\nabla := \{s \in \Gamma(U,E)$ *which are horizontal:* $\nabla s = 0\}$

List of flat connections

- *Gauss-Manin connection*
- *KZ connection*

Example 3.19 *(Perverse Sheaf)*
If X is a smooth complex algebraic variety and everywhere of dimension d, and F is a local system on X, then $F[d]$ is a perverse sheaf.

Let a category of sheaves $Sh_c(X)$ over X, which is an abelian category, and the local system $F \in Sh_c(X)$ could be an object, and we consider an embedding to its derived category $D_c^b(X)$, and indentify the local system as a chain complex $[\cdots \to 0 \to F \to 0 \to \cdots]$

$$\begin{aligned} Sh_c(X) &\to D_c^b(X) \\ F &\mapsto [\cdots \to 0 \to F \to 0 \to \cdots] \end{aligned} \tag{2}$$

and this chain complex as the mapped local system is $F[0]$ as an object of the derived category, where by abuse of notation we let $F = [\cdots \to 0 \to F \to 0 \to \cdots]$, and $F[d]$ is simply shifting of $F = F[0]$.

- *$dim(supp(H^{-i}(F))) \leqq i$ (support condition)*
- *$dim(supp(H^i(\mathbb{D}F))) \leqq i$ (cosupport condition)*

where $\mathbb{D}$ is Verdier duality. Verdier duality is a generalization of Poincare duality, which is $H_c^i(X, \mathbb{Z}) \cong H_{n-i}(X, \mathbb{Z})^*$ where H_c^i is a cohomology with a compact support and H_{n-i} is a homology, while Verdier duality consists of Verdier duality functor $\mathbb{D} : D^b(Sh(X)) \to D^b(Sh(X))$, that can be defined by $\mathbb{D}_X(\mathscr{F}) := RHom(\mathscr{F}, \omega_X^{\cdot})$, where $\mathscr{F}$ is a sheaf over X and $\omega_x^{\cdot}$ is a dualizaing complex. It satisfies $\cong \mathbb{D}(\mathbb{D}(\mathscr{F}))$ and $H_c^i(X, \mathscr{F}) \cong H^{-i}(X, D_X(\mathscr{F}))^{\cdot}$, and $D_Y(Rf_! F) \cong Rf_* D_X(\mathscr{F})$.

3.20 Microlocal Analysis

Definition 3.20.1 *(Grothendieck Six Operations)*
For a scheme morphism $f : X \to Y$, the six operations f^*, f_*, $f_!$, $f^!$, ${}^L\otimes$, and $RHom(-,-)$ are the only peculiar problems in sheaf cohomology, otherwise sheaf cohomology is the same as module cohomology. For example, pullback $f^* : D(Y) \to D(X)$ and pushforward $f_* : D(X) \to D(Y)$ are the invariants of the derived geometry as $f^* = Lf^* = L(f^*)$ and $f^* = Rf^*$, which are adjoint $f^* \dashv f_*$, and the derived tensor $- {}^L\otimes - : D(X) \times D(X) \to D(X)$ and the derived hom $RHom(-,-) : D^{op}(X) \times D(X) \to D(X)$ are another adjoint ${}^L\otimes G \dashv RHom(G,-)$. The remaining two operators are complicated, but as an extension from the usual pullback and pushforward by adding a compact support.

For shriek pushforward $f_! : D(X) \to D(Y)$, to minimize the complexity of $f_!$, let's say the $f : X \to Y$ is the composition of $X \xrightarrow{j} \overline{X} \xrightarrow{\overline{f}} Y$, where $j : U \hookrightarrow X$ is an inclusion, and $j_! F(V) = \begin{cases} F(V) \text{ if } V \subset U \\ 0 \text{ if } V \not\subset U \end{cases}$, or $f_!$ is simply $Rf_! F = R\overline{f_*}(j_! F)$ if we use it.

For shriek pullback $f^! : D(Y) \to D(X)$, $f^!G \cong f^*G \otimes \omega_{X/Y}[d]$ where $\omega_{X/Y}[d] := \wedge^d \Omega^1_{X/Y}[d]$. Here $\Omega_{X/Y}$ is a sheaf of relative differential form, and $[1]$ is a cohomological shfit, which is shifting the degree of chain complex by 1 with differential times -1, and $[d]$ is doing $[1]$ d-times, and its adjointness relations is $f_! \dashv f^!$, but the order of the upper/lower case are the opposite.

Definition 3.20.2 *(Support)*
Support of sheaf $\mathscr{F}$ is defined by $supp(\mathscr{F}) = \overline{\{x \in X | F_x \neq 0\}} \subset X$, and it's compact support if $supp(\mathscr{F})$ is compact, whose reason to take the closure is the same as all closed subset of an algebraic variety is an algebraic set. Combining the six operators, $supp(f^*G) = f^{-1}(supp(G))$, $supp(f_*F) \subset f(supp(F))$ where its subset is equal if f is proper, and $supp(f^!G) = f^{-1}(supp(G))$, $supp(f_!F) \subset f(supp(F))$ if proper, and $supp(F \otimes G) = supp(F) \cap supp(G)$, and $supp(Hom(F,G)) \subset supp(F) \cap supp(G)$.

Definition 3.20.3 *(Microlocal Support)*
In addition, the sheaf theory version of microlocal support is $SS(\mathscr{F}) \subset T^*X \backslash 0$ such that $SS(\mathscr{F}) = \{(x,\xi) \in T^*X\backslash 0 |$ $\mathscr{F}$ is not micro-locally flat in direction ξ at $x\}$, where micro-locally flat means $\exists$ functions ϕ such that $d\phi(x_0) = \xi_0$ and $H^*(\mathscr{F}_{\phi \geqq 0}) = 0$. $\mathscr{F}_{\phi \geqq 0} = R\Gamma_{\phi \geqq 0}(\mathscr{F})$ where $R\Gamma_Z(\mathscr{F}) \in D^b(\mathbb{Q} - mod)$, which is particularly $\Gamma_Z(\mathscr{F}) = \{s \in \Gamma(X, \mathscr{F}) | supp(s) \subset Z\}$.

Now, together with the six operators, $SS(f^*G) \subset f_d^{-1}(SS(G))$ where $f_d : T^*M \to T^*N$ $SS(f_*F) \subset f_\pi(SS(F))$ where $f_\pi : T^*M \to T^*N$ is a fiber-wise map, and $f_!$ and $f^!$ are similar. Also, $SS(F \otimes G) \subset SS(F) \hat{+} SS(G)$ and $SS(Hom(F,G)) \subset SS(G) \hat{-} SS(F)$ where $\hat{+}$ and $\hat{-}$ are microlocal sum/difference, that's given by the direction of synthesis over the cotangent bundle, meaning $A \hat{+} B = \{(x, \xi + \eta) | \exists x_n \to x, \xi_n \in A, \eta_n \in B \xi_n + \eta_n \to \xi + \eta\}$, and $A \hat{-} B = A \hat{+} (-B)$.

Example 3.21 *()*
For skyscraper sheaf k_x, $SS(k_x) = T^*_x X$ where X is a smooth complex algebraic variety and $\mathscr{F}$ is either D-module or constructible sheaf.

Definition 3.21.1 *(Perverse t-Structure)*
Perverse t-structure is t-structure (truncation-structucture) but extra dimensional conditions. First of all, the t-structure consists of a pair $(D^{\leqq 0}, D^{\geqq 0})$, where shift condition $D^{\leqq 0}[1] \subset D^{\leqq 0}$ and $D^{\geqq 0}[-1] \subset D^{\geqq 0}$, and $Hom_D(X,Y)$ $\forall X \in D^{\leqq 0}, Y \in D^{\geqq 1}$. For all $X \in D$, a triangle $A \to X \to B \to A[1]$ exists such that $A \in D^{\leqq 0}, B \in D^{\geqq 1}$. Perverse t-structure is as follows.

$D^{\leqq 0}_{perv} = \{K \in D^b_c(X) | dim(supp(H^i(K))) \leqq -i$ *for all* $i\}$
$D^{\geqq 0}_{perv} = \{K \in D^b_c(X) | dim(supp(H^i(DK))) \leqq -i$ *for all* $i\}$
$Perv(X) = D^{\geqq 0}_{perv} \cap D^{\leqq 0}_{perv}$ *The category of perverse sheaves $Perv(X)$ is an abelian category.*

Definition 3.21.2 *(Perverse t-Exact)*
A functor $F : D \to D'$ is called t-exact if it's both right/left t-exact, and it's right t-exact, if $F(D^{\geqq 0}) \subset D'^{\geqq 0}$ and left t-exact if $F(D^{\leqq 0}) \subset D'^{\leqq 0}$, and its main property is that it preserves t-structure. Moreover, a functor F is perverse t-exact, if in particular $D = Perv(X) = D^b(X)^{\heartsuit}_{perv}$.

Definition 3.21.3 *(Beilinson-Bernstein-Deligne)*
$D^b(Perv(X)) \cong D^b_c(X)$ where D_c means derived category of constructible sheaves. Construcible sheaf is a sheaf, that can be decomposable to finitely many locally constant sheaves.

Definition 3.21.4 *(Nadler-Zaslow)*
The microlocal support SS(F) is always conic Lagrangian if X is a complex variety, and conic means $(x, \xi) \in SS(F)$ means $(x, \lambda\xi)$, $\lambda > 0$, and there exists Fukaya objects that coresponds to it: constructible sheaves on $X \leftrightarrow$ Fukaya category of T^*X, or by formula $Sh_c(X) \cong Fuk_{conic}(T^*X)$ is an equivalence as A-∞ category.

Definition 3.21.5 *()*
Use microlocal analysis to relate perverse sheaf and Fukaya category.

Definition 3.21.6 *(DT Sheaf)*
DT sheaf is an improvement of classical DT invariant, since classical DT invariant is fragile, lacking BPS data and rich information, and computation is complicated. If we use Behrend function for weighted Euler characteristic, it'll make DT invariant as $DT(M) = \chi(M, \nu_M) = \sum_{x \in M} \nu(x)$, where Behrend function $\nu_M : M \to \mathbb{Z}$ takes value $\nu_M(x) = (-1)^{dim(M)}$ if it's smooth, or $|\nu_M(x)| > 1$ if it's singular. DT sheaf is just a perverse sheaf ϕ_f, called vanishing cycle perserve sheaf, which is locally defined by Behrend function ν_M, where $M \cong Crit(f) \subset U$ where $f : U \to \mathbb{C}$ and U is smooth ambient space, but it can be extended globally, so $DT(M) = \chi(M, \mathscr{F}_{DT})$, or using this, $\chi(\phi_f(\mathbb{Q}_U[dim(U)])_x) = \nu_M(x)$ and DT Sheaf$|_{crit(f)} := \phi_f(\mathbb{Q}_U[dim(U)])$.

Definition 3.21.7 *(Cohomological Integrality)*
For DT invariant $DT(\gamma) \in \mathbb{Z}$, $\sum_{\gamma} DT(\gamma) q^{\gamma} \in \mathbb{Z} = \prod_{\gamma} (1 - q^{\gamma})^{-\Omega(\gamma)}$ where $\Omega(\gamma)$ is a BPS invaraints.

Definition 3.21.8 *(Kontsevich-Soibelman)*
DT is a new type of quantum KZ.

Definition 3.21.9 *(Cotangent Complex)*
Cotangent complex $L_{\mathfrak{M}}$ is an object of derived category, given by $Hom_{D(A)}(L_{\mathfrak{M}} \otimes_A M, A) \cong$ Derived Deformation $GroupDer_A(A, M)$, and $\mathfrak{M} : Alg_k \to Groupoids$. Alternatively, cotangent complex $L_X \in D(M)$ is $H^0(L_X) = \Omega^1_X$ for a scheme X, or if Deligne-Mumford stack $\mathfrak{M} = [X/G]$, cotangent complex $L_{\mathfrak{M}} \in D(\mathfrak{M})$, $L_{[X/G]} \cong [L_X \to \mathfrak{g}^* \otimes \mathscr{O}_X]$.

For the case of closed embedding $i : X \hookrightarrow Y$, $L_{X/Y} \cong [I/I^2 \to i^*\Omega^1_Y]$.

Also, for $X \xrightarrow{f} Y \xrightarrow{g} Z$, $f^*L_{Y/Z} \to L_{X/Z} \to L_{X/Y} \to$ where $L_{X/Y}$ is a cotangent complex of $X \to Y$.

3.22 Cotangent Sheaf

Definition 3.22.1 *(Tangent Sheaf)*
The sheaf θ_X on a scheme X is for all $U = Spec(A)$, $\theta_X(U) = Der_k(A, A)$, or equivalently, it is a sheaf of the morphism $Hom(\Omega^1_{X/k}, \mathscr{O}_X)$

The stalk is Zariski tangent space.

Definition 3.22.2 *(Cotangent Sheaf)*
Let $\mathscr{F}$ be an $\mathscr{O}_X$ module.

Cotangent sheaf is a sheaf of $\mathscr{O}_X$ modules $\Omega_{X/S}$ that represents the S-derivation in the sense.

$Hom(\Omega^1_{X/S}, F) = Der_S(\mathscr{O}_X, F)$

where $\Omega_{X/S}$ is a Kähler differential.

Definition 3.22.3 *(Kodaira-Spencer Map)*
Let M be a complex manifold, and it has a transition maps

$$\begin{tikzcd} M \arrow[r, "="] \arrow[d] & M \arrow[d] \\ U_i \subset \mathbb{R}^n \arrow[r, "f_{ij}"] & U_j \subset \mathbb{R}^n \end{tikzcd}$$

whose deformation is

$$\begin{tikzcd} \mathscr{M} \arrow[r, "="] \arrow[d] & sheaf M \arrow[d] \\ B \arrow[r, "\tilde{f}_{ij}"] & B \end{tikzcd}$$

where $B = U_i \times Spec(k[\epsilon])$, and Kodaira-Spencer map is $KS : T_0B \to H^1(M, T_m)$.

In scheme theory, deformation of smooth manifold can be

$$\begin{tikzcd} X \arrow[r] \arrow[d] & \mathscr{X} \arrow[d] \\ Spec(k) \arrow[r] & Spec(k[\epsilon]) \end{tikzcd}$$

have a short exact sequence

$$0 \to \pi^* \Omega^1_{Spec(k[\epsilon])} \to \Omega^1_{\mathscr{X}} \to \Omega^1_{\mathscr{X}/S} \to 0$$

where $\pi : \mathscr{X} \to S = Spec(k[\epsilon])$. *If we tensored by the* $\mathscr{O}_{\mathscr{X}}$*-module* $\mathscr{O}_X$ *gives the short exact sequence*

$$0 \to \mathscr{O}_X \to \Omega^1_{\mathscr{X}} \otimes \mathscr{O}_X \to \Omega^1_X \to 0$$

Using derived categories, this can be

$$RHom(\Omega^1_X, \mathscr{O}_X[+1]) \cong RHom(\mathscr{O}_X, T_X[+1]) \cong Ext^1(\mathscr{O}_X, T_X) \cong H^1(X, T_X).$$

Of Ringed Topoi, this can be described more abstractly. For the composition of maps of ringed topoi,

$$X \to_f Y \to Z$$

Then, associated to this composition is a distinguished triangle

$$f^* L_{Y/Z} \to L_{X/Z} \to L_{X/Y} \to^{+1}$$

Definition 3.22.4 *(Flat topology)*
fppf cover of X *is a jointly surjective family of morphisms* $(\phi_a : X_a \to X)$ *where* X_a *affine* ϕ_a *flat, finitely presented. This generates pretopology.*

Definition 3.22.5 *(Flat Cohomology)*
Flat cohomology or etale cohomology??

Definition 3.22.6 *(Hilbert Scheme)*

If $f : X \to S$ *is flat, we have a canonically a commutative diagram*

$$\begin{array}{ccc} X & \longrightarrow & Hilb_S \\ \downarrow & & \downarrow \\ S & \longrightarrow & S \end{array}$$

where $Hilb_S$ *is a family of flat morphisms to* S. *Since* f *is flat, all fibers* $f_s : X_s \to s$ *for all* $s \in S$ *have the same Hilbert polynomial* Φ. *We could say* $Hilb_S^{\Phi} = Hilb_S$.

3.23 Derived Geometry

Category of coherent sheaves $Coh(X)$ is inconvenient in the derived language, since $A[1] \notin Coh(X)$ or $Cone(f) \notin Coh(X)$ in general case when $A \in Coh(X)$ or $f : A \to B$ where $A, B \in Coh(X)$.

Definition 3.23.1 *(Homological Perturbation)*
Homological perturbation is similar to deformation, but has different nuance.

For chain complex (C, d), we paraphrase the chain complex to its homology $(H, 0)$ to reduce the problems, and homological perturbation is an operation to define differential operator on its homology induced from the chain complex. Let $i : H \to C$, $p : C \to H$, and $h : C \to C[-1]$, satisfying the homological retract $pi = id_H$ and $ip + dh + hd = id_C$, and by using this, we define new operation on H $d_H = p(\sum_{n \geqq 0} (\delta h)^n \delta) i$.

4 Differential Geometry

Definition 4.0.1 *(Contact Manifold)*
(M^{2n+1}, ξ) is contact manifold, if n-dimensional submanifold $L \subset M$ is Legendrian submanifold such that $T_p L \subset \xi_p$ for all $p \in L$. Completely non-integrbale means $\alpha \wedge (d\alpha)^n \neq 0$.

Definition 4.0.2 *(Connection to Symplectic Geometry)*
existence of $\omega^n \neq 0$

Proposition 4.1 *()*
Contactification is prequantization.

Definition 4.1.1 *(Geometric Quantization)*
For a symplectic manifold (M, ω), and take $c_1(L) = \frac{[\omega]}{2\pi} \in H^2(M, \mathbb{Z})$ is integer valued. Prequantum line bundle $L \to M$ with connection ∇, whose curvature is $F_\nabla = -i\omega$, whose possible state ψ is $\psi(q, p)$. However, the state $\psi(q, p)$ has too much freedom, and we need to reduce the dimension to half, so that $\nabla_v \psi = 0$ where $v \in P$, where p is a polarization. Polarization P is n-dimensional Lagrangian submanifold $P \subset TM_{\mathbb{C}}$ (consider if M is symplectic, TM or $TM_{\mathbb{C}}$ is also symplectic), which satisfies integrability condition, means $\forall v, w \in P$, $\omega(v, w) = 0$, $[v, w] \in P$.

If we determine (M, ω), L, and P, there is a Hilbert space $\mathfrak{H} = \{\psi \in \Gamma(M, L) | \nabla_v \psi = 0 (v \in P)\}$, and hence generating Hilbert space from symplectic manifold.

Definition 4.1.2 *(Heisenberg Group)*
Heisenberg group H_n consists of x, p s.t. $[x, y] = i\hbar$. By Heisenberg action on Hilbert space $\mathfrak{H}$, so $(T(a)\psi)(x) = \psi(x + a)$ and $(M(b)\psi)(x) = e^{\frac{ibx}{\hbar}} \psi(x)$. Also, consider Stone-von-Neumann theorem for the uniqueness of irreducible unitary representation of $L^2(\mathbb{R}^n)$.

Definition 4.1.3 *(D-Hilbert Scheme)*
D-Hilbert scheme $D - Hilb^n(X)$ is moduli space of D_X-module of finite length and zero dimensional support. D-Hilbert scheme is just an analogy of Hilbert version of D_X-module.

4.2 Intersection of Analysis/Topology/Algebra

Definition 4.2.1 *(Topological Index Using K-Theory)*
For an elliptic operator $D : \Gamma(E) \to \Gamma(F)$, an analytic index is $dim(ker(D)) - dim(coker(D))$ is finite dimensional, if D is a Fredholm operator, while topological index is defined by a Thom isomorphism $\pi : K^0(M) \to K^0(pt) = \mathbb{Z}$, and topological index is precisely $\pi(\sigma(D)) \in \mathbb{Z}$. According to Atiyah-Singer index theory, (analytical index) = (topological index).

4.3 Geometric Aspect of PDE

Definition 4.3.1 *(Non-Linear PDE)*
List of Linear PDE is roughly speaking an eq $Lu = f$ where L is an operator. The famous example is n-variable 2nd order is $\sum_{|\alpha| \leqq 0} a_\alpha(x)\partial^\alpha u(x) = f(x)$, Laplace $\Delta u = 0$, heat eq $\partial_t u - \Delta u = 0$, and wave eq $\partial_t^2 u - \Delta u = 0$, and all of them has a linearity, while non-example is $\Delta u + u^2 = 0$.

Non-linear PDE can be transformed to linear PDE by deformation. For example, $\Delta u + u^2 = 0$ is non-linear, but we can replace $u = u_0 + \epsilon v$, and $\Delta(u_0 + \epsilon v) + (u_0 + \epsilon v)^2 = 0$ generates linear-PDE $\Delta v + 2u_0 v = 0$. How's Navier-Stokes?

In language of category, ordinary differential is a morphism $C^\infty(M, N) \to \Gamma(T^*M \otimes u^*TN)$ but the case of jet is $C^\infty(M, N) \to \Gamma(J^k(M, N))$, and PDE is a submanifold $E \subset J^k(M, N)$, where k-Jet space $J^k : Man \to Man$ is a functor, consists of a set of k-jet space $j_x^k u$, where $j_x^k u$ is a set of valued of u and kth-differential of u at x $(u, du, d^2u, d^nu)(x)$, and J^k is in fact ind-scheme.

4.4 Joyce Conjecture

Definition 4.4.1 *(Joyce)*
Let M be a G2-manifold, and this requires 3-form ϕ, that can define associative/coassociative submanifolds. Associative submanifold is $N^3 \subset M^7$ where $\phi|_{N^3} = vol_{N^3}$ Coassociative submanifold is $C^4 \subset M^7$ where $\star\phi|_{C^4} = vol_{C^4}$ where $\star\phi$ is Hodge dual of ϕ.

Joyce conjecture claims any compact G2-manifold can be generated by Joyce type operation (in other words, this question might be classification problem of G2-manifolds). Joyce's method is $\mathcal{T}^7$ quotient by finite group Γ. $M = \mathcal{T}^7/\Gamma$ is a orbifold not smooth, but we have resolution of singularities, so we can make smooth G2-manifold.

4.5 Integrability

Definition 4.5.1 *(Integrability)*
Consider $\omega = \Sigma_i a_i(x) dx_i$ is a 1-form, and we'll consider the existence of $\omega = 0$.

Try to paraphrase $\omega = \Sigma_i a_i(x)\frac{dx_i}{dt}dt$, and find a curve s.t. $\gamma(t) = (x_i)$. This is Phaffian integrable. The condition of integrability is $\omega \wedge d$

Definition 4.5.2 *(Fundamental Theorem of Calculus)*
1*-variable version of integrability is fundamental theorem of calculus.*

Definition 4.5.3 *(Foliation and Integrability)*
Integrability means the existence of foliation. A compact smooth manifold has 1-dim foliation iff Euler characteristic is zero.

4.6 Holonomy

Definition 4.6.1 *(Holonomy on Flatness)*
Möbius band is flat, but has holonomy. Or any topologically non-trivial case.

5 Commutative Algebraic Geometry

5.1 Complex Analysis

The very question of complex analysis is to classify the possible class of functions depending on the domains.

5.1.1 Singularity

In complex analysis, the singular points defined on the region is the clue to understand the behavior of the functions. Conversely, if is it possible to reconstruct the function from the singular points is the monodromy problem. For the future convenience, in the lagunge of algebraic geometry, such singular points is called divisor.

Definition 5.1.1 *(Divisor)*
Divisor is the singlar regions of the function in the domain. Typically codimension 1*.*

On extension of divisor argument, let's consider the intersection theory of the divisors in higher dimensions. Somehow, the elementary analogy of explanation is possible by using Euclidean topology.

Definition 5.1.2 *(Intersection in Euclidean Topology) For an Euclidean space $\mathbb{R}^n$, the basis of topology is open balls $B_\epsilon(x) = \{y|d(x,y) < \epsilon\}$ for $\epsilon > 0$ and $x \in \mathbb{R}^n$, so any open subset is a union of some open balls. Consider the dimension of an open ball $B_\epsilon(x)$ is n, namely $dim(B_\epsilon(x)) = n$, and the dimension of any open subsets are n, and the dimension of an open subset is invariant by union and intersection. On the other hand, a close subset is not always dimension n. For example, hyperplane is, by defintion, codimension* 1*, which means dimension $n-1$.*

Definition 5.1.3 *(Residue Theorem)*
Skipped

5.1.2 Mondromy

Definition 5.1.4 *(Linear and Non-Linear Equations)*

- *(Generalized Hypergeometric Function)*
  ${}_pF_q(a,b,c;z) = \Sigma_{n=0}^{\infty} \frac{(a_1)_n(a_2)_n\cdots(a_p)_n}{(b_1)_n\cdots(b_q)_n} \frac{z^n}{n!}$

  where $a_1, \cdots, a_p, b_1, \cdots, b_q \in \mathbb{C}$ *but* $b_i \neq 0, -1, -2, \cdots,$ *and* $z \in \mathbb{C}$.

- *(Hypergeometric Function)*

  If $p = 0$ *and* $q = 1$*, generalized hypergeometric function becomes ordinary hypergeometric function. Hypergeometric function is a solution of diffyq*

  $z(1-z)\frac{d^2f}{dz^2} + [c - (a+b+1)z]\frac{df}{dz} - abf = 0$

- *(Bessel Function)*

  If $p = 0$ *and* $q = 1$*, generalized hypergeometric function becomes Bessel function. That is,*

  ${}_0F_1(a,b,c;z) = \Sigma_{n=0}^{\infty} \frac{(-1)^n}{(b)_n} \frac{z^n}{n!}$

- *(Heun's Equation)*

  $\frac{d^2w}{dz^2} + [\frac{\gamma}{z} + \frac{\delta}{z-1} + \frac{\epsilon}{z-a}]\frac{dw}{dz} + \frac{\alpha\beta z - q}{z(z-1)(z-a)}w = 0$

- *(Theta Function)*

  Used for solitons and abelian varieties.

  $\vartheta(z;\tau) = \Sigma_{n=-\infty}^{\infty} exp(\pi i n^2\tau + 2i\pi nz)$

- *(Painleve)*

  Painleve equation is a transcendents are solution of non-linear second ODE, but it has an integrabl structure.

  $\frac{d^2y}{dt^2} = 6y^2 + t$

Definition 5.1.5 *(Monodromy Representation)*

Monodromy representation is a finite dimensional representation, which means $dim(V) < \infty$*, and* $\pi_1(X)$ *is a fundamental group. Monodromy representation is defined by*

$$\rho : \pi_1(X) \to GL(V)$$

We could define the complex functions whose domain is X*, and by choosing arbitrary dimension of* V*, we have several different solutions.*

Example 5.2 *(Monodromy Representation* $X = \mathbb{C}\backslash\{0\}$ *and* $dim(V) = 1$*)*
So there is only one singularity, and $f(z)$ *could be either* $f(z) = z^\alpha$ *where* $\alpha \in \mathbb{C}\backslash\mathbb{Z}$ *or* $f(z) = log(z)$*.*

Example 5.3 *(Monodromy Representation* $X = \mathbb{C}\backslash\{0\}$ *and* $dim(V) = 2$*)*
$f(z) = Az^\alpha + Bz^\beta$

Example 5.4 *(Monodromy Representation* $X = \mathbb{C}\backslash\{0,1\}$ *and* $dim(V) = 1$*)*
$\pi_1(\mathbb{C}\backslash\{0,1\}) = F_2$ *is a free group of 2-generators, and Jordan-Canonical form (or could be diagonalized)*

$$f(z) = z^\alpha(1-z)^\beta$$

Example 5.5 *(Monodromy Representation* $X = \mathbb{C}\backslash\{0,1\}$ *and* $dim(V) = 1$*)*
$\begin{bmatrix} e^\alpha & 1 \\ 0 & e^\beta \end{bmatrix}$*, and*
All the possible functions are the base solution $f(z) = z^\alpha(1-z)^\beta$ *and the matrix multiplication.*

Example 5.6 *(Monodromy Representation* $X = \mathbb{C}\backslash\{0,1\}$ *and* $dim(V) = 1$*)*

5.6.1 KZ Equation

Quasi-Hopf algebra for KZ equation.

Definition 5.6.1 *(Quasi-Hopf Algebra)*
Quasi-Hopf Algebra is a generalization of Hopf algebra. It's antipode with Quasi-bialgebra as

$$\Sigma_i S(b_i)\alpha c_i = \epsilon(a)\alpha$$
$$\Sigma_i b_i \beta S(c_i) = \epsilon(a)\beta$$

for all $a \in A$ *and where*

$$\Delta(a) = \Sigma_i b_i \otimes c_i$$

and

$\Sigma_i X_i \beta S(Y_i) \alpha Z_i = I$
$\Sigma_j S(P_j) \alpha Q_j \beta R_j = I$

where $\Phi = \Sigma_i X_i \otimes Y_i \otimes Z_i$ *and* $\Phi^{-1} = \Sigma_j P_j \otimes Q_j \otimes R_j$.

Definition 5.6.2 *(Knizhnik-Zamolodochikov Equation)*

$fr\hat{a}kg_k$ *be affine Lie algebra with level* k *and dual Coxeter number* h. $i, j = 1, 2, \cdots, N$, t^a *be basis of* $\mathfrak{g}$
$((k+h)\partial_{z_i} + \Sigma_{j \neq i} \frac{\Sigma_{a,b} \eta_{ab} t_i^a \otimes t_j^b}{z_i - z_j}) < \Phi(v_N, z_N) \cdots \Phi(v_1, z_1) >= 0$

We'd consider alternative approach of KZ equation.

Definition 5.6.3 *(Monodromy)*
An example of monodromy is a multi-valued function $f(z) = log(z)$, *since this function doesn't have the same value if we rorate around the singular point* $z = 0 \in \mathbb{C}$. *In fact, consider* $1 = e^{2\pi i}$, $log(z) = log(z * e^{2\pi i}) = log(z) + log(e^{2\pi i}) = log(z) + 2\pi i$, *thus the function can take multiple value even though it takes the same inputs. This is the idea of monodromy.*

These seemingly different functions $log(z)$ *and* $log(z) + 2\pi i$ *can be joined together by analytic continuation. Consider the domain of* $log(z)$ *is entire of the plane except the singular point, means that* $0 \leqq \theta < 2\pi$, *while the other* $log(z) + 2\pi i$ *for* $2\pi \leqq \theta < 4\pi$. *If we also consider intermediate function* $log(z) + \pi i$, *whose domain is* $\pi \leqq \theta < 3\pi$, *we have non-empty intersection with both of them, hence by identity theorem,* $log(z)$ *and* $log(z) + 2\pi i$ *are identified to be the same function.*

5.6.2 Configuration Space

Definition 5.6.4 *(Configuration Space)*
The configuraiton space is the domain of the solution of KZ equation. The configuration space is defined as

$Conf_n(\mathbb{C}) = \{(z_1, z_2, \cdots, z_n) \in \mathbb{C}^n | z_i \neq z_j \text{ for } i \neq j\}$

$Conf_n(\mathbb{C})$ *creates monodromy around the singular points. The monodromy is calculated by the monodromy representation of the fundamental group* $\pi_1(Conf_n(\mathbb{C})) = B_n$, *which is a braid group.*

Configuration space is a complex manifold.

Definition 5.6.5 *(Configuration Space)*
The surjective map of configuration spaces

$Conf_{n+N}(X) \to Conf_N(X)$

is a locally trivial fiber bundle with typical fiber $X \backslash \{x_1, \cdots, x_N\}$.

Definition 5.6.6 *(Horizontal Chord Diagrams)*
The monoid of horizontal chord diagram is the free monoid

$D_n^{pb} = FreeMonoid((ij)|1 \leqq i < j \leqq n)$

and each element (ij) *is a permutation of* i *&* j.

$2T$*-relation*
(ij) *and* (kl) *commutes if* i,j,k,l *are all distinct, namely,* $(ij)(kl) = (kl)(ij)$. *So* $2T$*-relation is*

$2T = \{(ij)(kl) - (kl)(ij)|i, j, k, l \text{ all distinct}\}$
$4T$*-relation*
Consider $(ik)(ij) = (ij)(jk)$ *are the same permutation but by using different elements, and also* $(jk)(ij) = (ij)(ik)$ *is. We join them together and* $(ik)(ij) + (jk)(ij) = (ij)(jk) + (ij)(ik)$.

$4T = \{(ik)(ij) + (jk)(ij) - (ij)(jk) - (ij)(ik)|i, j, k \text{ all distinct}\}$

$A_n^{pb} = Span(D_n^{pb})/4T$

Definition 5.6.7 *(KZ Connection)*
(KZ Differential Form)
$\omega_{KZ} \in \Omega(Conf(\mathbb{C}), A_n^{pb})$ *such that*

$\omega_{KZ} = \Sigma d_{dR} log(z_i - z_j) \otimes t_{ij}$

and KZ connection is flat since

$d\omega_{KZ} + \omega_{kz} \wedge \omega_{KZ} = 0$

(KZ Connection)
$\nabla_{KZ}\phi = d\phi + \omega_{kz} \wedge \phi$

and KZ equation is to find ϕ *such that*

(KZ Equation)
$\nabla_{KZ}\phi = 0$

KZ connection is flat, which means the result of integration doesn't change by the direction of path, so analytic continuation is well-defined.

Definition 5.6.8 *(Degeneration)*
Formally speaking, a degneration is a map

$$\pi : \mathfrak{X} \to S$$

where $\mathfrak{X}$ is the total space, and S is the base i.e. $\mathbb{A}^1$, and for each $s \in S$, $\mathfrak{X}_s = \pi^{-1}(s)$, together a degeneration at $s_0 \in S$ if the fiber $\mathfrak{X}_{s_0}$ is of idiosyncratic shape. For instance,

$$\mathfrak{X} = \{(x, y, t) \in \mathbb{A}^3 | y^2 = x^3 + tx\}$$

where elliptic curve if $t \neq 0$, and singular if $t = 0$.
The reason why we need to study degeneration is that if we compactify moduli spaces, we might add some singular fibers, by which we mean degeneration.

Definition 5.6.9 *(Degeneration of Moduli Space)*
Degeneration is part of a moduli problem, and we'll particularly cosider a moduli space arising from monodromy representation i.e. the monodromy representation defines complex functions, and we'll define moduli psace of the domain of the complex functions. Here, by context, degeneration of moduli space is a moduli space of a domain where Painleve trancendence is defined.

Example 5.7 *(Degeneration)*

- ${}_pF_q(a, b, c; z)$ *is a degeneration of* ${}_{p+n}F_{q+m}(a, b, c; z)$
- *Painleve trancendence is a degeneration of hypergeometric funcitons.*

5.7.1 Gauss-Manin Connection

Definition 5.7.1 *(Eheresmann Fibration Theorem)*

If a smooth mapping $X \to B$ where X and B are smooth manifolds is a submersion and a proper map, then it's a locally trivial fibration.

Note: locally trivial fibration means locally isomorphic to cartesian, which means locally each fiber is isomorphic (or diffeomorphic) to the same fibers.

The idea of Gauss-Manin connection is to track the family of complex algebraic varieties by the change of their parameters. By considering its cohomology class, .

Definition 5.7.2 *(Gauss-Manin Connection)*

Let a smooth morphism of $X \to B$. The fiber X_b is a smooth manifold and diffeomorphic to each other for all $b \in B$, and it's cohomology class $H^k(X_b)$ are all isomorphic, and consider that the cohomology classes are all vector spaces of

the same dimension, which makes a vector bundle (note each $H^k(X_b)$ is isomorphic to each other as vector spaces, but it has different basis), and it naturally defines Gauss-Manin connection, which defines a parallel transport of this vector bundle.

The explicit construction of Gauss-Manin connection is as follows. For $[\omega] \in H^k(\Omega^{\cdot}_{X/S})$ where $d_{X/S}\omega = 0$, and d_S is an exterior differential of S, we define conenction ∇ is a morphism

$\nabla : H^k \to H^k \otimes \Omega^1_S$ *(or* $\nabla : \Gamma(H^k) \to \Gamma(H^k) \otimes \Gamma(Omega^1_S)$ *),*

which satisfies

$\nabla[\omega] := [d_S\omega]$

Definition 5.7.3 *(Parallel Transport)*

Parallel transport is to move a vector on a geometry (the vector exists on the fiber of the point) along a curve, so that $\nabla_{\dot{\gamma}} v = 0$, and parallel transport is uniquely defined on a vector bundle using conneciton. Namely, if the cohomology group is finite dimension, the vector exists at least finitely many amount, but the connection which defines parallel transport is defined by the same connection.

For the path $\gamma : [0,1] \to S$, parallel transport means to find $v(t)$ such that $\nabla_{ga\dot{m}ma(t)} v(t) = 0$. In other words, if we parallely transport $v(0) \in H^k(X_b)$, it'll be $v(t)$ for each t.

Proposition 5.8 *(Gauss-Manin Connection is Flat)*

In order to prove flatness, the curvature needs to be zero, means $R_\nabla = \nabla^2 = \nabla \circ \nabla : H^k \to H^k \otimes \Omega^2_S$.

$\nabla \circ \nabla([\omega]) = [d_s(d_s\omega)] = 0$ *for all ω, since $d_S \circ d_S = 0$. Hence $\nabla^2 = 0$*

Proposition 5.9 *(Gauss-Manin and Monodromy)*

For a parallel transport along a curve S.
Monodromy is a holonomy defined on a flat connection, whose closed curve is defined as a loop around a singular point.

Also, consider Riemann-Hilbert correspondence claims correspondence of flat connections and monodromy representations.

Definition 5.9.1 *(Riemann-Hilbert Correspodence)*
RH correspondence is a 1-to-1 correspondence of flat connections and the monodromy representations. In other words, all the possible complex functions can be reconstructed from monodromy representations. Note that flat connection ∇

can define a system of diffyq $\nabla\phi = 0$, thus existence of ϕ, that can be derive from the representations.

Remark 5.10 *(Determining the Representation)*

The vector space V of the monodromy representation of $\rho : \pi_1(S) \to GL(V)$ is the dimension of the vector bundle. In other words, automorphism of fiber $GL(V)$ corresponds to the fundamental group $\pi_1(S)$.

5.11 Moduli Space

Moduli space is a space generated by geometric quotient of some objects, and each point corresponds to an isomorphism class of objects. For example, for a Mobius strip M and a circle S^1, a natural projection $M \to S^1$ is a vector bundle with fiber $\mathbb{R}^1$, and the moduli space is $M/\mathbb{R}^1$.

Remark 5.12 *Moduli space is more than a set of points. $M/\mathbb{R}^1 \cong S^1$ is an isomorphism, and $M/\mathbb{R}^1$ is a moduli space, but S^1 is not a moduli space, and these two are distinct only because each point corresponds to an isomorphism class.*

Remark 5.13 *Moduli space is not a fiber bundle, but what is the difference from the fiber bundle exactly? The moduli space has each point as an isomorphism class, not as an actual fiber. There are few different varieties of vector bundles over circle S^1, which is annulus or mobius strip, whose fibers are both $\mathbb{R}^1$, but moduli space cannot distinguish them.*

Remark 5.14 *Etymologically, Moduli stands for "mod"uli.*

5.14.1 $M_{g,n}$ Moduli Space

Definition 5.14.1 *(Configuration Space)*

Let X be a topological space, then the configuration space $Conf_n(X)$ is defined as

$$Conf_n(X) = \{(z_1, z_2, \cdots, z_n) \in X^n | z_i \neq z_j \text{ for } i \neq j\}$$

Definition 5.14.2 *($M_{g,n}$ Moduli Space)*

The $M_{g,n}$ moduli space is GIT quotient of configuration space of $\mathbb{P}^1$ by $SL^2(\mathbb{C})$, which is a geometry of genus g with n-marked points of the base space $\mathbb{P}^1$. In particular,

$$M_{0,n} = Conf_n(\mathbb{P}^1)/SL^2(\mathbb{C})$$

Remark 5.15 *($M_{0,n}$ Calculation)*

- *($n = 3$)*
  Consider that Möbius transformation $z \mapsto \frac{az+b}{cz+d}$ can put arbitary distinct 3-points z_1, z_2, and z_3 mapped to arbitray distinct 3-points w_1, w_2, and w_3.

  Since configuration space of 3-dimensions is a space of 3 independent distinct elements, each points could be arbitrarily mapped to any points of the space, which means by $SL^2(\mathbb{C})$ quotient, it'll be all identified. Hence,

  $M_{0,3} = \{pt\}$

- *($n = 4$)*

  Recall $SL^2\mathbb{C}$ fixed the 3-points, then the last 4th-point has some freedom, because kf we let $\lambda = \frac{(z_1-z_3)(z_2-z_4)}{(z_1-z_4)(z_2-z_3)}$, Cross ratio λ could take any values of $\mathbb{P}\backslash\{0,1,\infty\}$. $\lambda \neq 0$ or $\lambda \neq \infty$ because all points z_i are distinct, and none of the numerator or denominator could be zero, and if $\lambda \neq 1$, $\frac{(z_1-z_3)(z_2-z_4)}{(z_1-z_4)(z_2-z_3)} = 1$, then $(z_1 - z_3)(z_2 - z_4) = (z_1 - z_4)(z_2 - z_3)$, and by simple college algebra, it will be $(z_1 - z_2)(z_3 - z_4) = 0$, but it never happens since all points are distinct. Otherwise, λ could take any values. Hence,

  $M_{0,4} = \mathbb{P}\backslash\{0,1,\infty\}$

- *($n = 5$)*

  If we fix the mapped points w_1, w_2, and w_3 depending on z_1, z_2, and z_3, w_4 is cross ratio depending on z_4, while w_5 is another cross ration by z_5, which means

  - $\lambda = \frac{(z_1-z_3)(z_2-z_4)}{(z_1-z_4)(z_2-z_3)}$
  - $\mu = \frac{(z_1-z_3)(z_2-z_5)}{(z_1-z_5)(z_2-z_3)}$

  so we have two cross ratios λ and μ, so $M_{0,5}$ is two parameters. Hence,

  $M_{0,5} = \mathbb{P}\backslash\{0,1,\infty\}$

Finally, notice indeed $dim(M_{0,n}) = n - 3$

Definition 5.15.1 *($M_{1,1}$ Moduli Space)*
The very definition of $M_{1,1}$ is a moduli space of algebraic curves of genus 1 of

1-marked points, but the question is that are all the algebraic curves of genus 1 of 1-marked points elliptic curves and vice versa? Yes, and this is, in fact, equivalently to that it's a moduli space of elliptic curves quotient by j-invariants. Another question is how to explicitly classify all the elliptic curves, and this could be done by moduli space of torus, and only to prove that torus is equivalent to some elliptic curve.

$M_{1,1} = \mathbb{H}/SL_2(\mathbb{Z}) \cong \Lambda$ *where* $\mathbb{H}$ *is the upper-half plane.*

where $dim(M_{1,1}) = 1$

Or equivalently, $M_{1,1} = \{(E,p)|$ *genus 1 smooth curve,* $p \in E\ \}/\sim$

Remark 5.16 *($M_{1,1}$ Lattice and Curve Correspondence)*

In order to verify the previous definition. we'll start from the lattice correspondence with the elliptic curve.

- *(Step 1:)*
  Each lattice can be described by the standard form, which is

  $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$

  for some $\tau \in \mathbb{C}$

- *(Step 2:)*
  If E_τ *is a lattice, this* $\mathbb{C}/E_\tau$ *can be a domain of the Weierstrass* $\wp$ *function, which is in other words period, and Weierstrass* $\wp$ *function defines an elliptic function. To formulate, if we let* $Y = \wp'(z, \mathbb{C}/E_\tau)$ *adn* $X = \wp(z, \mathbb{C}/E_\tau)$*, we have the following identities:*

  $Y^2 = 4X^3 - g_2X - g_3$

  where g_2 *and* g_3 *are Eisenstein series.*

- *(Step 3: Legendre Transform)*

  What else do we have? This correspondence is actually not one-to-one, but it will be, if we take quotient of the set of elliptic curves by isomorphism classes by j-invariant, that is equivalent to say that if elliptic curves have the same j-invariant, then they have isomorphism as algebraic varieties, and Legendre transform is the step to defien j-invariant. The previous elliptic function can be canonically transformed to the Legendre form

  $y^2 = x(x-1)(x-\lambda)$ *where* $\lambda \in \mathbb{C}\backslash\{0,1\}$

- *(Step 4:j-Invariant)*

  By using λ arised from Legendre transform, if we let j-invariant as

  $j(\lambda) = 256 \cdot \frac{(1-\lambda+\lambda^2)^3}{\lambda^2(1-\lambda)^2}$

  $j(\tau) = j(\lambda(\tau))$

Note if two Weierstrass functions are distinct up to $SL_2(\mathbb{Z})$ transformation, g_2 and g_3 are always distinct.

Definition 5.16.1 *($M_{1,0}$ Moduli Space)*

Also, $M_{1,0}$ is $n = 0$, and it's also moduli space of elliptic curves $y^2 = x^3 + ax + b$,
For $n = 0$, $M_{1,0}$ is $M_{1,0} = M_{1,1}/Aut(C)$

Remark 5.17 *(Group Structure)*
Why the moduli space of elliptic curves are $M_{1,1}$ instead of $M_{1,0}$ is because each elliptic curve has a group structure, or in other words, if the point of the elliptic curve is depending on X and Y coordinates, whose value depends on the position of the lattice E_τ. Each element of the group is a point of the elliptic curve, and of course the unit element O is. The unit element O is point at infinity.

Remark 5.18 *(Elliptic Curve Is Not Torus)*

Definition 5.18.1 *($M_{2,0}$ Moduli Space)*

$M_{2,0} = \mathbb{H}_2/Sp_4(\mathbb{Z})$

where $dim(M_{2,0}) = 3$

Remark 5.19 *(Tree Structure)*

$M_{0,n}$ can be combinatorially constructed by the tree structure of multiple $M_{0,4}$.

5.19.1 Compactification of Moduli Space

Definition 5.19.1 *()*
The compactification $\overline{M_{0,4}}$ is $\overline{M_{0,4}} = \mathbb{P}^1$.
$\overline{M_{0,4}}^{GIT} \cong (\mathbb{P}^1)^4//PGL_2$

5.20 Representation

Definition 5.20.1 *(Verma Module)*
For a semi-simple Lie algebra $\mathfrak{g}$ and the triangle decomposition $\mathfrak{g} = \mathfrak{n}^- \oplus \mathfrak{h} \oplus \mathfrak{n}^+$, and let $\lambda \in \mathfrak{h}^*$ be a weight. Also, $\mathbb{C}_\lambda$ be a $\mathfrak{b}$-module, where $\mathfrak{b}$ is a Borel subalgebra $\mathfrak{b} = \mathfrak{h} \oplus \mathfrak{n}^+$, and $\mathbb{C}_\lambda$ is precisely $\mathbb{C}_\lambda = \mathbb{C}v_\lambda$, where the highest weight $v_\lambda = 1 \otimes 1$ satisfies $\mathfrak{n}^+ v_\lambda = 0$ and $hv_\lambda = \lambda(h)v_\lambda$. Finally, Verma module is defined to be $M(\lambda) := U(\mathfrak{g}) \otimes_{U(\mathfrak{b})} \mathbb{C}_\lambda$.

Definition 5.20.2 *(Harish-Chandra)*
Character $\chi_\lambda : Z(U(\mathfrak{g})) \to \mathbb{C}$ is 1-dim representation. Harish-Chandra claims $Z(U(\mathfrak{g})) \cong S(\mathfrak{h})^W$ where W is a Weyl group. $\chi_\lambda(z) = p_z(\lambda)$ where $p_z \in S(\mathfrak{h})^W$.

Definition 5.20.3 *(Twisted D-Module)*
D-module $\mathcal{D}_X$ is $\mathcal{D}_X(U) = \{\mathbb{C}$ linear maps $\mathcal{D} : \mathcal{O}_X(U) \to \mathcal{O}_X(U)|$ where $\mathcal{D}$ is a finite degree differential operat
Furthermore, twisted $\mathcal{D}$-module $\mathcal{D}_\lambda$ is defined by $\mathcal{D}_\lambda = \mathcal{L}_\lambda \otimes \mathcal{D}_X \otimes \mathcal{L}_\lambda^{-1}$, where $X = G/B$, and $\mathcal{L}_\lambda = G \times^B \mathbb{C}_\lambda$, and $\mathcal{L}_\lambda^{-1} = \mathcal{L}_\lambda^\vee = Hom_{\mathcal{O}_X}(\mathcal{L}_\lambda, \mathcal{O}_X)$.

Definition 5.20.4 *(D-Module)*
D-Module is a cohomology of Bessel functions (= BRST = dga).

Definition 5.20.5 *($\mathcal{D}_X$ Module)*
In the local coordinate $x_1, x_2, \cdots, x_n$, $\mathcal{D}_X(U) = \{\sum_{|\alpha| \leqq k} a_\alpha(x)\partial^\alpha\}$, and $\mathcal{O}_X \subset \mathcal{D}_X$.
$gr(\mathcal{D}_X) \cong \mathcal{O}_{T^*X}$.

Definition 5.20.6 *(Beilinson-Bernstein)*
Beilinson-Bernstein is an equivalence of categories $\mathcal{D}_\lambda - mod \cong U(\mathfrak{g})_\chi - mod$, or $\Delta_\lambda : \mathcal{D}_\lambda - mod(G/B) \cong U(\mathfrak{g})/(central\ ideal \chi_\lambda) - mod$. Here we call an operation localization, which maps a $\mathfrak{g}$-module $M \mapsto \Delta_\lambda(M)$. For example, a mapping $U(\mathfrak{g}) \mapsto \Gamma(G/B, \mathcal{D}_\lambda)$.

Definition 5.20.7 *(Beilinson Spectral Sequence)*
Beilinson claims the existence of spectral sequence $(\mathcal{O}, \mathcal{O}(1), \cdots, \mathcal{O}(n))$, that generates $D^b(Coh(\mathcal{P}^n))$. The category is "genenated", if any objects $X \in \mathcal{C}$ is generated by the collection $\{E_i\}$ by means of direct sum or extensions, while any morphisms are generated by identity morphisms.

Definition 5.20.8 *(Bernstein-Gelfand-Gelfand)*
For a finite dimensioanl irreducible representation $L(\lambda)$, BGG resolution is $\bigoplus_{l(w)=n} M(w \cdot \lambda) \to \cdots \to \bigoplus_{l(w)=1} M(w \cdot \lambda) \to M(\lambda) \to L(\lambda) \to 0$, where $M(\lambda)$ is a Verma module $M(\lambda) := U(\mathfrak{g}) \otimes_{U(\mathfrak{b})} \mathbb{C}_\lambda$, and use $[M(\lambda) : L(\mu)] \neq 0 \leftrightarrow \mu = w \cdot \lambda$ for some $w \in W$.

This resolution is defined in BGG category $\mathcal{O}$, which is an abelian category. BGG category $\mathcal{O}$ is a category of $\mathfrak{g}$-module of $\mathfrak{h}$ semisimple, each weight space is finite dimensional, $U(\mathfrak{n}^+)$ is locally nilpotent. This category $\mathcal{O}$ is an abelian category, and moreover abelian subcategory $\mathcal{O} \subset \mathfrak{g} - mod$.

Definition 5.20.9 *(D-Modules in Birational Geometry)*
In birational geometry, global section of canonical sheaf $H^0(X, \omega_X^{\otimes m})$ *is* $H^0(X, \omega_X^{\otimes m})$ *where* $\omega_X = wedge^{dim(X)}\Omega_X^1$ *is often a birational invariant.*

Definition 5.20.10 *(Takuro-Mochizuki)*

Definition 5.20.11 *(Q-Difference Module)*
The purpose of Q-difference module is to represent $\frac{d}{dx}f(x)$ *or* $f(x+1) - f(x)$ *over the module to describe diffyq algebraically. More precisely, an action* σ_q *is* $\sigma_q(k \cdot m) = \sigma_q(k) \cdot \sigma_q(m)$ *where* $f \in K$, $m \in M$, *and* $\sigma_q : f(x) \mapsto f(qx)$. *One of the applications is D-module.*

Definition 5.20.12 *(Riemann-Hilbert Correspondence/Non-Abelian Hodge Correspondence)*
Riemann-Hilbert correspondence claims the correspondence of flat connection (E, ∇) *and representation* $\rho : \pi_1(X) \to GL_n(\mathbb{C})$, *and category of flat connection (each object is flat bundle over* X*) and category of is equivalent.*

Similarly, non-abelian Hodge correspondence is a correpondence of flat bundles on X *and Higgs bundles on* X, *or correspondence of Higgs bundles on* X *and representation* $\rho : \pi_1(X) \to GL_n(\mathbb{C})$.

Definition 5.20.13 *(Mixed Twistor D-module)*
Mixed Twistor D-module is a generalization of mixed Hodge structure together with D-module. The problem of ordinary mixed Hodge structure is that it doesn't describe compactness and non-holomorphic singularity of the geometry, and it strongly relies on its complex structure. Mixed Hodge module is a mixed Hodge structure adding D-module. The pure twistor D-module consits $(M^{'}, M^{''}, C)$ *where* $M^{'}$ *is a D-module,* $M^{''}$ *is its dual,* C *is a pairing by Hermitian mean. Mixed twistor D-module is roughly speaking a mixed Hodge module with the pure twistor D-module.*

Definition 5.20.14 *(Kobayashi-Hitchin Correspondence)*
For a complex manifold X *and its complex vector bundle* $E \to X$. *Kobayashi-Hitchin claims the 1-to-1 correspondence:* E *is stable iff* E *has an Hermitian-Einstein connection.*

Definition 5.20.15 *(Non-Commutative Hodge Correspondence)*
Hodge correspondence over a geometry X *is defined to be the correspondence of cohomologies, but NC Hodge correspondence is the periodic cyclic homology* $HP_*(\mathcal{C})$, *where* $\mathcal{C}$ *is a dg-category or an* A_∞*-category, and noncommutativity is the outcome of categorification. Here, Kontsevich-Soibelman claims the exitence of Hodge structure* $HP_*(\mathcal{C})$. *In particular, if* $\mathcal{C} = D^b(Coh(X))$, *then* $HP_*(\mathcal{C}) \cong H^*_{dR}(X)$.

5.21 Algebraic Variety

Definition 5.21.1 *(Hilbert Scheme)*
For a algebraic variety (or scheme) X, $Hilb^P(X) = \{Z \subset X | p(Z) = P\}$, and Hilbert scheme is $Hilb(X) = \bigoplus_P Hilb^P(X)$. Below is a list of applications.

(Correspondence with Representations)

- *(Nakajima Quiver Variety)*
  Nakajima quiver variety has a correspondence with Hilbert scheme. For Jordan quiver, Nakajima quiver variety $M_0(V,W) := \{(B,i,j) | [B,B] = 0$, stability conditions$\}/GL(V)$, where $V = \mathbb{C}^n$ and $W = \mathbb{C}$, and $Hilb(\mathbb{C}^2) \cong M_0(V,W)$.

  The special case of quiver variety is Grassmannian. Grassmannian is $Gr(k,V) = \{W \subset V | dim(W) = k\}$, and this is an algebraic variety, and in particular $Gr(k,m)$ if $dim(V) = m$. There is a clsoed embedding of Hilbert scheme $Hilb^P(\mathbb{P}^n)$ to $Gr(dim(S_d) - P(d), dim(S_d))$, where S_d is a component of $S = k[x_0, x_1, \cdots, x_n]$. The natural question is if we can extend this notion to affine grassmannian.

(Hilbert-Chow Map)
Hilbert-Chow map is $\rho : Hilb^n(X) \to Sym^n(X) = X^n/S^n$, which maps $Z \mapsto \sum_{p \in X} length(Z_p) \cdot p$, and this is a resolution of singularity. If in particular $X = \mathbb{C}^2$, then $Hilb^n(\mathbb{C}^2)$ is a Nakajima quiver variety, and ρ is a crepant resolution.

Definition 5.21.2 *(Other Important Varieties)*

- *(Character Variety)*
  For an algebraic variety X and a lienar group G (e.g. SL_n or GL_n), the whole representation space $Hom(\pi_1(X), G)$, and character variety is $\mathfrak{X}(X,G) = Hom(\pi_1(X), G)//G$ where its equivalence class is $\rho \sim g\rho g^{-1}$, this name is derived from ρ is a character.

- *(Jacobian Variety)*
  For a given curve C, Jacobian variety is $J(C) = Pic^0(C)$. In particular, if C is a genus g compact Riemann surface, then $J(C) = \mathbb{C}^g/\Lambda$.

- *(Picard Variety)*
  Picard variety $Pic(C)$ is a Picard group but also an algebraic variety. In particular, if C is a smooth curve ,$Pic(C) = J(C)$

5.22 Geometric Representation

Definition 5.22.1 *(Quiver Grassmannians)*
Quiver grassmannian appears in calculation of cluster algebra. For a quiver

representation $M = (V_i, f_a)$, $Gr_e(M) = \{$subrepresentation $N \subset M | dim(N_i) = e_i\}$, where $e = (e_i)$ is a collection of dimension corresponding to each vertex.

Definition 5.22.2 *(Cluster Algebra)*
For $\mathbb{Q}(x_1, \cdots, x_n)$, let the cluster variables be $\{x_1, \cdots, x_n\}$ and exchange matrix $B = (b_{ij}) \in M_{n\times n}(\mathbb{Z})$, and the seed is the pair (x, B), and mutation of variable x_k generates $x_k^{'}$ by $x_k x_k^{'} = \prod_{b_{ik}>0} x_i^{b_{ik}} + \prod_{b_{ik}<0} x_i^{-b_{ik}}$. Cluster algebra is a subalgebra of $\mathbb{Q}(x_1, \cdots, x_n)$ generated by the mutations.

- *(Cluster Monoidal Categorification)*
  For a cluster algebra A and a monoidal category $\mathcal{C}$, take the Grothendieck group $K_0(\mathcal{C})$ as ring isomorphic to A, and notice $K_0(\mathcal{C})$ has a multiplication induced from $\mathcal{C}$, this $\mathcal{C}$ is called category of cluster monoidal categorification.

- *(Cluster Category)*
  Cluster category is a different notion from cluster monoidal categorification. For a finite dimensional hereditary algebra H, let $D^b(mod - H)$ and $F = \tau^{-1}[1]$ where τ is AR-translation. and $\mathcal{C}_H = D^b(mod - H)/F$ is a cluster category.

5.23 Scheme

Definition 5.23.1 *(Scheme)*
Affine scheme is defined as follows: classically $Spec(R)$ is a set of prime ideals of R with Zariski topology, and affine scheme is a pair $(Spec(R), \mathcal{O}_{Spec(R)})$. Otherwise, $Spec(R) : (Comm)^{op} \to Set$ is a functor, so $Spec(R)(A) = Hom_{comm}(R, A)$.

In general, scheme is a functor $X : (Comm)^{op} \to Set$, so $X(R) = Hom_{Sch}(Spec(R), X)$ is a natural transformations.

Definition 5.23.2 *(Thin Scheme)*
Association scheme is a pair of a finite set X and its relation $R_0, R_1, \cdots, R_d$ s.t. $R_i \subset X \times X$ and $R_0 \cup R_1 \cdots \cup R_d = X \times X$ and $R_i \cap R_j = \emptyset$ and $R_0 = \{(x, x) | x \in X\}$ is diagonal. Here, A_i is a $v \times v$-matrix $(A_i)_{ab} = 1 if (x_a, x_b) \in R_i 0 else$ where $v = |X|$ is the cardinality. $A_i A_j = \sum_k p_{ij}^k A_k$.

Thin scheme is an association scheme with extra conditions. For each R_i and $x \in X$, $|\{y \in X | (x, y) \in R_i\}| \leqq 1$, and its inverse relation R_i^T $|\{x \in X | (x, y) \in R_i\}| \leqq 1$, $p_{ij}^k \in \{0, 1\}$. Further, this is non-commutative thin scheme, if $A_i A_j \neq A_j A_i$. Anyway, the idea is similar to finite groupoid.

Definition 5.23.3 *()*

Definition 5.23.4 *(NAff)*
$NAff = N(Aff)$ is a category of affine scheme $Aff = (CRing)^{op}$, transformed to an ∞-category by the nerve functor N, which is defined as a n-simplicial set, and $N(\mathcal{C})_0 = \{X \in Ob(\mathcal{C})\}$, $N(\mathcal{C})_1 = \{f : X \to Y \in Mor(\mathcal{C})\}$,

$N(\mathcal{C})_2 = \{(f,g) : X \xrightarrow{f} Y \xrightarrow{g} Z \in Mor(\mathcal{C})\}$, *and* $N(\mathcal{C})_n = \{X_0 \xrightarrow{f_1} X_1 \xrightarrow{f_2} \cdots \xrightarrow{f_n} X_n \in Mor(\mathcal{C})\}$.

Also, $Shv(NAff)$ *is a category of sheaves defined over a Grothendieck site* $NAff$. Aff *naturally has a Grothendieck site, and this induces* $NAff$ *from above, so* $NAff$ *has also a Grothendieck site.*

Definition 5.23.5 *( )*

Definition 5.23.6 *(Thin Scheme)*
Thin scheme is a reduced scheme, whose local commutative ring A *doesn't have nilradical. Otherwise, for any commutative ring* A *have quotient by nilradical* $A_{red} = A/(nil)$.

In nc thin scheme, local ring is finite dimensional k-algebra, suffice to say Artinian k-algebra (in paritcular if k is a field, then f.d. k-algebra iff Artinian).

Definition 5.23.7 *(Affine Scheme)*
Categorical definition of a scheme $X : Aff^{op} \to Sets$ *is a representable functor s.t.* $X(R) = Hom(Spec(R), X)$, *which is also a sheaf (consider* Aff^{op} *has a Grothendieck topology).*

Definition 5.23.8 *(Ind Scheme)*
Ind-scheme is $X = \varprojlim X_i$ *for inclusion sequence* $X_1 \hookrightarrow X_2 \hookrightarrow X_3 \hookrightarrow \cdots$. *For example, ind-scheme is* ∞*loop group* $LG = G(\mathbb{C}((t)))$, *and* $Gr_\infty = \varprojlim Gr(n, N)$.

Also, ind-sheaf is in particular, $C^\infty(\mathbb{R}) \cong \varprojlim_{K \subset \mathbb{R}, K \text{ compact}} C_c^\infty(\mathbb{K})$. *We can also discuss distribution.*

$Ind(\mathcal{C})$ *is an Ind-category, which is in particular, if* $\mathcal{C}$ *is a category of finite-dimensional vector spaces, and* $Ind(\mathcal{C})$ *is a category of infinite-dimensional vector spaces.*

5.24 Deformation Theory

5.24.1 integrability

Definition 5.24.1 *(Frobenius Theorem)*
Frobenius theorem claims the integrability of the distribution $N \subset M$ *over a smooth manifold* M, *and for all the global section* $X, Y \in X(TN)$ *is closed under the distribution* $[X, Y] \in X(TN)$, *then the distribution is integrable. In other words, vector field is differential operator, which means that it's an element of the dgla.*

Definition 5.24.2 *(Newlander-Nirenberg Theorem)*
Newlander-Nirenberg theorem claims the condition of the integrability of a smooth

manifold M with an almost complex structure $J^2 = -Id$, and the theorem says the manifold has a vector bundle $TM \otimes \mathbb{C} = T^{1,0} \oplus T^{0,1}$, and each $T^{1,0}, T^{0,1} \subset TM \otimes \mathbb{C}$ As a distribution is integrable by means of Frobenius theorem, if Newlander-Nirenberg tensor $N_J(X,Y)$ vanishes, and this condition is equivalent to solving Maurer-Cartan equation but in a different language.

First, Newlander-Nirenberg tensor $N_J(X,Y)$ is defined by $N_J(X,Y) = [JX, JY] - J[JX,Y] - J[X,JY] - [X,Y]$ for $X, Y \in TM$, and if this tensor is $N_J(X,Y) = 0$, the manifold is integrable, because this makes $[T^{1,0}, T^{1,0}] \subset T^{1,0}$, where $TM \otimes \mathbb{C} = T^{1,0} \oplus T^{0,1}$. What is the problem?? In real smooth manifold, distribution $N = M$ satisfies Frobenius theorem trivially, thus naturally the manifold with almost complex structure is a real manifold, and thus $[X,Y] \in X(TM)$ for $X, Y \in X(TM)$. However, we don't guarantee that $[X,Y] \in X(T^{1,0} \cap TM)$ for $X, Y \in X(T^{1,0} \cap TM)$. However, it satisfies only if M is a complex manifold.

Definition 5.24.3 *(Frölicher-Nijenhuis Bracket)*
It makes graded Lie algebra. Similarly, Nijenhuis–Richardson bracket and the Schouten–Nijenhuis bracket or whatever.

Definition 5.24.4 *(Maurer-Cartan Equation)*
For a 1-form $\omega : TG \to \mathfrak{g}$, Maurer-Cartan equation is $d\omega + \omega \wedge \omega = 0$. By using $X, Y \in TG$, the equation is derived as follows. $d\omega(X,Y) = X(\omega(Y)) - Y(\omega(X)) - \omega([X,Y])$ where $\omega([X,Y]) = [\omega(X), \omega(Y)]$, and $X(\omega(Y)) = Y(\omega(X)) = 0$ for left invariant X and Y.

Conversely, let's consider what properties Maruer-Cartan equation shows. Recall Lie group G is by definition a smooth manifold, and we discuss integrability condition of distribution of vector bundles TG. Recall Frobenius theorem says integrability condition is $d\omega \wedge \omega = 0$. If $X, Y \in ker(\omega) = D$, $\omega(X) = \omega(Y) = 0$, and $d\omega(X,Y) = 0$, which means $d\omega|_D = 0$. This $D \subset TG$ is a distribution, which satisfies Frobenius condition.

In general, a smooth manifold M has a global section of tangent bundle $\Gamma(TM)$ where we can define the Lie bracket to define Lie algebra, and its corresponding Lie group exists, for which we discuss Maurer-Cartan equation. This claim only applies locally.

Definition 5.24.5 *(A_∞ Maurer-Cartan)*
Maurer Cartan is $\sum_{n \geqq 1} \frac{1}{n!} l_n(b, b, \cdots, b) = 0$, whose A_∞ version is $\sum_{n \geqq 1} m_n(b, b, \cdots, b) = 0$.

Definition 5.24.6 *(Deformation Functor)*
For the most of the case, a deformation functor is a functor $Def_X : Art_k \to Set$ from a category of artinian rings, satisfying the Schlessinger axiom: For $A \to C \leftarrow B$ as a diagram of the morphisms of artinian rings, Def_X is a deformation

functor, $Def_X(A \times_C B) \to Def_X(A) \times_{Def_X(C)} Def_X(B)$ is a natural morphism, and if, in addition, $B \to A$ is a small extension, the previous morphism is surjective, in other words $Def_X(B) \to Def_X(A)$ (H1). If $A = k$, then two different lift doesn't have different image (H2), $Def_X(k[\epsilon]/(\epsilon^2))$ is a finite dimensional vector space (H3), and the small extension can be properly controlled (H4). If the deformation functor Def_X satisfies the Schlessinger axiom, then Def_X is pro-representable as $Def_X \cong Hom_{Art_k}(R, -)$, where R is an artinian ring.

Deformation functor is, as an example, $Def_g(A) = MC(g \otimes m_A)/\sim$ where m_A is the maximal ideal of A.

Definition 5.24.7 *(dg-Manifold)*
M_0 is a dg-manifold, if there is a graded commutative ring $\mathcal{O}_M$ with differential d s.t. $d^2 = 0$. dg-manifold is not a scheme but a derived scheme because its ring doesn't show its spectrum. If $M = \mathbb{R}$, then $\mathcal{O}^{dg}_{M,p} = \mathbb{R}[[x_1, \cdots, x_n]][\epsilon_1, \cdots, \epsilon_m]$ where $d\epsilon_i = f_i(x_1, \cdots, x_n)$, while its noncommutative version for the same $M = \mathbb{R}$ is $\mathcal{O}^{nc-dg}_{M,p} = \mathbb{R} < x_1, \cdots, x_n > [\epsilon_1, \cdots, \epsilon_m]$ where $d\epsilon_i =$ nc formula.

If we are interested, not all dg-manifold is fp-manifold but if it's localized. Formal pointed manifold is a pointed manifold (M, p) for $p \in M$ with a nbhd data $\hat{\mathcal{O}_{M,p}} \cong \mathcal{R}[[x_1, \cdots, x_n]]$.

Definition 5.24.8 *(NC Thin Scheme)*
Thin scheme is a weaker notion of scheme, that is minimizing the information of the points of the scheme. If the minimized data is noncommutativized, then it NC thin scheme. Now here dg-manifold can be non-commutativized. Namely, replacing cdga to any (noncommutative) dga.

Ind-Scheme?

5.24.2 Fermat Last Theorem

Definition 5.24.9 *(Fermat Last Theorem)*
It proof mainly depends on Shimura-Taniyama conjecture, which claims flat deformation of Galois representation results modularity.

5.25 Formal Deformation

Definition 5.25.1 *(Formal Scheme)*
Formal scheme is a scheme.

5.25.1 Hopf Algebra

Definition 5.25.2 *(Algebra)*
We mean algebra by unitary associatve algebra (this means, linear algebra or Lie algebra is not algebra).

Definition 5.25.3 *(Coalgebra)*
Coalgebra C over K is a vector space with two structure morphisms

- $\Delta : C \to C \otimes C$
- $\epsilon : C \to K$

with the following identities

- $(id_C \otimes \Delta) \circ \Delta = (\Delta \otimes id_C) \circ \Delta$ *(coassociativity)*
- $(id_C \otimes \epsilon) \circ \Delta = id_C = (\epsilon \otimes id_C) \circ \Delta$

Definition 5.25.4 *(Hopf Algebra)*
Hopf algebra H is bialgebra with antipode, and bialgebra is coalgebra but also algebra. Antipode $S : H \to H$ is a k-linear map that commutes the diagram and

$S_{c_{(1)}} c_{(2)} = c_{(1)} S_{c_{(2)}} = \epsilon(c)1$ *for all* $c \in H$.

Definition 5.25.5 *(Representation of Hopf Algebra)*
Let A be Hopf algebra, and M and N are A-modules. Then $M \otimes N$ is also A-module, with

$a(m \otimes n) = \Delta(a)(m \otimes n) = (a_1 \otimes a_2)(m \otimes n) = (a_1 m \otimes a_2 n)$ *where* $m \in M$, $n \in N$, *and* $\Delta(a) = (a_1, a_2)$.

$a(m) = \epsilon(a)m$

$(af)(m) = f(S(a)m)$ *where* $f \in M^*$ *and* $m \in M$.

5.26 Unsorted

5.26.1 GIT

Definition 5.26.1 *(GIT and Reductive Group)*
In GIT, the question is if the quotient space has a nice property as a scheme, and it's known that a stable point $G \cdot x$ is not only a set theoretical point, but also a closed point in the Zariski topology of the scheme, but the problem is whether semistable point becomes a closed point in the Zariski topology of the scheme.

We claims if the quotient group G is reductive, we can relate semistale point to stable by relating $X//G$ and $Proj(R^G)$, since Hilbert theorem says R^G is finitely generated k-algebra if G is reductive, which makes $Proj(k[X]^G)$ scheme.

5.26.2 Hilbert Scheme

Definition 5.26.2 *(Hilbert Scheme)*

Hilbert Scheme is a scheme but also moduli space.
$Hilb(n)$ of $\mathbb{P}^n$ is a moduli space of closed subschemes of projective spaces.

$Hilb(n, P)$ is a moduli space of Hilbert polynomial P.

Let $\underline{Hilb}_{X/S} : (Sch/S)^{op} \to Sets$ be the functor sending a relative scheme $T \to S$ to the set of isomorphism classes of the set

$$\underline{Hilb}_{X/S}(T) = \{ \begin{tikzcd} Z \arrow[r, "incl"] \arrow[d] & X \times_S T \arrow[r] \arrow[d] & X \arrow[d] \\ T \arrow[r, "="] & T \arrow[r] & S \end{tikzcd} : Z \to T \text{ is flat}\}/\sim$$

Universality:
$H = Hilb(n, P)$ has a universal subscheme $W \subset X \times H$ flat over H such that

- *the fiber W_x over closed points $x \in H$ are closed subschemes of X. For $Y \subset X$ denote this point x as $[Y] \in H$.*
- *H is universal wrt all flat families of X having Hilbert polynomial P. That is, given a scheme T and a flat family $W' \subset X \times T$, there is a unique morphism $\phi : T \to W$ such that $\phi^* W \cong W'$.*

Tangent Space:
The tangent space at $[Y] \in H$ is given by the global sections of the normal bundle $N_{Y/X}$. That is,
$T_{[Y]}H = H^0(Y, N_{Y/X})$

Unobstrctedness of complete intersections:
For local complete intersections Y such that $H^0(Y, N_{Y/X}) = 0$, the point $[Y] \in H$ is smooth. This implies that the deformation of Y in X is unobstructed.

(coalgebra is smooth, if it has a lifting property. morphism $F(A) \to F(A/J)$ is means surjective.)

Complete Intersection:
The ideal of V is generated by precisely codim V elements.

Example 5.27 *(Hilbert Schemes)*

- *(Fano Schemes of Hypersurfaces)*

- *(Hilbert schemes of n-points)*
  X^n/S_n is the nice geometric interpretation where the boundary loci $B \subset H$ describing the intersection of points can be tought of parametrizing points along with their tagent vectors.

  $X^{[n]} = Bl_\Delta(X \times X)/S_2$.

  Alternatively,

  $X^{[n]} = Hilb_X^P$ *if the Hilbert polynomial P is constant $P(m) = n$ for all $m \in \mathbb{Z}$*

- *(Degree d hypersurfaces)*

- *(Hilbert schemes of curves and moduli of curves)*

Definition 5.27.1 *(Heisenberg Algebra)*
Hisenberg algebra is a vector space, whose bracket is $[x, p] = i\hbar$, or the infinite dimensional version if $[a_m, a_n] = m\delta_{m+n,0}$.

- *(The Connection with Kac-Moody Lie Algebra)*
  Affine Kac-Moody Lie algebra is $\hat{\mathfrak{g}} = \mathfrak{g}[t, t^{-1}] \oplus \mathbb{C}c \oplus \mathbb{C}d$ where $\mathfrak{g}[t, t^{-1}]$ is a loop algebra, c is the center, d is the dimension operator. the Cartan subalgebra $\hat{\mathfrak{h}} = \mathfrak{h}[t, t^{-1}] \oplus \mathbb{C}c$ is the multi-dimensional Heisenberg algebra.

- *(The Connection with Hilbert Scheme)*
  The graded vector space $\bigoplus_{n \geqq 0} H^*(Hilb^n(S))$ can naturally contain the Hisenberg group action, where $Hilb^n$ means the length n subscheme of S, which means $length(Z) = dim_{\mathbb{C}}\mathcal{O}_Z$.

- *(The Connection with CFT) In 2d CFT, the mode expansion $\partial\phi(z) = \sum_{n \in \mathbb{Z}} a_n z^{-n-1}$ where $\phi(z, \overline{z})$ is a field, and its bracket $[a_m, a_n] = m\delta_{m+n,0}$ is the infinite dimensional version if Heisenberg algebra.*

5.27.1 Symplectic Geometry

Definition 5.27.2 *(diff p-form and vector field)*

- *ex1:*

  $dx_i(\partial_j) = \delta_{ij}$

- *ex2:*
  $dx_{i_1} \wedge \cdots \wedge dx_{i_p}(\partial X_{j_1}, \cdots, \partial X_{j_p}) =$
  1 *if* (i_p *and* j_p *are even permutation)*
  -1 *if* (i_p *and* j_p *are odd permutation)*
  0 *(otherwise)*

- *ex3:*
  $\omega = d_x \wedge d_y$, $X = f(x)\partial_X$, $Y = g(x)\partial_X$,
  $\omega = f(x)g(x)(d_x(\partial_X) \wedge d_y(\partial_X) - d_y(\partial_X) \wedge d_x(\partial_X))$

Definition 5.27.3 *(interior product)*
$\iota_X : \Omega^p(M) \to \Omega^{p-1}(M)$

$$\iota_X \omega(X_1, \cdots, X_{p-1}) = \omega(X, X_1, \cdots, X_{p-1})$$

for any vector fields X *and* X_i.

Definition 5.27.4 *(Kähler manifold)*

- *(Symplectic Viewpoint)*
  $g(u, v) = \omega(u, Jv)$
- *(Complex Viewpoint)*
  Let h *be the Hermitian metric.*

  $$\omega(u, v) = Re(h(iu, v)) = Im(h(u, v))$$

  $$g(u, v) = Re(h(u, v))$$
- *(Riemannian Viewpoint)*

  Kähler manifold is Riemannian manifold X *of dimension* $2n$*, whose holonomy is contained in* $U(n)$.

  J *preserves metric i.e.* $g(Ju, Jv) = g(u, v)$*, and* J *is preserved by parallel transport.*

(Kähler potential)
Let ρ *be a smooth real valued function.*

$$\omega = \frac{i}{2}\partial\bar{\partial}\rho$$

is a Kähler potential.

Definition 5.27.5 *(Deligne-Mumford Space)*

Definition 5.27.6 *(Compactification of Moduli Space)*

6 Non-Commutative Algebraic Geometry

Definition 6.0.1 *(Non-Commutative Quotient)*
In classical AG, group quotient of X by G is a stack $[X/G]$, or equivalently Lie groupoid $G \times X \rightrightarrows X$, while non-commutative quotient is $A = C^\infty(X) \rtimes G$. These satisfy Morita equivalence $QCoh([X/G]) \cong Mod(C^\infty(X) \rtimes G)$, since $QCoh([X/G])$ is also category of modules.

6.1 unsorted

Definition 6.1.1 *( )*
$A \cong A^\vee[d]$ where $[d]$ is shifting of dga. $Hom(X,Y)^\vee \cong Hom(Y,X)[d]$

Definition 6.1.2 *( )*

Definition 6.1.3 *( )*

6.2 Deformation

Definition 6.2.1 *(Kadeishvili's Theorem)*
Kadeishvili's theorem claims A_∞ existence of A_∞ structure of $m_1 = 0$ generating from A_∞ structure. For an A_∞-algebra A, its homology $H^*(A)$ is just a graded vector spaces, but we can induce A_∞ structure from A, if we let the multiplication $m_2^{'}([x],[y]) := [x] \cdot [y] := [m_2(x,y)]$, as $m_2^{'} = p \circ m_2 \circ i$ where $p : A \to H^*(A)$ and $i : H^*(A) \to A$, which automatically deterines the higher degree m_n where $n \geqq 3$, and naturally $m_1 = 0$ since projectition p nullifies.

This newly generated A_∞ algebra $H^*(A)$ is quasi-isomorphic to A, which means the map $f : A \to H^*(A)$ has $f = (f_1, f_2, f_3, \cdots)$, so that $f_i \circ m_A = m_{H^((A)} \circ f_i$ commutes, and f_1 is cohomologically isomorphic. Then the derived category $D(A)$ is defined by category of A_∞-modules up to quasi-isomorphism for A_∞ algebra A, and $A \cong H^*(A)_{min}$ is identified because of quasi-isomorphism, so $D(A) \cong D(H^*(A))_{min}$.

Definition 6.2.2 *(Homotopy Deformation)*

Definition 6.2.3 *(Formal Deformaton)*

Definition 6.2.4 *(Kontsevch Formality)*

Definition 6.2.5 *(Flat Deformation)*
In flat deformation, each deformation parameter X_t from a geometry X over S smoothly changes. The famous example is $\mathfrak{X} = Spec(k[x,y,t])/(xy - t)$, and its fiber X_t if parabola if $t \neq 0$, but X_0 is two intersecting lines.

6.3 Springer Resolution

Definition 6.3.1 *()*
For semi-simple complex Lie group G and its lie algebra $\mathfrak{g}$, $\mathfrak{B} = G/B$ is a flag variety for a Borel subgroup B, and let $\mathfrak{N} \subset \mathfrak{g}$ be nilpotent subsets but also an algebraic variety. Flag variety is a variety, since it's generated by GIT. Springer resolution is $\tilde{\mathfrak{N}} := \{(x, gB) \in \mathfrak{N} \times \mathfrak{B} | Ad(g^{-1})x \in \mathfrak{b}\} \to \mathfrak{N}$, so $(x, gB) \mapsto x$, $\mathfrak{N}$ is typically not smooth, but $\tilde{\mathfrak{N}}$ is smooth.

Definition 6.3.2 *(Beilinson-Bernstein Correspondence)*

6.4 Hochschild Homology

Definition 6.4.1 *(Cunz-Quillen)*
As an analogy of de Rham complex $\Omega^1(A)$ for the case of commutative ring A, the non-commutative version of differential form is $\Omega^1_{nc} = ker(m : A \otimes A \to A)$, where $m(a \otimes b) = ab$. In general, n-form is $\Omega^n_{nc} = \Omega^1_{nc} \otimes_A \cdots \otimes_A \Omega^1_{nc}$, but the differential d is bit different from the case of de Rham. Further, $T_A(\Omega^1 A)$ satisfying Leibniz law with $d : A \to \Omega^1 A$. The important application of Cuntz-Quillen is X-complex $X(A) : A \underset{b}{\overset{B}{\Leftrightarrow}} to \Omega^1 A/[A, \Omega^1 A]$ where b is Hochschild differential and B is Conne's differential.

Definition 6.4.2 *(Cyclic Homology)*
The motivation of cyclic homology is to connect non-commutative algbera to de Rham cohomology. Cyclic homology is defined by $HC.(A) = H.(C.(A)[[u]], b + uB)$, where b is Hochschild differential and B is Conne's differential. By carefully seeing, this relation says cyclic homology can be constructed from de Rham cohomology, but it's not the other way around, meaning cyclic homology has a fewer data than de Rham cohomology, but it still contains many invariant as Atiyah-Singer index, quantization, integration, deformation, Chern character, and trace. In particular, if A is smooth, let $A = C^\infty(M)$, and $(C.(A)[[u]], b + uB) \cong (\Omega^{\cdot}(M)[[u]], ud)$ according to HKR theorem, thus $HC_n(A) \cong \underset{i \geqq 0}{\oplus} H^{n-2i}_{dR}(M)$.

Definition 6.4.3 *(Topological Hochschild Complex)*
Topological Hochschild complex if we treat A as a ring instead of a plain vector space. The homology $THH(A)$ is given by derived tensor product $THH(A) := A \otimes^{\mathbb{L}}_{A \otimes A^{op}} A$. In other words, it's $THH(A) \cong A \otimes S^1$ via stable homotopy theory.

Definition 6.4.4 *(Categorified Hochschild Homology)*
Hochschild chain complex $C.(\mathcal{C})$ is defined by $C_n(\mathcal{C}) = \underset{X_0, X_1, \cdots, X_n \in Ob(\mathcal{C})}{\oplus} Hom(X_0, X_1) \otimes Hom(X_1, X_2) \otimes \cdots \otimes Hom(X_n, X_0)$ with Hochschild differential b, adding Conne differential B s.t. $B^2 = 0$, $Bb + bB = 0$. Now, periodic cyclic homology $HP_*(\mathcal{C})$ is defined by $HP_*(\mathcal{C}) := H_*(C.(\mathcal{C})[[u]], b + uB)$, where u is degree 2 formal variable.

6.5 Derived Geometry

Instead of using commutative ring, derived geometry uses a functor whose value is commutative ring.

noncommutative algebra treats geometry as homotopy instead of point set, which is in other words, derived geometry is simply chain complex version of commutative geometry.

Definition 6.5.1 *(Derived Scheme(Stack))*
Affine derived scheme is $Spec(A.)$, where a simplicial ring is a functor $A. : \Delta^{op} \to CRing$. Derived scheme is locally isomorphic to affine derived scheme, and derived stack is a functor $F : (SCRings)^{op} \to \infty - Groupoids$, as an analogy of classical stack as a functor $F : (Schemes)^{op} \to Groupoids$. Derived scheme can be also given by cdga, and $Ho(SCR) \cong Ho(cdga)$ is homotopy equivalence where SCR short for simplicial commutative rings and cdga short for commutative dga.

Definition 6.5.2 *(-1 Symplectic Stack)*
$\omega : \mathbb{T}_M \wedge \mathbb{T}_M \to O_M[-1]$ for derived tangent complex $\mathbb{T}_M$ and a moduli stack M, and $O_M[-1]$ is a chain complex O_M with a cohomological shift by -1, which means $(O_M[-1])^k = O_M^{k-1}$, or in other words, $\omega \in \wedge^2 \mathbb{L}_M[1]$, as cotangent complex is the dual of tangent complex. The non-degeneracy of -1-Symplectic statck is that the morphism $\omega : T_X \to T_X^\vee[-1]$ is isomorphism.

If X is n-shifted Symplectic, and L_1 and L_2 are Lagrangian, then $L_1 \times_X L_2$ is $n-1$-shifted Symplectic, esp consider if $n = 0$.

Definition 6.5.3 *(Complex Morse Theory)*
Classical critical locus $Crit(f) = \{x \in X | df_x = 0\}$ for $f : X \to \mathbb{A}^1$, that's a notion used in complex Morse theory, and $df = \frac{\partial f}{\partial z_i}$ for all i due to Cauchy-Riemann equation. Derived version of critical locus is $dCrit(f) = Spec(A^\cdot)$ where $A^\cdot$ is a derived algebra encoding $df = 0$, and its cotangent complex is $\mathbb{L}_{dCrit(f)} \cong [TX \overset{Hess(f)}{\to} T^*X]$.

Definition 6.5.4 *(Derived Intersection)*
Derived intersection can be defined by derived fiber product. Derived fiber product is $A \otimes_R^L B$ is a chain complex $A \otimes_R^L B = P. \otimes_R B$ for $P. \to A$ is a projective resolution, which is a sequence of torsions $\cdots Tor^i(A, B) \to Tor^{i-1}(A, B) \cdots Tor^0(A, B) \to 0$, and it shows the sequences of intersection. In particular, $Tor^0(A, B)$ is the intersection in commutative algebraic geometry. If we let $R = k[x, y]$, $A = k[x, y]/f(x)$, and $B = k[x, y]/g(x)$, tensor product is $A \otimes_R B = k[x, y]/((f(x)) + (g(x)))$, which is in Zariski topology $V((f(x)) + (g(x))) = V((f(x))) \cap V((g(x))))$, thus this is intersection.

Definition 6.5.5 *(Derived Moduli Stack)*
Moduli stack is a functor $\mathcal{M} : (Schemes)^{op} \to Groupoids$. Derived moduli stack is a functor $\mathcal{M}^{der} : (SCRing)^{op} \to (\infty, 1) - Groupoids$. Derived analytic stack is a functor mapping $U \mapsto C^\infty(U)$.

Definition 6.5.6 *(Cotangent Complex)*

Definition 6.5.7 *(Quasi-Smooth Morphism)*
Classically, $f : X \to Y$ is a quasi-smooth morphism if $X \overset{i}{\hookrightarrow} Y \overset{g}{\to} Z$, where i is a regular immersion, and g is a smooth morphism, and $L_{X/Y} \in Perf(X)$ and tor amplitude is $[-1, 0]$. Tor amplitude is $[-1, 0]$ means $H^i(P^\cdot \otimes^L M) = 0$ for all $\mathscr{O}_X$-module M if $i \notin [-1, 0]$.

Definition 6.5.8 *(Koszul Complex)*
Originally used to calculate cohomology of Lie algebra.

For a commutative algebra A, its Koszul complex K_s is

$$\wedge^r A^r \to \wedge^{r-1} A^r \to \cdots \to \wedge^1 A^r \to \wedge^0 A^r \cong A^r$$

where the maps send

$$\alpha_1 \wedge \cdots \wedge \alpha_k \mapsto \Sigma_{i=1}^k (-1)^{i+1} s(\alpha_i) \alpha_1 \wedge \cdots \wedge \hat{\alpha}_i \wedge \cdots \wedge \alpha_k.$$

Or we can change A^r to any A-module.

Definition 6.5.9 *(Derived Scheme)*
Derived scheme is a homotopy theoretic generalization of a scheme in which commutative rings are replaced by dga An affine derived algebraic geometry is equivalent to the theory of commutative dg-rings.

$$(X, \mathscr{O}_\cdot) = RSpec(R/(f_1) \otimes_R^L \cdots \otimes_R^L R/(f_k))$$

where $f_i \in \mathbb{C}[x_1, \cdots, x_n] = R$

Then we get a derived scheme that

$$RSpec : (dga_{\mathbb{C}})^{op} \to DerSch$$

is the étale spectrum. Since we construct a resolution

$$\begin{tikzcd}
0 \arrow[r] & R \arrow[r, "\cdot f_i"] \arrow[d] & R \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & 0 \arrow[r] & R/(f_i) \arrow[r] & 0
\end{tikzcd}$$

a derived ring $R/(f_1) \otimes_R^L \cdots \otimes_R^L R/(f_k)$, a derived tensor product, is a Koszul complex $K_R(f_1, \cdots, f_n)$. The truncation of this derived scheme to amplitude $[-1, 0]$ provides a classical model motivating derived algebraic geometry.

Notice that if we have a projective scheme

$Proj(\frac{\mathbb{Z}[x_1,\cdots,x_n]}{(f_1,\cdots,f_k)})$

where $deg(f_i) = d_i$ *we can construct the derived scheme* $(\mathbb{P}^n, \mathscr{E}^{\cdot}, (f_1, \cdots, f_k))$ *where*

$\mathscr{E}^{\cdot} = [\mathscr{O}(-d_1) \oplus \cdots \oplus \mathscr{O}(-d_k) \to^{(\cdot f_1,\cdots,\cdot f_k)} \mathscr{O}]$

with amplitude $[-1, 0]$.

Definition 6.5.10 *(Cotangent Complex)*
Let $(A., d)$ *be a fixed dga over char* 0. *Then* $A.$*-dga* $(R., d_R)$ *is called semi-free if*

- $R.$ *is a polynomial algebra over* $A.$.
- *there exists a filtration* $= I_0 \subset I_1 \subset \cdots$ *such that* $\cup I_n = I$, *and* $s(x_i) \in A.[(x_j)_{j\in I_n}]$ *for any* $x_i \in I_{n+1}$.

The relative cotangent complex of an $(A., d)$*-dga* $(B., d_B)$ *can be constructed using semi-free resolution* $(R., d_R) \to (B., d_B)$, *which is defined as*

$\mathbb{L}_{B./A.} = \Omega_{R./A.} \otimes_{R.} B.$

Example 6.6 *(Cotangent Complexes)*
For the cotangnet complex of hypersurface $X = \mathbb{V}(f) \subset \mathbb{A}^n_{\mathbb{C}}$,

dga $K_R(f)$ *representing the derived enhancement of* X. *Its cotangent complex is*

$0 \to R \cdot ds \to^{\Phi} \to \oplus_i R \cdot dx_i to 0$

where $\Phi(gds) = g \cdot df$ *and* d *is the universal derivation. If we take complete intersection, then the Koszul complex*

$R^{\cdot} = \frac{\mathbb{C}[x_1,\cdots,x_n]}{(f_1)} \otimes^L_{\mathbb{C}[x_1,\cdots,x_n]} \cdots \otimes^L_{\mathbb{C}[x_1,\cdots,x_n]} \frac{\mathbb{C}[x_1,\cdots,x_n]}{(f_k)}$

is quasi-isomorphic to

$\frac{\mathbb{C}[x_1,\cdots,x_n]}{(f_1,\cdots,f_k)}[+0]$

6.7 Monoidal Category

Definition 6.7.1 *(Monoidal Category)*
Monoidal category $(\mathcal{C}, \otimes, I, \alpha, \lambda, \rho)$ *is a categoery* $\mathcal{C}$ *with binary operation* $\otimes$:

$\mathcal{C}\times\mathcal{C} \to \mathcal{C}$, and identity $I \in \mathcal{C}$ and associator $\alpha_{A,B,C} : (A\otimes B)\otimes C \to A\otimes(B\otimes C)$ and $\lambda_A : I \otimes A \to A$ and $\rho_A : A \otimes I \to A$ with pentagon coherence conditions, or this axioms can be interpreted as an isomorphism as $I \otimes A \cong A \cong A \otimes I$, and $(A \otimes B) \otimes C \cong A \otimes (B \otimes C)$. This monoidal category is often used to describe operadic structures. That case, for the operad O, the object of the monoidal category is O-algebra.

In particular, $\mathcal{C}$ is symmetric monoidal category, if $\sigma_{A,B} : A \otimes B \to B \otimes A$ is a commutator, and $\sigma_{A,B} \circ \sigma_{B,A} = id$, and this is for L_∞-operad, while braided monoidal category if $\sigma_{A,B} \circ \sigma_{B,A} \neq id$, and this is for for E_2-operad. However, many of operads are classified to the neither, so they are just plain monoidal. Also, modular category is a monoidal category with even stronger than symmetric monoidal category.

Definition 6.7.2 *(Koszul Duality)*

Definition 6.7.3

6.8 Morita Equivalence

Definition 6.8.1 *(Morita Equivalence)*
Morita equiavlence is an equivalence of categories $A-Mod \cong B-Mod$ where A and B are rings. The necessity and suffice condition is $B \cong End_A(P)$ where P is a projective module. In other words, there exists a functor $-\otimes_A P : A-Mod \to B-Mod$ is exact and equivalence of categories. Especially if $A = k \oplus k$ and $B = k[x]/(x^2)$, then $A_m$od and $B-$mod are equivalence of categories, but notice A and B are both short exact sequence $0 \to k \to A$ or $B \to k \to 0$, so intuitively Morita equivalence is an equivalence of short exact sequences.

The similar notions of Morita equivalence can be (higher) K-theory, and $K_n(A) \cong K_n(B)$ if $A \sim B$ is Morita equivalent, but it's not vice versa, where $K_n(A) = \pi_n(BGL_\infty(A)^+)$ for general $n \geqq 1$ where $+$ is a homotopy version of abelianization, and in particular if $n = 0$, Grothendieck group $K_0(A)$ for a ring A, and if $n = 1$, $K_1(A) = GL_\infty(A)^{ab}$ is also Morita equivalent, where ab is abelianization, given by $GL_\infty(A)/[GL_\infty(A), GL_\infty(A)]$.

Definition 6.8.2 *(Derived/A_∞ Morita Equivalence)*
Let A and B are dga, then A and B are derived Morita equivalent, if $D(A) \cong D(B)$ where $D(A) = DG-Amod[qiso^{-1}]$, and this is just a localization of DG-category $DG-Amod$. but more explicitly, it's given by homotopy category as $D(A) = H^0(Ho(DG-Amod[qiso^{-1}]))$, where $\mathscr{C}$ is a DG-category and $Ho(\mathscr{C})$ is the homotopy category. In addition, $H^0(Hom_{\mathscr{C}}(X,Y))$ describes its equivalence class of homotopy $f : X \to Y$, in other words, $H^0(Hom_{\mathscr{C}}(X,Y)) \cong (Hom_{Ho(\mathscr{C})}(X,Y))$.

A_∞ Morita equivalence is $D_\infty(A) \cong D_\infty(B)$ is ∞-equivalence where D_∞ means category of A_∞-modules, and this is important in homological mirror symmetry $D_\infty(\mathscr{F}(Y)) \cong D_\infty(Coh(X))$.

Definition 6.8.3 *(Hochschild and Morita Equivalence)*
If $A \sim B$ is Morita equivalent, then $HH^{\cdot}(A) \cong HH^{\cdot}(B)$.

Definition 6.8.4 *(Monoidal Category)*

Definition 6.8.5 *()*

6.9 Groupoid

Definition 6.9.1 *(Groupoid)*
Groupoid is a category $\mathscr{C}$, whose morphisms are all invertible with $s, t : Mor(\mathscr{C}) \to Ob(\mathscr{C})$ s.t. if $s(g) = t(h)$ for $g, h \in Mor(\mathscr{C})$, the composition $g \circ h$ is defined. Or $\mathscr{C}$ is Lie groupoid only if we replace $G = Mor(\mathscr{C})$ and $M = Ob((C))$ are smooth manifolds, and in particular $s, t : G \to M$, $i : G \to G$, $e : M \to G$, $m : G \times_M G \to G$ where $G \times_M G = \{(g, h) \in G \times G | s(g) = t(h)\}$.

- *(Deligne Groupoid)*
  It's Deligne groupoid, if $Ob(\mathscr{C})$ is a solution set of Maurer-Cartan equations $MC(\mathfrak{g})$ where $\mathfrak{g}$ is dgla, then there is an action $\alpha \mapsto g \cdot \alpha$ where $g \in exp(\mathfrak{g}^0)$, so we can let $Mor(\mathscr{C}) = exp(\mathfrak{g}^0)$, and the category $\mathscr{C}$ is a groupoid. Explicitly, $\{\alpha \in \mathfrak{g}^1 | d\alpha + \imath 2[\alpha, \alpha] = 0\}$. Homotopy deformation of a group is a groupoid, which is often a Deligne groupoid. However, deformation of groupoid is ∞-groupoid by using L_∞-algebroid, and very complicated.

- *(Fundamental Groupoid)*
  For a topological space X, let $Ob(\Pi_1(X)) = X$ and $Mor(\Pi_1(X))$ consists of $Mor(x, y) = \{[\gamma] | \gamma : [0, 1] \to X \ \gamma(0) = x$ and $\gamma(1) = y\}$ where $x, y \in X$, and just be careful, x, y are objects of X, so $Mor(x, y)$ doesn't mean morphism of point x to point y, x, y are just symbolic objects. Then $\Pi_1(X)$ is a fundamental groupoid.

Definition 6.9.2 *(Lie Algebroid)*
Lie algebroid is a global generalization of Lie algebra. For a vector bundle $A \to M$, the bracket $[-, -]$ is defined over the section as $[s_1, fs_2] = f[s_1, s_2] + (\rho(s_1)f)s_2$. Now graded bundle $A^{\cdot} \to M$ for L_∞ algebroid. This is important since L_∞ algebroid corresponds to local data of Fukaya category (since Fukaya is Ainfty category, we have categorified version of Koszul duality).

Definition 6.9.3 *(∞-Groupoid)*
∞-groupoid is not a category, but ∞-category. But as analogy to groupoid, ∞

version of fundamental groupoid consists of the object $\pi_0(X), \pi_1(X), \pi_2(X), \pi_3(X), \cdots$, *and the* 1*-morphism is* $\pi_1(X), \pi_2(X), \pi_3(X), \pi_4(X), \cdots$, *and the* 2*-morphism is* $\pi_2(X), \pi_3(X), \pi_4(X), \pi_5(X), \cdots$, *and its higher orders and so on.*

Definition 6.9.4 *(L_∞ Algebroid)*

Proposition 6.10 *(Stack From Groupoid)*
Given a groupoid $G_0 \rightrightarrows G_1$, *we can generate stack* $[G_0/G_1]$. *As a set theoretical quotient, its equivalence* $x \sim y$ *is given if* $\exists g$ *s.t.* $s(g) = t(g)$, *or in other words, if* x *has an arrow to* y, *they are considered to be equivalent. Moreover, it's a stack, since* $G_0 \to [G_0/G_1]$ *is a fibration. Conversely, we can generates groupoid from stack.*

Definition 6.10.1 *()*

6.11 Orbifold

Definition 6.11.1 *Groupoid Groupoid is a small category in which all morphisms are isomorphism (invertible).*

Example 6.12 *Groupoid*

- *(Symplectic Groupoid)*
  In Poisson geometry, a Sympletic groupoid is a Lie groupoid.

Definition 6.12.1 *Orbifold definition using Lie groupoid Let* $G_1 \rightrightarrows G_0$ *be a morphism where* G_0 *is a set of objects and* G_1 *is a set of arrows, and the structural maps* $s, t : G_1 \rightrightarrows G_0$ *and other structural maps i.e. unit, invert maps. It's called Lie groupoid if* G_1 *and* G_0 *are smooth manifolds, all maps are smooth, and* s, t *are submersions.*

The intersection of fibers $G_1(x) = s^{-1}(x) \cap t^{-1}(x)$ *is a Lie group called isotropy group of* G_1 *at* x.

A Lie groupoid is proper if $(s, t) : G_1 \to G_0 \times G_0$ *is a proper map.*

A Lie groupoid is Etale if both the source and target morphisms are local diffeomorphisms.

An orbifold groupoid is either

- *a proper etale Lie groupoid*
- *a proper Lie groupoid whose isotropies are discrete spaces*

Let $M \rightrightarrows G$ be a groupoid, and $|M/G|$ be the orbit space of the Lie groupoid G i.e. quotient of M by equivalence $x \sim y$ if there exists $g \in G$ such that $s(g) = x$ and $t(g) = y$. This shows that orbifolds are particular kind of differentiable stack.

An orbifold structure on Hausdorff space X is an Morita equivalence class of homeomorphism $|M/G| \cong X$.

6.13 Lie Gropoid/Algebroid

Definition 6.13.1 *(Fibered Category)*
For a functor $p : \mathscr{S} \to \mathscr{C}$, the category $\mathscr{S}$ is a fibered category if the morphism $f : V \to U \in Mor(\mathscr{C})$ induces $f^*x \to x$ for some $x \in \mathscr{S}$, and they make the strongly cartesian product.

$$\begin{tikzcd} f^*x \arrow[r, "p"] \arrow[d] & V \arrow[d, "f"] \\ x \arrow[r, "p"] & U \end{tikzcd}$$

A strong cartesian product is .

$$\begin{tikzcd} z \arrow[r] \arrow[d] & ? \arrow[r] \arrow[d, "p"] & x \arrow[d, "p"] \\ p(z) \arrow[r] & V \arrow[r, "f"] & U \end{tikzcd}$$

If such a morphism $V \times_U x \to x$ exists, then it's called a strongly cartesian morphism.

Lie group/algebra are tools to study geometry of smooth manifold, but Lie group/algebra only define local property of the manifold, so I'm interested in Lie groupoid/algebroid generalization to discuss global situation. We will use category theoretic language for this.

Definition 6.13.2 *(Groupoid)*

Definition 6.13.3 *(A category fibered in groupoid)*
Let $p : \mathscr{F} \to \mathscr{C}$ be a functor of categories $\mathscr{F}$ and $\mathscr{C}$. For $x \in \mathscr{C}$, $p^{-1}(x) \in \mathscr{F}_c \subset \mathscr{F}$ be a subcategory, which we name by fiber category.

Definition 6.13.4 *(Lie Groupoid)*
Let G and M be smooth manifolds. Lie groupoid $G \rightrightarrows M$ consists of several morphisms.

- *(Two arrows)*
  $s, t : G \rightrightarrows M$ *be smooth submersions.*

- *(Multiplication map)*
  m: $G^{(2)} := \{(g, h) | s(g) = t(h)\} \subset G \times G \to G$

- *(Unit map)*
  $u : M \to G$
  Using unit map, identity map 1_x *is defined as*
  $1_x = u(x)$

- *(Inverse map)*
  $i : G \to G$
  $g^{-1} := i(g)$

Example 6.14 *(Lie Subgroupoid)*
For a Lie groupoid $G \rightrightarrows M$*, its Lie subgroupoid is* $H \rightrightarrows N$ *where* $H \subset G$ *is an immersed submanifold. Some examples of Lie subgroupoid might be below*

- *(Unit Lie subgroupoid)*
  $u(M)$ *is a sub Lie groupoid*

- *(Inner subgroupoid)*
  $IG = \{g \in G | s(g) = t(g)\}$

Example 6.15 *(Notable Example of Lie Groupoid – Trivial and Extreme)*

- *(Group)*
  A Lie groupoid $G \rightrightarrows *$ *is the same as Lie group.*

- *(Pair groipoid)*
  $M \times M \rightrightarrows M$ *with precisely one morphism from one another (i.e.* $s = t$*).*

- *(Trivial Groupoid)*
  $M \times G \times M \rightrightarrows M$ *with structure maps*

  - $s(x, g, y) = y$
  - $t(x, g, y) = x$
  - $m((x, g, y), (y, h, z)) = (x, gh, z)$
  - $u(x) = (x, 1, x)$
  - $i(x, g, y) = (y, g^{-1}, x)$

- *(Unit Groupoid)*
  $u(M) \rightrightarrows M$

Lie algebroid is an analogy of Lie algebra.

Definition 6.15.1 *(Lie Algebroid)*

Lie algebroid $(A, [\cdot,\cdot], \rho)$ *consisting of*

- *A vector bundle* A *over a manifold* M*.*
- *A Lie bracket* $[\cdot,\cdot]$ *on its space of sections* $\Gamma(A)$
- *A morphism of vector bundles* $\rho : A \to TM$*, called an anchor, where* TM *is the tangent bundle of* M*.*

with product rule

$$[X, fY] = \rho(X)f \cdot Y + f[X, Y]$$

where $X, Y \in \Gamma(A)$*,* $f \in C^\infty(M)$*, and* $\rho(X)f$ *is an image of* f *via derivation* $\rho(X)$*.*

Example 6.16 *(Example of Lie Algebroid)*

- *(Tangent Lie Algebroid)*
  $TM \to M$ *and* $\rho = id_{TM}$ *is an identity.*
- *(Lie algebroid to a point)*
  $TM \to *$ *is just a Lie algebra.*
- *(Any bundle of Lie algebra)*
  Yes, and $\rho = id$*.*

6.17 Differentiable Stacks

Differentiable stack is an analogue of algebraic stack, and it is always Morita equivalent to Lie groupoid.

Definition 6.17.1 *(Differentiable Stacks)*

Definition 6.17.2 *(Quotient Stack)*
Let G *be an affine group scheme over* S*, and* X *be an* S*-scheme on which* G *acts. A quotient stack* $[X/G]$ *is a stack where for each* T *defines a category* $[X/G](T)$*, and an object* $P \to T \in Ob([X/G](T))$ *is a principal* G*-bundle together with an equivariant map* $P \to X$*.*

Definition 6.17.3 *(Categorical Quotient)*
X/G *A categorical quotient* X/G *is a morphism* $\pi : X \to Y$ *that coequalizes* $\sigma, pr_2 : G \otimes X \to X$*, and that satisfies universal property: the other equalizer can be factored by this. Note* π *need not to be surjective.*

Notable example of categorical quotients are geometric quotient G/H *and GIT* $X//G$*.*

symplectic reduction (symplectic quotient)
phase space cotangent bundle

6.18 Category Fibred in Gropoids

Definition 6.18.1 *(Categorical Quotient)*

Let X be an object of a category $\mathscr{C}$, G be a group. A categorical quotient X/G is a morphism $\pi : X \to Y$ such that the group action and projection $\sigma, p_2 : G \times X \to X$ is coequalized by $\pi \circ \sigma = \pi \circ p_2$, and any other morphisms $X \to Z$ factors by it.

Definition 6.18.2 *(Quotient Stack)*
Let X be a S-scheme and G be an affine group S-scheme, and G acts on X. A quotient stack $[X/G]$ is a category over the category of S-schemes such that

- *an object is a pricipal G-bundle $P \to T$ together with an equivariant map $P \to X$.*
- *a morphism is a bundle map from $P \to T$ to $P' \to T'$ compatible with equivariant maps $P \to X$ and $P' \to X$*

A quotient stack is always an algebraic stack.

ex:

- *(trivial case)*
  $[pt/pt] = Sch_S$
- *()*
  For $[X/pt]$, the affine group pt is trivial, so the principal G-bundle is trivial bundle $T \to T$, that can be identified to T itself. Hence, the object is identified with a morphism $T \to X$, but since the group G is trivial, equivarint map could be any morphism $T \to X$.
- *()*
  For $[pt/G]$, an equivariant map $X \to pt$ is always trivial, which means unique, so it's naturally 1-to-1 to category of pricipal G-bundle of any basement S-scheme T.
- *(inclusion relation)*
  If $Y \subset X$ is a subscheme and $H \subset G$ is a subgroup scheme, then we have an inclusion relation

  $[Y/H] \subset [Y/G] \subset [Y/G] \subset [X/G]$

Definition 6.18.3 *(Algebraic Stack)*
An algebraic stack is a fibred category

$$p : \mathscr{X} \to (Sch/S)_{fppf}$$

over $(Sch/S)_{fppf}$*, which a category of scheme added with Grothethendieck topology, and*

- $\mathscr{X}$ *is a stack in groupoid over* $(Sch/S)_{fppf}$*.*
- $\Delta : \mathscr{X} \to \mathscr{X} \times \mathscr{X}$ *is representable*
- *there exist a scheme* $U \in (Sch/S)_{fppf}$ *and a* 1*-morphism* $(Sch/S)_{fppf} \to \mathscr{X}$ *which is surjective and smooth.*

Diagonal morphism is representable ??
flat and smooth morphism ??

Definition 6.18.4 *(Stack)*
Prestack $\mathscr{S}$ *over a presite* X *is a* 2*-category. For each object* $U \in \mathscr{C}_X$*,* $\mathscr{S}(U)$ *is a category.*

Stack is a separated prestack.

Definition 6.18.5 *(Morita Equivalence)*
Let $G \rightrightarrows M$ *and* $H \rightrightarrows N$ *be Lie groupoids. They are Morita equivalent if there exists* P *such that*

$P \to M$ *is a principal* H*-bundle while* $P \to N$ *is a principal* G*-bundle such that two actions on* P *commutes.*

Definition 6.18.6 *(Differentiable Stack)*

Definition 6.18.7 *(Homological Perturbation)*
Homological perturbation is similar to deformation, but has different nuance. For chain complex (C, d)*, we paraphrase the chain complex to its homology* $(H, 0)$ *to reduce the problems, and homological perturbation is an operation to define differential operator on its homology induced from the chain complex. Let* $i : H \to C$*,* $p : C \to H$*, and* $h : C \to C[-1]$*, satisfying the homological retract* $pi = id_H$ *and* $ip + dh + hd = id_C$*, and by using this, we define new operation on* H $d_H = p(\sum_{n \geqq 0} (\delta h)^n \delta)i$*.*

Definition 6.18.8 *(Normal Bundle)*

- *(Normal Bundle)*

- *(Tangent Bundle)*

- *(Cotangent Bundle)*

Definition 6.18.9 *(Cotangent Complex)*

Definition 6.18.10 *(d-Critical Structure)*
A d-critical structure is a morphism $s : \mathbb{C}_X \to (O_X \overset{d_{dR}}{\to} \mathbb{L}_X)$*, where the function underlying* s *vanishes on* $X^{red} \subset X$*, and there is an open cover of* X *by* $Crit(f)$ *with* s *determined by* f

Proposition 6.19 *()*
if $\mathfrak{X}$ *is a derived scheme with a (-1)-shifted symplectic structure, then* $X = t_0(\mathfrak{X})^{an}$ *is a d-critical locus with* $K_X^{vir} = det(\mathbb{L}_{\mathfrak{X}})|_{X^{red}}$

Definition 6.19.1 *(Odd Tangent Bundle)*
ΠTM *is an odd tangent bundle consisting* (x^i, v^i) *where* Π *can map functorially* $v^i \mapsto \theta^i$*, and* $\theta^i\theta^j = -\theta^j\theta^i$*. This is interesting because odd tangent bundle satisfies* $C^\infty(\Pi TM) \cong \Omega^{\cdot}(M)$ *by a morphism* $\theta^i \leftrightarrow dx^i$*.*

non-commutative gometry is to consider geometry as homotopy than an individual point set. L_∞ algebra is a homotopy, which describes the local property of the geometry.

dgla isn't coalgebra, but more or less. $V = \oplus V^i$ isn't coalgebra, but $S(V[1])$ is coalgebra.

Definition 6.19.2 *(Smooth Coalgebra)*
Smooth coalgebra is a coalgebra, whose dual algebra is smooth algebra, and smoothness is of finite dimensionality.

Definition 6.19.3 *(Unital/Non-Unital* A_∞ *Algebra)*

Definition 6.19.4 *(dg Calabi-Yau)*
Calabi Yau variety is smooth compact complex manifold X *and* $K_X = \Omega_X^{dim(X)}$ *is flat, and* $H^i(X, O_X) = 0$ *for* $0 < i < dim(X)$*. The algebraic version of Calabi-Yau is given, if in addition dga* A *is dg Calabi Yau, if* A *is homologically smooth, and the existence of duality* $RHom_{A^e}(A, A \otimes A) \cong A[d]$*, or categorically* $\mathcal{C}$ *is dg Calabi-Yau, if for* $X, Y \in \mathcal{C}$*,* $Hom(X, Y) \cong Hom(Y, X)^*[d]$ *for some* d*.*

It satisfies $RHom_{\mathcal{C}\otimes\mathcal{C}^{op}}(\mathcal{C}, \mathcal{C}\otimes\mathcal{C}) \cong \mathcal{C}[d]$*, or Hochschild homology has duality* $HH^*(\mathcal{C}) \cong HH_*(\mathcal{C})^*[d]$*. The categorical tensor means the objects are* $(X, Y) \in \mathcal{C}$ *and morphisms are* $Hom_{\mathcal{C}\otimes\mathcal{C}^{op}}((X_1, Y_1), (X_2, Y_2)) = Hom_{\mathcal{C}}(X_1, X_2) \otimes Hom_{\mathcal{C}^{op}}(Y_1, Y_2)$*.*

Definition 6.19.5 *(Deformation Functor)*
A deformation functor is a functor $Def_X : Art_k \to Set$. Or deformation functor is constructible from DGLA $(\mathfrak{g}, d, [\cdot,\cdot])$, where $d : \mathfrak{g}^i \to \mathfrak{g}^{i+1}$, and $[\cdot,\cdot] : \mathfrak{g}^i \times \mathfrak{g}^j \to \mathfrak{g}^{i+j}$ are graded morphisms. $Def_{\mathfrak{g}}(A) = \frac{x \in \mathfrak{g}^1 \otimes \mathfrak{m}_A | dx + \frac{1}{2}[x,x]=0}{gauge\ equivalence}$. In fact, any geoemtric deformation is all about dgla.

Definition 6.19.6 *(Homotopy Action)*
An L_∞-algebra L is homotopy invariant by action of a group G, means the any action $g \cdot L \sim L$ is homotopy invariant. This G is often a gauge group.

Definition 6.19.7 *(Deformation Quantization)*
For a Poisson manifold $(M, \{\cdot,\cdot\})$, we'll generate $\star$ product $f \star g = fg + \hbar B_1(f,g) + \hbar^2 B_2(f,g) + \cdots$ from the oridinary product $f \cdot g$ where B_i is a bidifferential operator, and the $\star$ product should satisfy $f \star g - g \star f = i\hbar\{f,g\} + O(\hbar^2)$. This $\star$-product is associative. An example of $\star$-product is Moyal product.

Definition 6.19.8 *(L_∞-Module)*
L_∞-module is an abelian group, that cosists of actions $\mu_1 : M \to M$, $\mu_2 : L \otimes M \to M$, $\mu_3 : L^{\otimes 2} \otimes M \to M$, $\mu_4 : L^{\otimes 3} \otimes M \to M$, $\cdots$, and L_∞-module is $S(L[1]) \otimes M$ and its coderivation is $D = \overset{i}{\Sigma} \mu_i$ s.t. $D^2 = 0$. In particular, for Lie algebra $\mathfrak{g}$, Lie algebra module M is given by action $\rho : \mathfrak{g} \times M \to M$, such that $[x,y] \cdot m = x \cdot (y \cdot m) - y \cdot (x \cdot m)$. The famous example of L_∞-module is Hochschild chain $C.(A,A)$, as a $C^{\cdot}(A,A)$-module as considered L_∞-module, since Hochschild cochain $C^{\cdot}(A,A)$ is dgla.

Definition 6.19.9 *(Kontsevich Formality)*
Kontsevich formality is a theorem that connects Poisson geometry and deformation quantization, and it claims the existence of quasi-isomorphism $T_{poly}(M) \cong D_{poly}(M)$ for the case of $M = \mathbb{R}^n$, where $T_{poly}(M)$ is a multi-vector field complex over a manifold, and $D_{poly}(M)$ is a polydifferential operator Hochschild complex. Both $T_{poly}(M)$ and $D_{poly}(M)$ are DGLA, not only L_∞.

$T_{poly}(M) = \overset{\infty}{\underset{k=-1}{\oplus}} T^k_{poly}(M)$ is defined by $T^k_{poly}(M) = \Gamma(M, \wedge^{k+1} TM)$, and $D_{poly}(M)$ is defined by $D^k_{poly}(M) = \{$(k+1)th-term polydifferential operators$|\}$ where $A = C^\infty(M)$, so in particular, $D^{-1}_{poly}(M) = C^\infty(M)$, $D^0_{poly}(M) = \{$differential operators $A \to A\}$, and $D^1_{poly}(M) = \{$bidifferential operators$B(f,g)\}$ where $B(f,g) = \Sigma a^{ij}(x) \partial_i f \partial_j g$.

Definition 6.19.10 *(Quantum Torus)*
An ordinary torus is algebraically a commutative Laurant polynomial ring $\mathbb{C}[x_i^{\pm 1}]$. Its quantum version is $\mathbb{C} < x_i^{\pm 1} > /(x_i y_j = q_{ij} x_j x_i)$, where $q_{ij} \in \mathbb{C}^\times$ $q_{ij} q_{ji} = 1$, and if $q_{ij} \to 1$, then the quantum torus becomes the ordinary torus.

Definition 6.19.11 *(Koszul Duality)*
The Koszul duality is a duality of operads. For an operad P, it is given by $P = F(E)/(R)$, where E is an operad where $E(n) = span\{\mu(\cdot, \cdots, \cdot)\}$, where

μ is an n-array operations on P, and $R \subset E \otimes E$, and $F(E)$ is a free operad. Now the dual operad P is $P = F(E^*)/(R^\perp)$ where E^* is the dual operad $E^* = Hom(E, k)$, and $R^\perp \subset E^* \otimes E^*$, and $R^\perp = \{f \in E^* \otimes E^* | f(r) = 0 \ \forall r \in R\}$. n example of Koszul duality is $Ass = Lie$, whose dual is $Lie = Com$, and moreover, $Ass_\infty = Lie_\infty$, whose dual is $Lie_\infty = Com_\infty$.

Definition 6.19.12 *(Gerstenhaber Algebra)*
Gerstenhaber algebra is a graded commutative algebra with a Lie bracket of degree -1 satisfying the Poisson identity, satisfying

- $(ab)c = a(bc)$
- $ab = (-1)^{|a||b|}ba$
- $|ab| = |a| + |b|$
- $|[a, b]| = |a| + |b| = -1$
- $[a, bc] = [a, b]c + (-1)^{(|a|-1)|b|}b[a, c]$ *(Poisson Identity)*
- $[a, b] = -(-1)^{(|a|-1)(|b|-1)}[b, a]$
- $[a, [b, c]] = [[a, b], c] + (-1)^{(|a|-1)(|b|-1)}[b, [a, c]]$

Definition 6.19.13 *(Graded Symmetric Algebra)*
Graded symmetric algebra M as a R-module is a quotient of tensor algebra by an ideal I, so $M = T(R)/I$, where I is generated by $xy - (-1)^{|x||y|}yx$.

Definition 6.19.14 *(E_n-Algebra)*
E_n-algebra is the higher dimensional generalization of E_2-algebra, and its n-array operator $[-, \cdots, -]$ is degree $1-n$, so naturally its homology has $[-, \cdots, -]$-bracket, and its homology is n-Gerstanhaber algebra. It appears in higher product in Fukaya category and deformation quantization. Also, E_n-operad is alternatively called the configuration space operad, since $E_n(k) \cong Conf_k(\mathbb{R}^n)$. The perverse sheaf $\mathcal{F}_k \in Perv(Conf_k(\mathbb{R}^n))$ has fusion operation $\mathcal{F}_k \otimes \mathcal{F}_l \to \mathcal{F}_{k+l}$ when the points collide, while $\mathcal{F}_k \boxtimes \mathcal{F}_l \to \mathcal{F}_{k+l}$ if the points are separated, which corresponds to the operadic composition. Furthermore, geometric Langlands is $Perv_{G(\mathcal{O})}(Gr_G) \cong Rep(G^\vee)$ where $\mathcal{O}$ is $\mathbb{C}[[t]]$.

Definition 6.19.15 *(E_∞-Algebra)*
E_∞-algebra is actually homotopy-commutative and homotopy-associative algebra. E_∞-operad is an topological operad quasi-isomorphic to Com-operad with each $E_\infty(n)$ is contractible, with free Σ_n action. Or more simply, E_∞-operad is a just colimit $E_\infty = colimit E_n$.

Definition 6.19.16 *(Picard-Lefschetz)*
For $f : X \to \mathbb{C}$, the fiber $X_t = f^{-1}(t)$ is smooth most of the case, but let there is some $t_0 \in \mathbb{C}$ is singular. homology of fiber is twisted around t_0, that's monodromy. Also, let vanishing cycle be $\delta \in H_n(X_t)$, which vanishes as $t \to$

t_0. The purpose of Picard-Lefschetz is to describe how this δ affects to the other cycles. The monodromy action $T : H_n(X_t) \to H_n(X_t)$ is $T(\gamma) = \gamma + (-1)^{\frac{n(n+1)}{2}} < \gamma, \delta > \delta$ where γ is any cycle, and $< \cdot, \cdot >$ is an intersection pairing.

Definition 6.19.17 *(Quantum Torus)*
$x_\gamma x_{\gamma'} = q^{<\gamma,\gamma'>} x_{\gamma+\gamma'}$

Definition 6.19.18 *(Wall Crossing Formula)*
For a lattice $\Gamma \cong \mathbb{Z}^n$ with a bracket $< \cdot, \cdot >: \Gamma \times \Gamma \to \mathbb{Z}$, central charge $Z : \Gamma \to \mathbb{C}$, and the phase order for stability is $arg(Z(\gamma))$. $\Omega(\gamma) \in \mathbb{Z}$ is BPS invariant.

The fundamental automorphism $\mathcal{S}_\gamma : x_{\gamma'} \mapsto (1-x_\gamma)^{<\gamma,\gamma'>} x_{\gamma'}$. The Kontsevich-Soibelman formula is $\prod_{left} S_\gamma^{\Omega_-(\gamma)} = \prod_{right} S_\gamma^{\Omega_+(\gamma)}$.

Definition 6.19.19 *(Fukaya-Seidel)*
Fukaya-Seidel is an A_∞ category equipping KS wall crossing geometrically, or we could say the Lagrangian Symplectic version of complex Morse theory, and it contains spherical twist $T_L(K) = cone(CF^*(L, K) \otimes L \to K)$, which is a twist for the vanishing cycles, where the Lefschetz fibration $W : X \to \mathbb{C}$ is defined. This is not Fukaya categeory, but it's still an analogue. Each object is Lagrangian submanifold same as Fukaya category, but the Lagrangian submanifold doesn't need to be compact. For KS formula, electoric lattice $\Gamma = K_0(FS(W))$ with Euler pairing, and central charge is $\int_\gamma \Omega$. In mirror symmetry, Fukaya-Seidel is A-model, and its corresponding B-model is matrix factorization.

Definition 6.19.20 *(Ooguri-Vafa)*
The notion of Ooguri-Vafa is TQFT perspective of BPS counting. It is interpreting the 3d knot to the boundary condition of IIA/IIB D-brane. the free energy of open topological string corresponds to a knot K is $F = \sum_{d,m} \frac{N_{d,m}}{m}(q^{m/2} - q^{-m/2})e^{-mdt}$ where $N_{d,m}$ is a BPS integer.

References

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